

Syllabus

(Session: 2020/2021; 2021/2022 and 2022/2023)

B.Sc. (Honors)

in

Information and Communication Technology (ICT)



Institute of Information Technology
Jahangirnagar University
Savar, Dhaka-1342.

About the Institute

Institute of Information Technology of Jahangirnagar University (IIT-JU) started its journey on October 2009 to create efficient Information & Communication Technology professionals. The erstwhile Computer and Information Technology Institute (CITI) previously (known as Computer Center) of JU has been taken as the backbone of the institute. IIT-JU currently offers Bachelor of Science in Information and Communication Technology (BICT), Master of Science in Information and Communication Technology (MICT), Master of Philosophy (MPhil) in Information and Communication Technology, Doctor of Philosophy (Ph.D.) in Information and Communication Technology, Professional Master in Information Technology (MIT) and Post Graduate Diploma in Information Technology (PGDIT). This institute also offers short certificate courses and all modules of CCNA. In addition, IIT is conducting state-of-the-art research works in collaboration of industries and foreign universities.

Vision

The vision of the IIT-JU is to enlighten the students through need-based academic innovation and research works and prepare them for the real-life challenges in the field of ICT.

Mission

The mission of IIT-JU is to create splendid students by enhancing knowledge and competitiveness through excellent education, hands-on experience, state-of-the-art research and industry collaboration.

Program objectives

The Program Educational Objective (PEO) of IIT-JU are as follows:

PEO1: Provide students with a strong theoretical foundation as well as industry relevant skills in various technologies related to ICT to be able to adapt to the evolving technical challenges and changing career opportunities.

PEO2: Instill analytic and critical thinking skills in order to exhibit analytical decision-making and problem-solving abilities when dealing with complex real-time challenges.

PEO3: Indoctrinate professional and ethical aspects of modern computing technology so that students can apply their skills to design, create and use computing artifacts that are economically feasible and socially acceptable through their innovations.

PEO4: Providing an education that will prepare graduates to create, sustain and improve ICT infrastructure in organizations, as well as manage ICT resources to support a variety of organizational functions in order to confront contemporary issues in a broader social context.

Program Learning Outcomes (PLO):

A graduate student of IIT-JU will have an ability to

PLO1: Professional Skills - Recognize, interpret, design, model, build, evaluate, and manage complex software and information management systems.

PLO2: Problem Analysis – Identify, formulate, study, and analyze complex Information and Communication Technology related problems using mathematical principles and natural sciences to draw demonstrable conclusions.

PLO3: Design/Development of Solutions – Design and develop software projects given their specifications considering performance and cost constraints by identifying, formulating and solving software engineering problems and understanding the software project management principles.

PLO4: Investigation – Conduct investigations into complex problems while demonstrating creativity and using evidence-based skills and research methods such as experiment design, data analysis and interpretation and information synthesis pertaining to Information and Communication Technology to draw relevant conclusions.

PLO5: Modern Tool Usage – Create, select and apply appropriate techniques, resources, modern tools, technologies and the fundamental principles that underlie the students within the context of ICT with an understanding of the limitations.

PLO6: The Engineer and Society – Apply reasoning based on contextual knowledge to become software engineers who are eager to work on projects that are both relevant and have a significant impact on people's lives concerning environmental, economic and social issues both in the local and global perspective.

PLO7: Environment and Sustainability – Examine the global and environmental impact of information and communication technology based solutions and apply the learning for sustainable development.

PLO8: Ethics – Recognize professional obligations and use legal and ethical standards to make informed decisions in computing practice.

PLO9: Communication – Extend entrepreneurship and leadership skills by communicating efficiently with the technical, management, user and academic audiences in oral, written and graphic form making effective presentations and give and receive clear instructions including documentation of hardware and software systems.

PLO10: Individual and Teamwork – Ability to function independently or as part of a team with responsibility for multidisciplinary settings.

PLO11: Lifelong Learning – Recognize the need for adapting to emerging technology and continuously upgrading their abilities with a desire towards lifelong learning.

PLO12: Successful Career and Entrepreneurship– Ability to create contemporary career pathways as an entrepreneur using modern programming languages, environments, and networks as well as a desire to pursue higher education by conducting research that could help uplift ICT with a sense of nationalism and social responsibility.

Mapping PEO vs. PLO:

Program Learning Outcomes (POs)	Program Educational Objectives			
	PEO1	PEO2	PEO3	PEO4
1. PLO1 Professional Skills	√			√
2. PLO2 Problem Analysis		√		
3. PLO3 Design/Development of Solutions	√		√	
4. PLO4 Investigation	√	√		
5. PLO5 Modern Tool Usage			√	√
6. PLO6 The Engineer and Society		√	√	
7. PLO7 Environment and sustainability				√
8. PLO8 Ethics			√	

9. PLO9 Communication	√	√		
10. PLO10 Individual and Teamwork	√	√		√
11. PLO11 Life-long Learning	√	√		
12. PLO12 Successful Career and Entrepreneurship	√	√	√	

Mapping PEO vs. Types of Courses

First Year First Semester					
Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT 1101	Introduction to ICT	√	√		
ICT 1103	Structured Programming Language	√	√	√	
ICT 1105	Electrical Circuits	√	√		√
ICT 1107	Physics	√	√		
ICT 1109	Differential and Integral Calculus	√	√		
ICT 1111	Communicative English	√		√	
ICT 1104	Structured Programming Language Lab	√	√		
ICT 1106	Electrical Circuits Lab		√	√	
ICT 1100	Course Viva	√	√	√	

First Year Second Semester					
Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT 1201	Electronic Devices and Circuits	√	√		√
ICT 1203	Object Oriented Programming	√	√	√	
ICT 1205	Linear Algebra and Analytical Geometry	√	√		

ICT 1207	Discrete Mathematics	√	√		
ICT 1209	Bangladesh Studies	√			
ICT 1202	Electronic Devices and Circuits Lab		√	√	
ICT 1204	Object Oriented Programming Lab	√	√		
ICT 1200	Project Work - I and Course Viva	√	√	√	

Second Year First Semester

Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT 2101	Data Structures	√	√		
ICT 2103	Digital Logic Design	√	√		√
ICT 2105	Numerical Analysis	√	√		
ICT 2107	Statistics and Probability for Engineers	√	√	√	
ICT 2109	Data Communication		√	√	
ICT 2102	Data Structures Lab	√	√		
ICT 2104	Digital Logic Design Lab	√	√	√	
ICT 2106	Internet and Web Technology Lab	√	√		
ICT 2100	Course Viva	√	√	√	

Second Year Second Semester

Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT 2201	Algorithm Analysis and Design	√	√		
ICT 2203	Database Management System	√	√		

ICT 2205	Analog and Digital Communication		√	√	
ICT 2207	Matrices, Vector, Fourier Analysis and Laplace Transforms	√	√		
ICT 2209	Financial and Managerial Accounting	√	√	√	
ICT2202	Algorithm Analysis and Design Lab	√	√		
ICT 2204	Database Management System Lab	√	√		
ICT 2206	Analog and Digital Communication Lab		√	√	
ICT 2200	Project Work - II and Course Viva	√	√	√	

Third Year First Semester

Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT 3101	Operating System	√	√		√
ICT 3103	Computer Networks	√	√	√	√
ICT 3105	ICT Business Analytics and Data Visualization	√	√	√	
ICT 3107	Information and Data Security	√	√	√	
ICT 3109	Principles of Economics	√		√	
ICT 3102	Operating System Lab	√	√	√	√
ICT 3104	Computer Networks Lab	√	√		
ICT 3100	Special Study/Industrial Attachment and Viva		√	√	√

Third Year Second Semester

Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT 3201	Software Engineering	√	√	√	√
ICT 3203	Computer Architecture and Microprocessor	√	√	√	

ICT 3205	Signals and Systems	√			
ICT 3207	Server Administration and Management	√	√	√	√
ICT 3209	Smart Sensors and Internet of Things		√	√	√
ICT 3202	Software Engineering Lab	√	√		
ICT 3204	Computer Architecture and Microprocessor Lab		√	√	
ICT 3200	Project Work – III and Course Viva	√	√	√	√

Fourth Year First Semester					
Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT4100	Internship	√	√	√	√
Or					
ICT 41XX	From Option – I				
ICT 41XX	From Option – I Lab				
ICT 41XX	From Option – I				
ICT 41XX	From Option – I Lab				
ICT 41XX	From Option – I				
ICT 41XX	From Option – I				
ICT 41XX	From Option – I				

Option I					
Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT 4101	Digital Signal Processing	√	√	√	
ICT 4102	Digital Signal Processing Lab	√	√		

ICT 4103	Mobile Application Development			√	√
ICT 4104	Mobile Application Development Lab	√	√		
ICT 4105	Cyber Security	√	√		√
ICT 4107	Cryptography and Network Security	√	√	√	
ICT 4109	Big Data Analytics and Application	√	√		√
ICT 4111	Optical Fiber Communication	√	√	√	
ICT 4113	IT Professional and Ethics	√		√	√
ICT 4115	IT Project and Service Management	√		√	√
ICT 4117	IT Risk Management	√	√		√
Fourth Year Second Semester					
Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT 4201	Artificial Intelligence	√	√	√	
ICT 4203	Parallel and Distributed System	√	√		√
ICT 4205	Wireless and Cellular Networks	√	√	√	√
ICT 42XX	From Option – II				
ICT 42XX	From Option – II				
ICT 4202	Artificial Intelligences Lab	√	√		
ICT 4200	Research Project	√	√	√	

Option – II					
Course Code	Course Title	PEO1	PEO2	PEO3	PEO4
ICT 4211	Data Mining and Knowledge Discovery	√	√		
ICT 4213	Digital Image Processing	√	√		

ICT 4215	Bio-informatics	√	√		
ICT 4217	Data Science	√	√	√	
ICT 4219	Simulation and Modeling	√	√		
ICT 4221	Machine Learning	√	√	√	
ICT 4223	Embedded System Design	√	√	√	√
ICT 4225	Research Methodology	√		√	
ICT 4227	Digital Forensic		√		√
ICT 4229	Communication Management	√		√	√
ICT 4231	Microwave Engineering and Satellite Communication	√	√	√	√
ICT 4233	Multimedia Communication	√		√	√
ICT 4235	Contemporary Course on Information and Communication Technology	√	√	√	√

COURSE CURRICULUM FOR B. Sc. (Hons.) in ICT**Program Overview:**

B. Sc. (Hons.) in Information and Communication Technology shall extend over a period of four academic years and will consist of eight semesters. The program hereinafter is called as Undergraduate Program. Each year will divide into two semesters. Each semester will have a duration of six months. Students shall be evaluated in each semester. A semester will be segmented into Class-weeks, Preparatory leave and Semester-end examination. The total time distribution for completing a semester will be as follows:

Sl.	Segment	Length
I.	Classes	14 Weeks
II.	Preparatory leave before semester-end examination	1 Weeks

III.	Semester-end examination	3 Weeks
IV.	Result Publishing & Semester Break	4 Weeks
Total		22 Weeks

During class-weeks, if classes do not held in any particular week due to the reason beyond the control of the university, the week shall deem to be an effective class-week, if number of working days is equal to or more than three.

Admission:

Admission of students and examination of courses to the B.Sc. (Honors) program shall be guided by the Admission Ordinance and the Examination Ordinance of the University.

Eligibility:

Eligibility of students for taking part into the admission test shall be determined and guided as per rules of the University.

Admission Test:

Procedures for admission test shall be guided by the rules of the University. Along with that, a student will be considered eligible for admission if he/she obtains minimum 30% marks in Physics and minimum 40% marks in Mathematics in the admission test. Information relating to the detail syllabus, type and format of questions, date, time and place of the admission test will be found in the admission brochure, daily newspapers and on the web site, <http://www.juniv.edu/iit/> or <http://iit.juniv.edu>.

Selection Procedure:

Selection procedure shall be guided as per rules of the University.

Rules for Admission:

Procedures for admission shall be guided as per rules of the University.

Tuition & Other Fees:

Tuition fees and the mode of payment of four years program shall be guided as per rules of the university.

Course Offering and Instruction:

The courses to be offered in a particular semester are announced and published in the Registration Package along with the tentative semester schedule before the end of the previous semester. The courses to be offered in any semester will be decided by the Committee of Courses for Undergraduate Program. Each course is conducted by a course teacher who is responsible for maintaining the expected standard of the course and for the assessment of students' performance. One of the course teachers or any other member of the teaching staff of the Institute will be designated as course coordinator for each semester. He/she has the full responsibility for coordinating the work of the other members of the Institute involving in that semester.

Course Pattern and Credit Structure:

The undergraduate program is covered by a set of theoretical courses along with a set of laboratory courses to support them.

Course Designation and Numbering System:

A course will be represented by course number, course title, credit hours and contact hours per week (Theory or Lab). Each course is designated by a three two letter code identifying the B. Sc. program offered followed by a four-digit number having the following interpretation: The first and second digits correspond to the year and the semester in which the course is normally taken by the students. The third digit is reserved for maintaining continuity. The last digit is an odd number for theoretical courses and an even number for laboratory courses. The following example illustrates a course representation system:

Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
ICT 1101	Introduction to ICT	3.0	3 hrs.

Assignment of Credits:

The assignment of credits to a theoretical course follows a different rule from that of a practical or laboratory course. Courses of study for the B.Sc. (Hons.) in Information & Communication Technology are defined as per rules of the University.

Credit Hour Requirement:

The total contact hours for each 3 credits Theoretical course is 42 hours and for each 1.5 credits Lab oriented course is 42 hours. Marks allocated for each course either theoretical or practical is 100. The evaluation of a course will be carried by taking tutorial examination and a final examination. A student for the B.Sc. (Hons.) in Information & Communication Technology will have to complete a total of 158.5 credit-hours during the four years' undergraduate program.

Industrial/Professional Training Requirements:

There shall be an Industrial/Professional training requirements at the end of the fourth semester. The objective of this training program is to enlighten the students with practical orientation and give them an opportunity to make use of their theoretical concepts and practical skills in real life situations. All students will be placed in public and private sectors, particularly those organizations that are engaged in activities having direct relevance to the Information & Communication Technology and likely to enhance the knowledge and skill of the students. The training program shall extend over a period of minimum three weeks. The outcome of this program will be an Industrial/Professional training report as prescribed in the syllabus. Training program shall be equivalent to a two credit hours laboratory course and be evaluated by this final report accordingly. The credit earned in this training program will not contribute in any GPA/CGPA calculation of the student's result but the student has to obtain a satisfactory (S) grade in this course to get promotion in the next semester.

Placement of Students for Industrial /Professional Training Requirement:

The academic committee of the Institute shall arrange for the placement of students and shall nominate internal and external supervisor(s) of the students going for Industrial Attachment. The Director of the Institute will send the names of the internal and external supervisors to the Director Controller of the Examination office for appointment.

Research Project:

Research Project work is required for the partial fulfillment of the completion of bachelor degree. A committee shall be formed for monitoring the project works for undergraduate students. This committee will finalize the placement of students for research project and shall nominate supervisor, internal and external members. The Director of the Institute will send the names of the internal and external members to the Controller of the Examination Office for appointment.

Assessment:

For the purpose of assessment, 100 marks shall be assigned to each three-credit hours' course. Assessment of a student in a course shall be based on marks obtained in the course-end examination (written) and class assessments/continuous assessment. Marks allotted for class assessment/continuous assessment shall be 40% of the total earned marks for each theoretical course and 60% for each practical course.

Assessment Criteria	Theoretical Course	Practical Courses
Class/Continuous Assessment	40%	60%
Final examination	60%	40%
Total	100%	100%

Class Assessment / Continuous Assessment and Submission of Assessment:

Class assessment/continuous assessment will consist of class attendance, written class tests, quizzes, presentation, project works, case studies, assignments, term papers and discussion sessions. For assessment of class test in theoretical courses there shall be a minimum of three tutorial tests (declared/undeclared) for each three-credit hours course. For assessment of class test in practical courses there shall be a minimum of three declared written tutorial tests for each three-credit hours' course. The distribution of marks for each theoretical course shall be as follows:

Theoretical Courses	Marks (%)
Class participation / Attendance	10%
Assignments	10%
Tutorial tests/Class tests	20%
Semester-end Examination	60%
Total	100%

The distribution of marks for each practical course will be as follows:

Practical Courses	Marks (%)
Class participation/Attendance	10%
Class test/Tutorial	20%
Experiment Evaluation	10%
Report	10%
Quiz/Viva	10%
Semester-end Examination	40%
Total	100%

The distribution of marks for internship will be as follows

Internship	Marks (%)
Midterm Presentation	10%
Final presentation	20%
Final Report	10%
Company Evaluation	60%
Total	100%

Examinations:

Final examination for each semester will be conducted as per Examination Ordinance for semester system in the university and controlled by Office of the Controller of Examination.

Grading System:

The Universal Grading System introduced by the University Grant Commission (UGC) of Bangladesh will be followed which is given below. The total numerical marks obtained by a student in each course will be converted into Letter Grade (LG) and Grade Point (GP). According to the Grade Point, the GPA (Grade Point Average) and CGPA (Cumulative Grade Point Average) will be calculated. The conversion of Letter Grade and Grade Point will be as follows:

Numerical Grade	Letter Grade	Grade Points
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Incomplete	I	
Satisfactory or Unsatisfactory	S or U	Industrial/ Professional Tanning
Continuation	X	Industrial Attachment etc.

Earned Credits:

- The grades of the courses, in which a student has obtained minimum qualifying pass grade, shall only be counted as credits earned by him/her. Other grades shall not be counted for Grade Point Average (GPA) calculation.
- If a student obtains an F grade in any course in any semester, he/she will have to repeat the course(s), whenever offered within his/her total duration of academic years. In that case his/her earned credit shall not be more than B.
- If a student obtains I(incomplete) grade in one or more courses in any semester, he/she will have to repeat the course(s), whenever offered within his total duration of academic years.

Performance Evaluation:

The performance of a student will be evaluated in terms of two indices: (i) semester grade point average (GPA) and (ii) Cumulative Grade Point Average (CGPA) which is the grade point average for all the semester completed.

Students will be considered to be making normal progress toward a degree if their Cumulative Grade Point Average (CGPA) for all work attempted is 2.00 or higher. Students who regularly maintain a GPA of 2.00 in each semester or better are making good progress toward the degrees and are in good standing with the University. Students who fail to maintain this minimum rate of progress will not be in good standing. This can happen when any one of the following conditions exists. The earned GPA in each semester falls below 2.00, or The Cumulative GPA falls below 2.00, or the earned number of credits falls below 15 times the number of semester attended. All such students can make up their deficiencies in GPA and credit requirements by completing courses in the subsequent semester(s) and backlog courses, if there are any, with better grades. When the minimum GPA and credit requirements are achieved, the student is again returned to good standing.

Class Attendance:

To sit for the class assessment and course-end examination, a student must have to have minimum class attendance of 75% which will be guided by the rules of the university.

Qualifying Marks:

- The qualifying pass grade in a particular course will be determined by the rules of the University. If any student gets F grade in one or more courses, he/she has to cover it within the time limit which is mentioned in section 19 of this ordinance.

- If a candidate remains absent in a course-end Examination for a course for reasons like - serious illness, accident or any other valid reason, his/her course may be graded I (Incomplete). With subject to the approval of the concerned authority of the University, he/she may get a chance to recover it like section 19.

Promotion to next semester:

A student must secure the minimum qualifying grade in each of the courses in the semester-end examination in order to be considered “pass” in that semester. However, for promotion to the next semester, a candidate shall have to obtain a minimum GPA, which will be followed as per University rules.

Referred Examination:

Matters relating to referred examination shall be guided by the rules of the University.

Student Adviser:

One adviser is normally appointed for a group of students by the Director of the Institute. The adviser advises each student about the academic program of that particular semester. However, it is also the student's responsibility to keep regular contact with his/her adviser who will review and eventually approve the student's specific plan of study and monitor subsequent progress of the student. The adviser is also authorized to permit the student to drop one or more courses based on his/her previous academic performance and corresponding categorization.

Time Limit:

How long a student shall be permitted to continue as a Bachelor's Degree candidate will be decided by the rules of the University.

COURSE CURRICULUM

First Year First Semester				
Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT 1101	Introduction to ICT	3.0	2 hrs.
2	ICT 1103	Structured Programming Language	3.0	3 hrs.
3	ICT 1105	Electrical Circuits	3.0	3 hrs.
4	ICT 1107	Physics	3.0	2 hrs.
5	ICT 1109	Differential and Integral Calculus	3.0	3 hrs.

6	ICT 1111	Communicative English	3.0	2 hrs.
7	ICT 1104	Structured Programming Language Lab	1.5	3 hrs.
8	ICT 1106	Electrical Circuits Lab	1.5	3 hrs.
9	ICT 1100	Course Viva	0.5	
		Total Credit	21.5	

First Year Second Semester

Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT 1201	Electronic Devices and Circuits	3.0	3 hrs.
2	ICT 1203	Object Oriented Programming	3.0	3 hrs.
3	ICT 1205	Linear Algebra and Analytical Geometry	3.0	3 hrs.
4	ICT 1207	Discrete Mathematics	3.0	3 hrs.
5	ICT 1209	Bangladesh Studies	3.0	2 hrs.
6	ICT 1202	Electronic Devices and Circuits Lab	1.5	3 hrs.
7	ICT 1204	Object Oriented Programming Lab	1.5	3 hrs.
8	ICT 1200	Project Work - I and Course Viva	1.0	
		Total Credit	19.0	

Second Year First Semester

Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT 2101	Data Structures	3.0	3 hrs.

2	ICT 2103	Digital Logic Design	3.0	3 hrs.
3	ICT 2105	Numerical Analysis	3.0	3 hrs.
4	ICT 2107	Statistics and Probability for Engineers	3.0	3 hrs.
5	ICT 2109	Data Communication	3.0	3 hrs.
6	ICT 2102	Data Structures Lab	1.5	3 hrs.
7	ICT 2104	Digital Logic Design Lab	1.5	3 hrs.
8	ICT 2106	Internet and Web Technology Lab	1.5	3 hrs.
9	ICT 2100	Course Viva	0.5	
		Total Credit	20.0	

Second Year Second Semester				
Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT 2201	Algorithm Analysis and Design	3.0	3 hrs.
2	ICT 2203	Database Management System	3.0	3 hrs.
3	ICT 2205	Analog and Digital Communication	3.0	3 hrs.
4	ICT 2207	Matrices, Vector, Fourier Analysis and Laplace Transforms	3.0	3 hrs.
5	ICT 2209	Financial and Managerial Accounting	3.0	2 hrs.
6	ICT2202	Algorithm Analysis and Design Lab	1.5	3 hrs.
7	ICT 2204	Database Management System Lab	1.5	3 hrs.
8	ICT 2206	Analog and Digital Communication Lab	1.5	3 hrs.
9	ICT 2200	Project Work - II and Course Viva	1.0	

		Total Credit	20.5	
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Third Year First Semester				
Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT 3101	Operating System	3.0	3 hrs.
2	ICT 3103	Computer Networks	3.0	3 hrs.
3	ICT 3105	ICT Business Analytics and Data Visualization	3.0	3 hrs.
4	ICT 3107	Information and Data Security	3.0	3 hrs.
5	ICT 3109	Principles of Economics	3.0	3 hrs.
6	ICT 3102	Operating System Lab	1.5	3 hrs.
7	ICT 3104	Computer Networks Lab	1.5	3 hrs.
8	ICT 3100	Special Study/Industrial Attachment and Viva	1.0	
		Total Credit	19.0	

Third Year Second Semester				
Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT 3201	Software Engineering	3.0	3 hrs.
2	ICT 3203	Computer Architecture and Microprocessor	3.0	3 hrs.
3	ICT 3205	Signals and Systems	3.0	3 hrs.
4	ICT 3207	Server Administration and Management	3.0	3 hrs.

5	ICT 3209	Smart Sensors and Internet of Things	3.0	3 hrs.
6	ICT 3202	Software Engineering Lab	1.5	3 hrs.
7	ICT 3204	Computer Architecture and Microprocessor Lab	1.5	3 hrs.
8	ICT 3200	Project Work – III and Course Viva	1.0	
		Total Credit	19	

Fourth Year First Semester

Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT4100	Internship	18.0	----
		Total Credit	18.0	
Or				
1	ICT 41XX	From Option – I	3.0	3 hrs.
2	ICT 41XX	From Option – I Lab	1.5	3 hrs.
3	ICT 41XX	From Option – I	3.0	3 hrs.
4	ICT 41XX	From Option – I Lab	1.5	3 hrs.
5	ICT 41XX	From Option – I	3.0	3 hrs.
6	ICT 41XX	From Option – I	3.0	3 hrs.
7	ICT 41XX	From Option – I	3.0	3 hrs.
		Total Credit	18.0	

Option – I				
Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT 4101	Digital Signal Processing	3.0	3 hrs.
2	ICT 4102	Digital Signal Processing Lab	1.5	3 hrs.
3	ICT 4103	Mobile Application Development	3.0	3 hrs.
4	ICT 4104	Mobile Application Development Lab	1.5	3 hrs.
5	ICT 4105	Cyber Security	3.0	3 hrs.
6	ICT 4107	Cryptography and Network Security	3.0	3 hrs.
7	ICT 4109	Big Data Analytics and Application	3.0	3 hrs.
8	ICT 4111	Optical Fiber Communication	3.0	3 hrs.
9	ICT 4113	IT Professional and Ethics	3.0	3 hrs.
10	ICT 4115	IT Project and Service Management	3.0	3 hrs.
11	ICT 4117	IT Risk Management	3.0	3 hrs.

Fourth Year Second Semester				
Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT 4201	Artificial Intelligence	3.0	3 hrs.
2	ICT 4203	Parallel and Distributed System	3.0	3 hrs.
3	ICT 4205	Wireless and Cellular Networks	3.0	3 hrs.
4	ICT 42XX	From Option – II	3.0	3 hrs.

5	ICT 42XX	From Option – II	3.0	3 hrs.
6	ICT 4202	Artificial Intelligence Lab	1.5	3 hrs.
7	ICT 4200	Research Project	5.0	10 hrs.
		Total Credit	21.5	

Option – II				
Sl.	Course Code	Course Title	Credit Hrs.	Class Hrs./ Week
1	ICT 4211	Data Mining and Knowledge Discovery	3.0	3 hrs.
2	ICT 4213	Digital Image Processing	3.0	3 hrs.
3	ICT 4215	Bio-informatics	3.0	3 hrs.
4	ICT 4217	Data Science	3.0	3 hrs.
5	ICT 4219	Simulation and Modeling	3.0	3 hrs.
6	ICT 4221	Machine Learning	3.0	3 hrs.
7	ICT 4223	Embedded System Design	3.0	3 hrs.
8	ICT 4225	Research Methodology	3.0	3 hrs.
9	ICT 4227	Digital Forensic	3.0	3 hrs.
10	ICT 4229	Communication Management	3.0	3 hrs.
11	ICT 4231	Microwave Engineering and Satellite Communication	3.0	3 hrs.
12	ICT 4233	Multimedia Communication	3.0	3 hrs.
13	ICT 4235	Contemporary Course on Information and Communication Technology	3.0	3 hrs.

1st Year 1st Semester**ICT 1101 Introduction to ICT**

Course Objectives:	The objective of this course is to • Provide the concepts of computer and programming fundamental. • Describe the IT fundamentals and its applications. • Outline the basics about e-commerce, information security, multimedia and Management Information System (MIS).	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the concept of input and output devices, processors and memory of computers and how it works and recognize the basic terminology used in computer programming.	PLO1
	CLO2- To design programs connecting decision structures, loops and functions.	PLO2, PLO5
	CLO3- To understand the concepts of e-commerce, information security, multimedia and Management Information System (MIS).	PLO2, PLO5

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	HISTORY AND HISTORICAL EVALUATION OF COMPUTERS: 1. Counting devices used before the advent of computers 2. Characteristics of early computers 3. Generations of computer 4. Outstanding characteristics of each generation	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise

CLO1	INTRODUCTION TO COMPUTER & COMPUTER SYSTEM: 1. Computer & it's working principles 2. Classification of computer 3. Outstanding characteristics of computers 4. Block diagram of a typical computer 5. Basic components of a computer 6. Hardware vs. Software	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO1	CPU USED IN PERSONAL COMPUTERS: 1. Elements of CPU and their functions 2. Types of processor: RISC Vs. CISC 3. Bus system and port in microcomputer 4. Factors affecting the processing speed of a computer	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO1	INPUT AND OUTPUT DEVICES: 1. Definition and classification 2. Functions and applications of popular input and output devices 3. Common I/O devices 4. Speech recognition Vs. Speech synthesis	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO1	MEMORY FUNDAMENTALS: 1. Definition and functions 2. Primary Vs. Secondary memory 3. Semiconductor, optical and magnetic memories 4. RAM Vs. ROM 5. CPU-Memory interconnection	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation
CLO1, CLO2	SOFTWARE AND THEIR USAGE: 1. Software and its classification 2. Application Vs. System software 3. Operating system Vs. Utility programs 4. Functions, objectives, and features of an operating system 5. Software development life-cycle	Lecture Discussion Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

CLO1, CLO2	BASIC PROGRAMMING CONCEPTS: 1. High-level Vs. Low-level programming language 2. Translator programs 3. Flowchart and algorithm 4. Data types, variables and operators 5. Control statements: Branching and looping 6. Errors while programming 7. Number system and conversions	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Report
CLO3	BASIC INFORMATION TECHNOLOGY: 1. Data vs. Information 2. Significance of information in daily life 3. Information Technology vs. Information System 4. Components and functions of a computer-based information system 5. Management Information System 6. Ethical issues in information system 7. Overview of information security	Lecture Group Assignment Demonstration	Essay Type Answer Report Exercise Presentation
CLO3	DATA COMMUNICATION, COMPUTER NETWORK AND THE INTERNET: 1. Telecommunication Vs. Data Communication 2. Components and modes of a Data Communication System 3. Types of computer network 4. Merits and demerits 5. Network topologies and protocols 6. History of the Internet 7. Features of Internet 8. Providing security of Internet and network	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

CLO3	CONCEPTS OF E-COMMERCE: 1. E-commerce Vs. E-Business, 2. Merits, demerits and outstanding features of e-commerce 3. Various e-commerce models 4. Payment system in e-commerce 5. Status of e-commerce in Bangladesh	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO3	CONCEPTS OF E-GOVERNANCE: 1. Why e-governance 2. Components of e-governance 3. Evolution of e-governance in Bangladesh 4. Factors for implementing Digital Bangladesh 5. Key development sectors in Bangladesh 6. E-government initiatives from government entities	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO3	CYBER SECURITY AND CYBER LAW: 1. Cyber world vs. Real world 2. What is cyber security 3. The state of security today 4. Why security matters today 5. Cybercrime vs. Cyber law	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√			√	√
CLO2	√	√			√
CLO3	√			√	√

Suggested Texts:

- Introduction to Information Technology, 2nd Edition by ITL Education Solutions Ltd, Publisher: Pearson Education.
- Computer and Information Processing, 4th Edition by William M. Fouri.
- Introduction to Computer, 7th Edition by Peter Norton.
- Computer Fundamentals and ICT, 1st Edition by M Lutfar Rahman.
- Computers Today, 1st Edition by Suresh K Basandra.
- Fundamentals of Computing, 2nd Edition by Allen B.Tucker et.al, Publisher: TMH New Delhi.
- Fundamentals of Computers, 6th Edition by V.Rajaraman, Publisher: Prentice-Hall of India.

Intended Learning Outcomes:

- Knowledge and understanding
 - ▯ Understand different terms associated with ICT.
 - ▯ Understand different terms associated with the Internet and World Wide Web.
 - ▯ Understand the enabling/pervasive features of ICT.
- Cognitive skills (thinking and analysis)
 - ▯ Be able to identify various components of a computer system.
 - ▯ Be able to gather knowledge behind e-commerce and its applications.
- Communication skills (personal and academic)
 - ▯ Be able to define terms associated with communication and networking.
- Practical and subject specific skills (Transferable Skills)
 - ▯ Be able to use various web tools including Web Browsers, E-mail clients and search utilities.
 - ▯ Be able to use text processing, spreadsheets and presentation tools.
 - ▯ Be able to identify the various categories of software and their usage.

ICT 1103 Structured Programming Language

Course Objectives:	The objective of this course is to • Introduce fundamental programming concepts by using various concepts and structures for writing programs. • Distinguish language definition from implementation, syntax and parsing from semantics and evaluation, understand how program state maps to memory (global/statics, locals/stack, heap/instances) and understand the implications of heap reachability for memory management. • Provide students with opportunities to develop, understand, test, and evolve substantial programs using a modern IDE and associated configuration tools with respect to program design and development.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the basic building blocks of structured programming	PLO1
	CLO2- To analyze structured program.	PLO2, PLO5
	CLO3- To develop structured program for solving computational problems.	PLO3, PLO5
	CLO4- To design structured program for complex computational applications.	PLO3, PLO4

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction to C Programming (input, output, variables, data type, operators, and expressions).	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise Presentation
CLO1	Memory organization, memory access, precision of decimal, binary and floating point number, program execution in memory, memory addressing.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise Presentation

CLO1, CLO2	Structured Program Development in C: Control Statements 1(if, if...else, while, top-down and stepwise refinement).	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise Presentation
CLO1, CLO2	Program Control: Control Statements 2 (for, do...while, switch, break and continue).	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise Presentation
CLO2, CLO3	Introduction to Functions: Math Library Functions, Function Definitions, Function Prototypes and Argument, Header Files, Random Number Generation.	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3	Introduction to Functions: Recursive functions, References and Reference Parameters, passing arguments to functions and passing arguments by reference.	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3	Introduction to Arrays: Arrays, Declaring Arrays, Examples Using Arrays, Passing Arrays to Functions, arrays of strings).	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3	Introduction to Arrays: Case study, Searching Arrays, Sorting Arrays, Multidimensional Arrays.	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3	Structure and Union: Definition of structure, Union, Structure union applications, Self-referential structure, Structure and Union: Link List, Array of Structure.	Lecture Discussion Problem Based Learning (PBL)	Exercise

CLO2, CLO3, CLO4	Pointers: Pointer Variable Declarations and Initialization, Pointer Operators, Passing Arguments to Functions by Reference with Pointers. Pointer Expressions and Pointer Arithmetic, Arrays, Pointers, Function Pointers.	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3, CLO4	Characters and Strings: String Conversion, String Manipulation, Comparison Functions, Search Functions, and Memory Functions.	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO4	File Processing: Files and Streams, Creating a Sequential File, Reading Data from Sequential File, Writing data to File, and Updating Sequential Files.	Lecture Discussion Problem Based Learning (PBL)	Exercise

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√		√	√
CLO3	√			√	√
CLO4	√		√		√

Suggested Texts:

- C programming language, 2nd Edition by Kernigham & Ritchie.
- Teach Yourself C, 3rd Edition by Herbert Schildt.
- Programming in Ansi C, 8th Edition by Balagurushwami.
- Turbo C/C++. The Complement Reference, 2nd Edition by Herbert Schildt.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the fundamental programming constructs.
 - Understand and write searching and sorting techniques.
 - Understand a typical C-like program environment.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analyze any problem and derive its solution.
 - Be able to develop and design algorithms.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to write C-like programs to build programming skills, designing skills and problem solving skills.

ICT 1105 Electrical Circuits

Course Objectives:	The objective of this course is to ● Describe the concept of circuit elements lumped circuits, waveforms, circuit laws and network reduction. ● Outline the electrical network using mesh and nodal analysis by applying network theorems. ● Provide the concept of Energy storage elements. ● Analyze the transient response of RC, LC and RLC circuits.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To learn concepts of DC and AC circuit analysis.	PLO1
	CLO2- To explain laws and methods of circuit analysis.	PLO1, PLO2
	CLO3- To apply laws and theorems for solving electrical circuits.	PLO3, PLO6, PLO10

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Fundamental electrical concepts and measuring units. Active and passive components: Resistor, properties of resistors, types of resistors.	Lecture Assignment	Short Answer Exercise/
CLO2	Ohm's law, Power and Energy: Ohm's law, DC-circuits, conductance and resistance.	Lecture Assignment	Short Answer Exercise
CLO1, CLO2	Series and Parallel DC Circuit: Kirchoff's current law, current divider rule, open circuit, short circuit, etc. Analysis of series-parallel network. Methods of analysis for DC networks, current source, source conversion. branch-current analysis, mesh analysis, nodal analysis, bridge network, star delta, delta star.	Lecture Assignment Case Study Group Discussion	Short Answer Exercise MCQ
CLO2, CLO3	Network theorems (DC): superposition theorem, Thevenin's theorem, Norton's theorem, maximum power transfer theorem. Millman's theorem, substitution theorem, reciprocity theorem. Sinusoidal alternating waveforms. The sine wave, general format for the sinusoidal voltage or current. phase relations, etc. Response of basic R, L and C elements to a sinusoidal voltage or current, frequency response, rectangular and polar form, conversion between forms, phasors.	Lecture Assignment Problem Based Learning (PBL)	Short Answer Identification Group Exercise
CLO1, CLO2, CLO3	Analysis of series and parallel AC circuit: Impedance and phasor diagram, voltage divider rule, frequency response of the R-C circuit, admittance and susceptance. Current divider rule, frequency response of the parallel R-L network, etc. Analysis of series-parallel AC circuit. Methods of analysis for AC network, independent and dependent controlled source and source conversions, mesh analysis, nodal analysis, bridge network.	Lecture Assignment Case Study Group Discussion	Short Answer Discussion Exercise MCQ
CLO2	Introduction to transformers: Single phase and three phase transformers, Applications of transformer.	Lecture Assignment	Short Answer Discussion
CLO2, CLO3	Introduction to polyphase system: Three-phase generator, Y-connected generator, Phase Sequence.	Lecture Assignment	Short Answer Assignment MCQ

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√			√	√
CLO3	√	√			√

Suggested Texts:

- Introductory Circuit Analysis, 13th Edition by R. L. Boylestad.
- Basic Electric Circuit Analysis, 5th Edition by David E. Johnson, J.L. Hilborn & J.R. Johnson.
- Alternating Current Circuits, 4th Edition by R. M. Kerchner, G. F. Corcoran.
- Electric Machines, 5th Edition by J. Nagarath and D. P. Kothari.
- Lesson's in Electrical Circuit, 5th Edition by Tony R. Kuphaldt.
- Introductory Circuitry for Electrical and Computer Engineering, 1st Edition by Nilson.

Intended Learning Outcomes:

- Knowledge and understanding
 - Acknowledge the principles of operation and the main features of electric machines and their applications.
 - Understand the impact of engineering solutions in a global, economic, environmental and societal context.
- Cognitive skills (Thinking and analysis)
 - Be able to identify, formulate, and solve engineering problems.
 - Be able to apply nodal analysis, mesh analysis, superposition and source transformation to solve DC circuits.
 - Be able to compute time response of current and voltage in second order RLC circuits.
 - Be able to analyze electric circuit using simulation software.
- Communication skills (Personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable skills)

- Be able to apply electrical circuit analysis skill to build efficient and constructive devices.

ICT 1107 Physics

Course Objectives:	The objective of this course is to • Provide basis of heat and thermodynamics, wave and oscillation, optics. • Provide knowledge of basic theories of the optical fiber. • Describe physical behavior of matter and their characteristics.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To explain various concepts and principles in the field of physics.	PLO1
	CLO2- To apply experimental statistics to determine the precision of a series of measurements.	PLO1, PLO2
	CLO3- To measure properties of a variety of electrical and optical systems.	PLO2, PLO6
	CLO4- To determine critically and to use appropriate concepts to analyze qualitatively problems or situations involving the fundamental principles of physics.	PLO2, PLO4, PLO6
	CLO5- To represent these physics concepts and principles mathematically in those situations; to apply those concepts and principles in particular situations through the analysis and solution of practical physics problems.	PLO6, PLO7

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Heat and Thermodynamics: Principle of temperature measurements: platinum resistance thermometer, thermoelectric thermometer, pyrometer; Kinetic theory of gases: Maxwell's distribution of molecular speeds, mean free path, equipartition of energy, Brownian motion, Van der Waal's equation of state, review of the First Law of thermodynamics and its application, reversible and irreversible processes, Second Law of thermodynamics,	Lecture Assignment	Short Answer Exercise

	Carnot cycle; Efficiency of heat engines, Carnot's Theorem, entropy and disorder, thermodynamic functions, Maxwell relations, Clausius-Clapeyron Equation, Gibbs Phase Rule, Third Law of thermodynamics.		
CLO1, CLO2	Waves and Oscillations: Differential equation of a simple harmonic oscillator, total energy and average energy, combination of simple harmonic oscillations, Lissajous' figures, spring-mass system, calculation of time period of torsional pendulum, damped oscillation, determination of damping coefficient, forced oscillation, resonance, two-body oscillations, Reduced mass, differential equation of a progressive wave, power and intensity of wave motion, stationary wave, group velocity and phase velocity, architectural acoustics, reverberation and Sabine's formula.	Lecture Assignment	Short Answer Presentation
CLO3, CLO4, CLO5	Physical Optics: Theories of light; Interference of light, Young's double slit experiment; Displacements of fringes and its uses; Fresnel Bi-prism, interference at wedge shaped films, Newton's rings, interferometers; Diffraction of light: Fresnel and Fraunhofer diffraction, diffraction by single slit, diffraction from a circular aperture, resolving power of optical instruments, diffraction at double slit & N-slits-diffraction grating; Polarization: production and analysis of polarized light, Brewster's law, Malus law, Polarization by double refraction, retardation plates, Nicol prism, optical activity, polarimeters, polaroid.	Lecture Assignment Problem Based Learning (PBL)	Short Answer Group Discussion Exercise

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√				√
CLO3	√				√
CLO4	√	√		√	√
CLO5	√	√			√

Suggested Texts:

- Physics Principles with Applications, 7th Edition by Douglas C. Giancoli, Publisher: Pearson and Prentice Hall.
- Fundamentals of Physics, 10th Edition by David Halliday, Jearl Walker, and Robert Resnick.
- The Physics of Vibrations and Waves, 6th Edition by H. J. Pain, Publisher: John Wiley and Sons, 2013.
- Principle of Optics, 6th Edition by M. Born and E. Wolf, Publisher: Pergamon Press, 1980.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand fundamental ideas of differential and integral calculus of functions of one and more variables.
 - Understand relation within Computer Science and Calculus.
- Cognitive skills (thinking and analysis)
 - Be able to solve first and second order linear homogeneous and non-homogeneous differential equations.
 - Be able to apply differential equations to problems in engineering, physics, biology and economics.
 - Be able to identify different types of series and determine whether a particular series converges.
 - Be able to perform integration by using the appropriate method of integration.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to use Differentiation and Integration concepts to comprehend and construct mathematical arguments.

ICT 1109 Differential and Integral Calculus

Course Objectives:	The objective of this course is to • Provide the basic concepts of limits, continuity, function of single variable and several variables, calculus and applications of calculus in engineering. • Understand the applications of calculus to science and real life are numerous.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping

	CLO1- To understand basic concept of functions, limits and continuity.	PLO1
	CLO2- To learn with differential calculus and expose the concepts of integral calculus.	PLO1, PLO2
	CLO3- To apply the applications of differential and integral calculus especially in science and engineering field.	PLO3, PLO6
	CLO4- To analyze the IT related and engineering applications and real-life problems.	PLO2, PLO6

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2, CLO3	Differential Calculus: Functions, Limit, Continuity, Tangent and Normal, Subtangent and subnormal in Cartesian and polar co-ordinates, Derivative and technique of differentiation, Successive differentiations, Leibnitz's theorem; Roole's theorem, Mean value theorem in finite and infinite forms; Expansion of functions. Indeterminate forms, Maximum and minimum values of functions of single variable.	Lecture Assignment	Short Answer Exercise
CLO2, CLO3	Partial differentiations: Basic of Partial differentiations, Partial differentiations for a transformation of variables, Euler's rule on homogenous functions.	Lecture Assignment Discussion	Short Answer Identification Exercise
CLO2, CLO3	Integral Calculus: Techniques of integration, Integration by the method of substitutions; Integration by parts; Standard integrals; Integration by the method of successive reduction.	Lecture Assignment	Short Answer Discussion
CLO2, CLO4	Definite Integrals: Fundamental theorem of integral calculus, Properties of definite integral and its use in summing series, Walli's formula, Beta function and Gamma function, Numerical integration; Trapezoidal approximation, Simpson's rule, Improper integrals.	Lecture Assignment	Short Answer Presentation

CLO3, CLO4	Applications of Definite Integrals: Area under a curve, area between two curves, volumes, length of a plane curve, area of a surface of revolution in Cartesian and polar co-ordinates.	Lecture Assignment Demonstration	Short Answer Assignment MCQ
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Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√				√
CLO3		√			√
CLO4	√			√	√

Suggested Texts:

- Calculus, 10th Edition by Howard Anton, Irl C. Bivens, Stephen Davis.
- Calculus, 4th Edition by Robert T Smith and Roland Minton.
- Thomas' Calculus, 14th Edition by Joel R. Hass, Christopher E. Heil, Maurice D. Weir.
- Calculus (Single and Multivariable), 6th Edition by Deborah Hughes-Hallett, Andrew M. Gleason.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the mathematical functions.
 - Understand the use of differential and integral calculus in Science and Engineering.
- Cognitive skills (thinking and analysis)
 - Be able to explain at high levels concepts and implement basic operations in Differential and Integral Calculus.
 - Be able to analyze the optimum value of a quantity using concept of Differential Calculus, able to analyze the area of a region or volume of solid using knowledge of Integral Calculus.
- Communication skills (personal and academic)

- ▯ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
- ▯ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - ▯ Be able to use the concepts of Differential and Integral Calculus to comprehend and construct mathematical arguments.

ICT 1111 Communicative English

Course Objectives:	The objective of this course is to ● Enhance skills of students about presentations skills and public speaking skills. Effective presentations and public speaking skills are important in business, sales and selling, training, teaching, lecturing and generally feeling comfortable speaking. ● Developing the confidence and capability to deliver good presentations, and to stand up in front of an audience and speak well, are also extremely helpful competencies for self-development and social situations. The formats and purposes of presentations can be very different, for example: oral (spoken), multimedia using various media - visuals, audio, etc), PowerPoint presentations, short impromptu presentations, long planned presentations, educational or training sessions, lectures and simply giving a talk on a subject to a group on a voluntary basis.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand importance verbal and nonverbal communication.	PLO1
	CLO2- To understand the norms of academic writing and be able to produce different genres of writing with cohesion and accuracy.	PLO1, PLO2
	CLO3- To develop required presentation skill and ability of organizing information for presentation.	PLO2, PLO5
	CLO4- To master the art of handling the questions from the audience.	PLO9, PLO10
	CLO5- To attract and hold an audience's attention during meetings and presentations.	PLO9
	CLO6- To master common grammatical knowledge necessary to function in a real world.	PLO9, PLO10

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO6	LANGUAGE: imperative verbs- common past simple verbs- basic question forms and Short Answers- can for ability and possibility- like and would like- subject - verb - object word order- possessive pronouns and 's'- adverbs of frequency (sometimes, always)- like +ing- irregular past simple verbs- the present perfect simple with ever- going to for future plans- Modal verbs: have to for obligation, should for advice, could / might / must / can't for possibility- too / (not) enough- subject questions (e.g. Who gave it to you?)- the present continuous for future arrangements- can / can't for permission- the present perfect simple and continuous with for and since- the first and second conditional- used to- so / neither- reported speech with say and tell the past continuous for unfulfilled promises (I was going to.. but..)- Reported speech - statements, commands and questions, say vs. tell- tag questions- if only / I wish + past perfect- if and unless.	Lecture Exercise	Short Answer Exercise Assignment Quiz
CLO1, CLO3	SPEAKING: Commenting on still picture, seeking permission, speaking against listening, talking about preferences, making appointment, expressing opinion, talking about current news, expressing possibilities, conversations over the phone, seeking suggestion and advice, agreement, asking for clarification, talking about social networking sites, expressing probabilities, complaining, expressing regret.	Lecture Discussion Practice	Presentation Conversation
CLO4	LISTENING: listening exercise to enhance the ability to understand announcements, lectures and other interactive messages in audio format.	Lecture Audio Exercise	Short Answer
CLO5	READING: Reading comprehension - reading articles taken from the journals, newspapers –answering different types of questions based on comprehensions – helping in learning to read English texts easily and with full understanding.	Lecture Practice Exercise	Short Answer Presentation

CLO2	WRITING: Learning how to compose an academic writing with proper development of the theme, how to develop ideas in essay, paragraph, letter, application, CV, email and dialogue writing.	Lecture Assignment Demonstration	Short and Broad Answer Assignment Quiz
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Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√			√
CLO3				√	
CLO4	√				√
CLO5				√	√
CLO6	√				√

Suggested Texts:

- Cliff's TOEFL (for practice of different skills), 5th Edition.
- Practical English Usage, 4th Edition by Michael Swan.
- Barman, Binoy. Basic Language Skills Practice, 1st Edition by M. Maniruzzaman. Publisher: Dhaka Friends' Book Corner, 2004.
- Intermediate English Grammar, 2nd Edition by Murphy, Raymond, Publisher: UK Cambridge University Press, 1994.

Intended Learning Outcomes:

- Knowledge and understanding
 - Gathering information and applying in writing skill.
 - Understanding the use of grammar in communication.
- Cognitive skills (thinking and analysis)
 - Thinking against listening.
 - Analyzing and speaking in the same time.

- Communication skills (personal and academic)
 - Speaking both formal and informal in personally and academically.
 - Group discussion - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to use the concepts of reading, speaking and writing in practical life.

ICT 1104 Structured Programming Lab

Course Objectives:	The objective of this course is to • Provide students with the skills needed to effectively design develop, implement, debug, test, and maintain programs a more generally to solve problems using a computer • Solve various problems in C on a regular basis to increase the programming ability. • Prepare themselves for national and international competitions ro • Develop beginner level software.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the basic building blocks of structured programming.	PLO1
	CLO2- To analyze structured program.	PLO2
	CLO3- To develop a structured program for solving computational problems.	PLO3, PLO5
	CLO4- To design a structured program for complex computational applications.	PLO3, PLO6

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy

CLO1	Introduction to C Programming (input, output, variables, data type, operators, and expressions)	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO1	Memory organization, memory access, precision of decimal, binary and floating point number, program execution in memory, memory addressing	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO1, CLO2	Structured Program Development in C: Control Statements 1(if, if...else, while, top-down and stepwise refinement)	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO1, CLO2	Program Control: Control Statements 2 (for, do...while, switch, break and continue)	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3	Introduction to Functions: Math Library Functions, Function Definitions, Function Prototypes and Argument, Header Files, Random Number Generation	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3	Introduction to Functions: Recursive functions, References and Reference Parameters, passing arguments to functions and passing arguments by reference	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3	Introduction to Arrays: Arrays, Declaring Arrays, Examples Using Arrays, Passing Arrays to Functions, arrays of strings)	Lecture Discussion Problem Based Learning (PBL)	Exercise

CLO2, CLO3	Introduction to Arrays: Case study, Searching Arrays, Sorting Arrays, Multidimensional Arrays	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3	Structure and Union: Definition of structure, Union, Structure union applications, Self-referential structure, Structure and Union: Link List, Array of Structure	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3, CLO4	Pointers: Pointer Variable Declarations and Initialization, Pointer Operators, Passing Arguments to Functions by Reference with Pointers. Pointer Expressions and Pointer Arithmetic, Arrays, Pointers, Function Pointers.	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO3, CLO4	Characters and Strings: String Conversion, String Manipulation, Comparison Functions, Search Functions, and Memory Functions	Lecture Discussion Problem Based Learning (PBL)	Exercise
CLO2, CLO4	File Processing: Files and Streams, Creating a Sequential File, Reading Data from Sequential File, Writing data to File, and Updating Sequential Files.	Lecture Discussion Problem Based Learning (PBL)	Exercise

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√			√
CLO3	√		√		√
CLO4	√		√		√

Intended Learning Outcomes:

- Knowledge and understanding
 - ▢ Understand the fundamental programming constructs.
 - ▢ Understand and write searching and sorting techniques.
 - ▢ Understand a typical C-like program environment.
- Cognitive skills (thinking and analysis)
 - ▢ Be able to understand and analysis any problem and derive its solution.
 - ▢ Be able to develop and design algorithms.
- Communication skills (personal and academic)
 - ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - ▢ Be able to write C-like programs to build programming skills, designing skills, problem solving skills.

ICT 1106 Electrical Circuits Lab

Course Objectives:	The objective of this course is to ● Introduce with basic electrical components such as capacitor Introduce inductors and electronic devices. ● Demonstrate and understand different electrical circuits components. ● Design different electrical circuits. ● Apply different circuit network theorem. ● Construct real time embedded circuit.	
Course Learning	On completion of the course students will be able:	CLO-PLO

Outcomes (CLOs):		Mapping
	CLO1- To learn concepts of DC and AC circuit analysis.	PL01
	CLO2- To explain laws and methods of circuit analysis.	PL02
	CLO3- To apply laws and theorems for solving electrical circuits.	PL03, PL05

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Familiar with register color code.	Demonstration Case studies Lecture	Practical Exam Report Viva-Voce
CLO1	Series and parallel registers.	Inquiry Based Learning (IBL) Lecture	Practical Exam Report Observation
CLO2, CLO3	Verification of Ohm's and Kirchhoff's Laws.	Demonstration Problem Based Learning (PBL)	Practical Exam Viva-Voce Observation
CLO2, CLO3	Verification of Superposition, Thevenin's, Norton's and Maximum power transfer theorem.	Group assignment Lecture Problem Based Learning (PBL)	Observation Report Viva-Voce
CLO2, CLO3	RC Circuit Analysis	Inquiry Based Learning (IBL) Lecture Exercise	Practical Exam Viva-Voce

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				
CLO2	√				√
CLO3	√	√	√		√

Intended Learning Outcomes:

- Knowledge and understanding
 - Acknowledge the principles of operation and the main features of electric machines and their applications.
 - Understand the impact of engineering solutions in a global, economic, environmental, and societal context.
- Cognitive skills (Thinking and analysis)
 - Be able to identify, formulate, and solve engineering problems.
 - Be able apply nodal analysis, mesh analysis, superposition and source transformation to solve DC circuits.
 - Be able to compute time response of current and voltage in second order RLC circuits.
 - Be able to analyze electric circuit using simulation software.
- Communication skills (Personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable skills)
 - Be able to apply electrical circuit analysis skill to build efficient and constructive devices.

ICT 1100: Course Viva**Course Content:**

The content of the viva includes the syllabus of all major courses.

ICT-1201: Electronic Devices and Circuit

Course Objectives:	The objective of this course is to • Understand the basic of Electronics on which analysis and design of electrical and electronic circuits and systems are based, including lumped circuit, digital and operational amplifier abstractions. • Enable students to use abstractions to analyze and design simple electronic circuits. • To understand how complex devices, such as semiconductor diodes and field-effect transistors works. • Understand how the models are used in the design and analysis of useful circuits. • Enable students to design and construct electronics circuits, take measurements of circuit behavior and performance.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the operation of electronic circuit elements.	PLO1
	CLO2- To analyze responses of electronic circuits.	PLO2, PLO3
	CLO3- To design amplifier and instrumental circuits.	PLO2, PLO3

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction: Introduction to semiconductors, p-type and n-type semiconductors. p-n junction diode characteristics.	Lecture Assignment	Short Answer Exercise Tutorial
CLO1, CLO2	Diode applications: half and full wave rectifiers with and without different filter, clipping and clamping circuits, regulated power supply using zener diode.	Lecture Assignment	Short Answer Assignment Exercise

CLO1, CLO2, CLO3	Bipolar Junction Transistor (BJT): principle of operation, I-V characteristics; Transistor circuit configurations (CE, CB, CC). BJT biasing, load lines; BJTs at low frequencies; Hybrid model, h parameters, simplified hybrid model; Small-signal analysis of single and multi-stage amplifiers, frequency response of BJT amplifier.	Lecture Assignment	Short Answer Presentation
CLO2, CLO3	Field Effect Transistors (FETs): Principle of operation of JFET and MOSFET; Depletion and enhancement type NMOS and PMOS; Biasing of FETs; Low and high frequency models of FETs, Switching circuits using FETs; Introduction to CMOS.	Lecture Assignment	Exercise Interview
CLO2, CLO3	Operational Amplifiers (OP-AMPs): Linear applications of OP-AMPs, gain, input and output impedances, active filters, frequency response and noise. Introduction to feedback, Oscillators, Silicon Controlled Rectifiers (SCR).	Lecture Assignment	Short Answer Assignment MCQ
CLO3	TRIAC, DIAC and UJT: Characteristics and applications; Introduction to IC fabrication processes.	Lecture Assignment	Exercise Interview

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√			√	√
CLO3	√	√	√		√

Suggested Texts:

- Electronic Devices and Circuit, 5th Edition by David. A. Bell.
- Integrated Electronics, 2nd Edition by Millman & Halkiasa.
- Basic Electronics, 1st Edition by Bagdi and Sing.
- Electronic Devices and Circuit Theory, 11th Edition by R. L. Boylestad, L. Nashelsky.
- Principle of Electronic Circuits, 1st Edition by R.K. Mozumder.

Intended Learning Outcomes:

- Knowledge and understanding
 - ▯ Understand about constant gain, summing and buffering amplifiers.
 - ▯ Understand conduction using electron and hole theory.
- Cognitive skills (thinking and analysis)
 - ▯ Be able to explain different types of controlled sources.
 - ▯ Be able to develop an understanding of what common mode operation is.
 - ▯ Be able to acquaint with the frequency response of BJT and FET amplifier.
 - ▯ Be able to calculate ac, dc and average ac resistance of a diode from the characteristics.
- Communication skills (personal and academic)
 - ▯ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▯ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - ▯ Be able to use mathematical reasoning to comprehend and construct mathematical arguments.

ICT 1203: Object Oriented Programming

Course Objectives:	The objective of this course is to ● Understand fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries, etc. ● Be aware of the important topics and principles of software development. ● Have the ability to write a computer program to solve specified problems. ● Be able to use the Java SDK environment to create, debug and run simple Java programs. ● To build themselves as a professional developer in Java.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To demonstrate the important features of object oriented programming.	PLO1, PLO2

	CLO2- To analyze a real time problem and develop an object oriented program that solves it.	PLO2, PLO3, PLO5
	CLO3- To understand the basic principles of creating Java applications with graphical user interface (GUI).	PLO1, PLO5
	CLO4- To understand the basic approaches to the design of software applications.	PLO1, PLO5
	CLO5- To test a Java application of medium complexity, consisting of multiple classes.	PLO2, PLO7

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	Introducing C++: Definition of OOP, C++ Console I/O, Introduction to Classes, Basic concept of Object Oriented Programming, Difference between Structured Programming and Object Oriented Programming, Difference between C/C++, Introducing Function Overloading, Benefits of OOP, Characteristics of Procedure Oriented Programming, Characteristics of Object Oriented Programming and Application of Object Oriented Programming.	Lecture Group Assignment	Short Answer Report Exercise
CLO2, CLO3	Introducing Classes, Arrays, Pointers and References: Constructor and Destructor functions, Constructors with parameters, Introducing Inheritance, Object Pointers, Relation between Classes, Structures and unions, In-line functions, Automatic in-line functions, Assigning objects, Passing objects to functions, Returning objects from function, Defining member functions, Friend functions, Static member functions. Array of objects, Pointer to objects, the pointer, using new and delete, passing references, returning references independent references.	Lecture Group Assignment	Essay Type Answer Report Exercise
CLO3, CLO4	Function Overloading and Operator Overloading: Constructor Overloading, Copy constructor, Default arguments, Overloading ambiguity, Address of overloaded function. Binary operator overloading, Unary operator	Lecture Demonstration Group Assignment	Short Answer Report Exercise

	overloading, Relational and logical operator overloading, Operator overloading using friend functions, Limitations of operator overloading.		
CLO3, CLO4	Inheritance: Defining derived classes, Single inheritance, multiple inheritance, multilevel inheritance, Hierarchical inheritance, Virtual base classes, Constructors in derived classes, nesting of classes.	Lecture Group Assignment	Essay Type Answer Exercise Report Presentation
CLO3, CLO4	C++ I/O System: Streams, Stream classes, Unformatted I/O, Binary I/O, formatted I/O, I/O manipulators, Inserters, Extractors, File I/O streams, Opening and closing files, Random access files, I/O status checking, Customized I/O and files.	Lecture Assignment Demonstration	Short Answer Report Exercise
CLO3, CLO4	Virtual Functions: Pointers to derived classes, Applying Polymorphism using virtual functions, Polymorphic class, Pure Virtual functions, Abstract classes, early binding, and late binding.	Lecture Group Assignment	Essay Type Answer Report Exercise Presentation
CLO4, CLO5	Template Exception Handling and Standard Template Library: Generic functions, Generic classes, Exception handling, Throwing mechanism, Catching mechanism, Rethrowing mechanism, Specifying exceptions Templates, Components of STL, Container, Algorithms.	Lecture Assignment	Essay Type Answer Exercise Report
CLO3	Introduction to Java: History of Java, Java Class Libraries, Introduction to Java Programming, A simple Program, Memory Concepts, Decision Making.	Lecture Demonstration	Short Answer Report Exercise
CLO3	Methods: Introduction to Program Module in Java, Static Methods, Math Class Methods, Declaring Methods with multiple parameters, Java API Packages, Automatic Variables, Recursion, Method Overloading, Method of the Applet Class.	Lecture Group Assignment	Short Answer Exercise Report Presentation
CLO3, CLO4	Inheritance and Polymorphism: Introduction to Superclasses and Subclasses, Protected Members, Using Constructors and Finalizers in Subclasses, Composition vs. Inheritance, Introduction to polymorphism, Dynamic method building, Final Methods and Classes.	Lecture Demonstration Group Assignment	Essay Type Answer Short Answer Exercise Report Presentation

CLO3	GUI Components: Overview of Swing Components, Introduction to Event Handling with Nested Classes, Common GUI Event Types and Listener Interfaces, Mouse Event Handling,	Lecture Demonstration Group Assignment	Short Answer Exercise Presentation
CLO3, CLO4	Graphics and Java 2D: Graphics Contexts and Graphics Objects, Color Control, Font Control, Drawing different Objects like Lines, Rectangles, Polygons, Java 2D API.	Lecture Demonstration Group Assignment	Short Answer Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√			√	√
CLO3			√		√
CLO4	√		√		√
CLO5	√	√			√

Suggested Texts:

- The C++ Programming Language, 4th Edition by Bjarne Stroustrup.
- Object-Oriented Programming with C++, 4th Edition by Robert Lafore.
- C++: The Complete Reference, 4th Edition by H. Schildt.
- C++: How to program. 10th Edition by Deitel H M and Deitel P J.
- Java The Complete Reference, 9th Edition by Herbert Schildt.
- The Java Programming Language, 4th Edition by Ken Arnold, James Gosling, David Holmes.
- JAVA How To Program, 10th Edition by Deitel and Deitel, Publisher: Prentice/Hall International.
- Java, Java, Java™: Object-Oriented Problem Solving, 3rd Edition by Ralph Morelli, Ralph Walde, Publisher: Prentice Hall.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic features of object oriented programming.

- Understand the fundamental object oriented programming constructs.
- Understand and write basic OOP based program in computer.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any problem and derive its solution.
 - Be able to develop and design algorithms.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to identify the real time problems to solve based on object oriented programming concepts.

ICT 1205: Linear Algebra and Analytical Geometry

Course Objectives:	The objective of this course is to • Provide the knowledge of basic techniques of mathematics such as matrix algebra, Fourier analysis and so on. • Use of mathematical tools are of a variety of uses especially in the field of information and communication Engineering. • Utilize mathematics in analyzing engineering theories.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand fundamental properties of matrices including determinants, inverse matrices, eigenvalues and linear transformations	PLO1
	CLO2- To solve system of linear equations.	PLO1, PLO2
	CLO3- To comprehend vector spaces and subspaces.	PLO1
	CLO4- To interpret vectors in two and three-dimensional spaces both algebraically and geometrically.	PLO4
	CLO5- To analyze real life problems using coordinate axis.	PLO6

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2, CLO3	Theory of Matrices: Concepts of matrices, types of matrices, transposes, determinant, Sarrus' rule, Laplace expansion, inverse matrix, elementary row operations of matrices (echelon form and row reduced echelon form of matrices), LU-Decomposition and rank. Systems of linear equations and its solution by Cramer's Rule, Gaussian elimination method and inverse matrix method, Kronecker-Capelli theorem. Vector spaces: Definition and examples of vector spaces, subspaces, linear combination, linear independence, basis and dimension. Linear transformations on vector spaces: Linear transformation, the kernel and image of a linear transformation, rank and nullity of a linear transformation. Diagonalization of matrices: Introduction, eigenvalues and eigenvectors, diagonalization of matrices and the Cayley-Hamilton theorem.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO4, CLO5	Two-Dimensional Geometry: Change of axes, Transformation of co-ordinates, Pair of straight lines, Circles, System of circles, The general equation of second degree, Identification of curves.	Lecture , Discussion Group Assignment Demonstration	Report Exercise Presentation
CLO4, CLO5	Three-Dimensional Geometry: Co-ordinate system, Direction cosines and direction ratios, Plane, Straight line and Sphere.	Lecture Discussion Group Assignment Demonstration	Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1					√
CLO2	√	√			√

CLO3				√	√
CLO4	√				√
CLO5	√	√			√

Suggested Texts:

- Elementary Linear Algebra, 11th Edition by Howard Anton, Chris Rorres.
- Elementary Linear Algebra, 6th Edition by Ron Larson, David C. Falvo, Publisher: Houghton Mifflin Harcourt Publishing Company, Boston, New York.
- Linear Algebra and Its Applications, 5th Edition by David C. Lay.
- Analytical Geometry: 2D and 3D, 1st Edition by P. R. Vittal, Publisher: Pearson Education India, 2013.
- Introduction to Linear Algebra, 5th Edition by Gilbert Strang, Publisher: Wellesley-Cambridge Press.
- Linear Algebra, Concepts and Methods, 1st Edition by Martin Anthony and Michele Harvey.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of linear algebra and analytical geometry.
 - Understand relation within Computer Science and Coordinate geometry.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis the different mathematical problems related to the matrix, vectors and transformations.
 - Be able to analyze Mathematical model using Coordinate geometry concepts.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to understand and analysis the important theorem and ideas of linear algebra and analytical geometry to solve the complex problems, particularly related to the computer science and engineering fields.

Course Objectives:	The objective of this course is to • Apply formal mathematical, logical reasoning techniques and basic counting techniques to solve a variety of problems. • Apply algorithms to problems including searching algorithms, base conversion algorithms in the design and analysis of other algorithms, computability theory, software engineering, and computer systems. • To solve discrete probability problem sets in combinatorics and probability theory to media applications. • Apply Graph and Tree.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand mathematical reasoning.	PLO1
	CLO2- To learn theorem proving.	PLO1, PLO2
	CLO3- To analyze combinatorial algorithmic problems.	PLO1, PLO2
	CLO4- To apply discrete structures to model real life problems.	PLO3, PLO4

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	Set theory, Propositional logic and Mathematical Induction: Set Theory, Mathematical reasoning and proof techniques, Propositional calculus and predicate calculus.	Lecture Assignment	Short Answer Exercise Tutorial Assignment
CLO1, CLO2	Relations: Relations and Their Properties, -binary Relations and Their Applications, Representing Relations, Closures of Relations, Equivalence Relations, Partial Orderings, Function and algebraic Structure	Lecture Assignment	Short Answer Assignment Exercise

CLO1, CLO2	Graph and Tree: Basic terminologies of graph and tree, Handshaking theorem, Bipartite graphs, Graph Isomorphism, Hamilton circuits and path, Euler circuit and path, Shortest path problem, Chromatic number, Properties of tree, Binary search tree, Tree traversal, BFS and DFS, Graph coloring, N-queens problem, Minimum Spanning Tree and Prim's and Kruskal's algorithm.	Lecture Assignment	Exercise Interview
CLO2, CLO3, CLO4	Number Theory: Divisibility, Primes, Prime Examples, Generating functions, Factorial factors, Relative Primality, 'mod': The Congruence Relation, Independent Residues, Binomial Coefficient, Special Numbers and Chinese remainder theorem,	Lecture Assignment	Exercise Interview Assignment
CLO3, CLO4	Discrete Probability and Algorithms: Definitions, Mean and Variances, Probability Generating Functions, Flipping Coins, Hashing. Growth of functions, complexity of algorithms.	Lecture Assignment	Short Answer Assignment MCQ
CLO3, CLO4	Graph and Tree: Basic terminologies of graph and tree, Handshaking theorem, Bipartite graphs, Graph Isomorphism, Hamilton circuits and path, Euler circuit and path, Shortest path problem, Chromatic number, Properties of tree, Binary search tree, Tree traversal, BFS and DFS, Graph coloring, N-queens problem, Minimum Spanning Tree and Prim's and Kruskal's algorithm.	Lecture Assignment	Exercise Interview
CLO2, CLO4, CLO4	Counting and Boolean Algebra: The Basics of Counting, The Pigeonhole Principle, Permutations and Combinations, Binomial Coefficients and Identities, Linear Recurrence Relations, Divide-and-Conquer Algorithms and Recurrence Relations, Generating Functions, Inclusion-Exclusion, Boolean Functions, Representing Boolean Functions, Logic Gates, Minimization of Circuits	Lecture Assignment	Exercise Assignment Tutorial MCQ

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√

CLO2	√				√
CLO3	√			√	√
CLO4		√			√

Suggested Texts:

- Discrete Mathematics and its Applications, 7th Edition by Rosen, K.H.
- Discrete Mathematics, 4th Edition by Olympid Nicodemi.
- Number Theory, 4th Edition by S.G. Telang.
- Elements of Discrete Mathematics, 3rd Edition by C. L. Liu.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the mathematical reasoning behind a program.
 - Understand relation within a common set of numbers.
- Cognitive skills (thinking and analysis)
 - Be able to explain at high levels concepts and implement basic operations in discrete mathematics.
 - Be able to perform combinatorial analysis to solve counting problems.
 - Be able to develop mathematical models through relations, combinatory, graphs, and trees.
 - Be able to apply graph theory and other mathematical methods to both data structures and analysis of algorithms, and some other problems in computer sciences.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to use mathematical reasoning to comprehend and construct mathematical arguments.

ICT 1209: Bangladesh Studies

Course Objectives:	The objective of this course is to • Equip students with factual knowledge and analytical skills that will enable them to learn and critically appreciate history, politics, and economy of Bangladesh. • Trace the historical root of Bangladesh as an independent state focusing on the social, economic and political developments that have taken place since its independence. • Identify the major socio-economic, political, environmental and developmental issues that have arisen during this period, before assessing the progress over time.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To identify specific stages of Bangladesh's political history, through the ancient, medieval, colonial and post-colonial periods and critically analyses plurality of cultural identities of Bangladesh.	PL01, PL02
	CLO2- To explain the economy and patterns of economic changes through qualitative and quantitative analysis. This will increase their awareness on global issues of development processes and the nature of environmental challenges including ways to address them effectively.	PL02
	CLO3- To appreciate the role of NGOs and civil society in developing new models and pathways to resolve the range of development challenges that the country is currently facing.	PL04, PL07

Course Contents:

CLO	Course Content	Teaching Learning Strategy	Assessment Strategy
CLO1	Lecture-1 Introduction to Course	Brainstorming Conversation Storytelling	Quiz Peer Assessment Evaluation

		Sharing Audio-Video Sharing Real Life Experience	
CLO1	Lecture-2, 3, 4 Bangladesh: Geography, Demography, Social Stratification, Social Class, Ethnicity & Cultural Traits	Open Discussion Peer Study Sharing Audio-Video	Class Lecture Quiz Peer Assessment
CLO1	Lecture 5,6,7 Origin of the name of Bangladesh, Bangla Language & People of Bangladesh	Class Lecture Open Discussion	Online Activities e.g. Q/A Peer Assessment
CLO1	Lecture 8 Class Test/Quiz-1, Review of the previous Classes	Open Book Exam	Lecture Q/A Peer Assessment
CLO1	Lecture-9-13 Historical Studies of Bangladesh: Ancient time, Muslim and British colonial rules, Pakistani movement and birth of Pakistan(1947), Language movement(1952), six point movement(1966), Mass upsurge(1969) and Independence of Bangladesh	Lecture Peer Discussion Debate Story-Telling Question-Answer, Sharing Audio-Video Pictures	Lecturer Q/A Peer Assessment
CLO1	Lecture 14,15,16 Constitution of Bangladesh	Debate Case Studies Sharing audio-video Sharing Real Life Experience Online Assignment	Lecture Open Discussion Q/A Peer Assessment
CLO1, CLO3	Lecture 17,18 Government of Bangladesh	Debate Case Studies Poster Presentation Sharing Real Life Experience Sharing Audio-Video	Lecturer Q/A Peer Assessment Assignment

CLO1, CLO3	Lecture 19 Class Test -2, Review of the previous Classes	Written Exam	Rational
CLO2	Lecture 20-22 Economy of Bangladesh (Historical Background, Nature, important economic sectors), significance of IT sector in Bangladesh economy, SDG, Vision 2021,	Debate Group Problem Solving Sharing real life Experience	Lecture Open Discussions Q/A Peer Assessment
CLO2	Lecture 23 Environmental degradations & Climate Change: Bangladesh perspective	Q/A Session Debate Brainstorming Poster Presentation Assignment Topic Presentation	Lecture Open Discussion Q/A Peer Assessment Presentation Assignment
CLO2	Lecture 24 Class Test-3, Review of the previous Classes	Written Exam (Classroom and Online)	Rational
CLO2, CLO3	Lecture 25 Urbanization	Q/A Session Debate Brainstorming Poster presentation	Lecture Open Discussion Q/A Peer Assessment Presentation Assignment
CLO2, CLO3	Lecture-26 Class Test/Quiz-3, Review of the whole syllabus, Final Examination	Written Exam (Classroom and Online)	Rational

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√			√	√
CLO3	√	√			√

Suggested Texts:

- Banglapedia: National Encyclopedia, Asiatic Society of Bangladesh, Dhaka, 2nd Edition by Islam, S., 2003.
- Bangladesh at the Crossroads. University Press Ltd, 1st Edition by Kibria, S. A., 1999.
- Bangladesh: A Political History since Independence. London: IB Taurus, 1st Edition by Riaz, A., 2016.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the historical, archeological and anthropological journey of Bengal people and their development and present status of the nation.
- Cognitive skills (thinking and analysis)
 - Critically analyze how different constitutional bodies and socio-political institutions operate and how their behavior impact on political governance.
 - Critically analyze and present cogent argument on why tensions and contestations between and among social groups may emerge within and among states both in written and oral form.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - The students attending this course are expected to be made aware of the past and present of Bangladesh for the enlargement of their knowledge to face 21st century.

ICT 1202: Electronic Devices and Circuits Lab

Course Objectives:	The objective of this course is to Understand Basic characteristic and operation of electronic devices; Design amplifier, oscillator, filter etc. and construct real time embedded circuit.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To verify the working of different diodes, transistors, CRO probes and measuring instruments. Identifying the procedure of doing the experiment.	PL01, PL02
	CLO2- To design the circuits with basic semiconductor devices (active & passive elements), measuring instruments and power supplies that serves many practical purposes.	PL03, PL05
	CLO3- To apply measure and record the experimental data, analyze the results, and prepare a formal laboratory report.	PL03, PL04
	CLO4- To design structured program for complex computational applications.	PL03, PL06

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO3	Verification of P-N junction diode characteristics.	Demonstration Case studies Lecture	Practical Exam Report Viva-Voce
CLO1, CLO3	Verification of Zener diode characteristics.	Inquiry Based Learning (IBL) Lecture	Practical Exam Report Observation
CLO1, CLO3	Verification of Rectifiers (with and without c-filter).	Demonstration Problem Based Learning (PBL)	Practical Exam Viva-Voce Observation

CLO2, CLO3	Verification of BJT characteristics (CE configuration).	Group assignment Lecture Problem Based Learning (PBL)	Observation Report Viva-Voce
CLO2, CLO3	Verification of FET characteristics (CS configuration).	Inquiry Based Learning (IBL) Lecture Exercise	Practical Exam Viva-Voce Completion
CLO2, CLO3	Verification of SCR characteristics.	Demonstration Problem Based Learning (PBL)	Observation Report Viva-Voce
CLO2, CLO3	Verification of UJT characteristics.	Group assignment Lecture Problem Based Learning (PBL)	Practical Exam Viva-Voce Completion
CLO2, CLO3	Verification of CRO Operation and its Measurements.	Inquiry Based Learning (IBL) Lecture, Exercise	Observation Report Viva-Voce
CLO2, CLO3	Verification of BJT-CE Amplifier.	Demonstration Problem Based Learning (PBL)	Practical Exam Viva-Voce Completion
CLO2, CLO3, CLO4	Verification of FET-CE Amplifier.	Group assignment Lecture Problem Based Learning (PBL)	Practical Exam Report Observation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√

CLO2	√				√
CLO3		√			
CLO4			√		√

Intended Learning Outcomes:

- Knowledge and understanding
 - ▢ Understand about constant gain, summing and buffering amplifiers.
 - ▢ Understand conduction using electron and hole theory.
- Cognitive skills (thinking and analysis)
 - ▢ Be able to explain different types of controlled sources.
 - ▢ Be able to develop an understanding of what common mode operation is.
 - ▢ Be able to acquaint with the frequency response of BJT and FET amplifier.
 - ▢ Be able to calculate ac, dc and average ac resistance of a diode from the characteristics.
- Communication skills (personal and academic)
 - ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
 - ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - ▢ Be able to use mathematical reasoning to comprehend and construct mathematical arguments.

ICT 1204: Object Oriented Programming Lab

Course Objectives:	The objective of this course is to ● Provide understanding of programming language using C++ and Java. Its main objective is to teach the basic concepts and techniques which form the object oriented programming paradigm.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To analyze the various important features of object oriented programming.	PL01, PL05

	CLO2- To design and implement the object oriented program for developing the industrial level applications.	PLO3, PLO6
	CLO3- To apply the knowledge to solve real time problem.	PLO7, PLO11
	CLO4- To know in preparing a professional looking package for each business project using javadoc.	PLO11, PLO12

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction to Java: Basics of Java programming, Data types, Variables, Operators, Control structures including selection, Looping, Java methods, Overloading, Math class, Arrays in java.	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report
CLO1, CLO2	Objects and Classes: Basics of objects and classes in java, Constructors, Finalizer, Visibility modifiers, Methods and objects, Inbuilt classes like String, Character, String Buffer, File, this reference..	Problem Based Learning (PBL) Lecture Group Assignment	Practical Exam Exercise Report
CLO1, CLO2, CLO3	Inheritance and Polymorphism: Inheritance in java, Super and sub class, Overriding, Object class, Polymorphism, Dynamic binding, Generic programming, Casting objects, Instance of operator, Abstract class, Interface in java, Package in java, UTIL package.	Problem Based Learning (PBL) Lecture Group Assignment	Practical Exam Exercise Report
CLO2, CLO3, CLO4	Event and GUI programming: Event handling in java, Event types, Mouse and key events, GUI Basics, Panels, Frames, Layout Managers: Flow Layout, Border Layout, Grid Layout, GUI components like Buttons, Check Boxes, Radio Buttons, Labels, Text Fields, Text Areas, Combo Boxes, Lists, Scroll Bars, Sliders, Windows, Menus, Dialog Box, Applet and its life cycle, Introduction to swing.	Lecture Demonstration	Exercise Presentation Report

CLO3, CLO4	I/O programming: Text and Binary I/O, Binary I/O classes, Object I/O, Random Access Files.	Lecture Demonstration Group Assignment	Short Answer Exercise Presentation Report
CLO2, CLO4	Multithreading in java: Thread life cycle and methods, Runnable interface, Thread synchronization, Exception handling with try-catch-finally, Collections in java, Introduction to JavaBeans and Network	Lecture Problem Based Learning (PBL) Project	Practical Exam Report Exercise

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√			√	√
CLO3		√			√
CLO4			√		√

Intended Learning Outcomes:

- Knowledge and understanding
 - ▯ Understand the basic concepts of object oriented programming.
 - ▯ Understand the fundamental object oriented programming constructs.
 - ▯ Understand and write basic OOP program.
- Cognitive skills (thinking and analysis)
 - ▯ Be able to understand and analysis any problem and derive its solution.
 - ▯ Be able to develop and design algorithms.
- Communication skills (personal and academic)

- ▯ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
- ▯ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - ▯ Be able to realize the real time problems and develop the solution with the principle of OOP.

ICT 1200: Project Work - I and Course Viva

Course Contents:

The content of the viva includes the syllabus of all major courses. Each group of student must have to do a project work based on the courses taught in the 1st Year's 1st and 2nd Semester. A faculty member or representative from other dept./institute/industry will supervise the project.

2nd Year 1st Semester

ICT 2101: Data Structures

Course Objectives:	The objective of this course is to ● Solve problems using data structures such as linear lists, stacks, queues, hash tables, binary trees, heaps, binary search trees, graphs and writing programs for these solutions.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To describe how arrays, records, linked structures, stacks, queues, trees, and graphs are represented in memory and used by algorithms to solve a problem.	PL01
	CLO2- To develop algorithms for manipulating stacks, queues, linked lists, hash tables, trees, and graphs.	PL02, PL05
	CLO3- To analyze and select the most suitable and effective algorithm for solving certain problem and case study.	PL02, PL05
	CLO4- To use programming tools to write and debug codes for abstract data types.	PL04, PL05

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO4	Introduction to Data Structures & Algorithms: Types of Data Structure, Basic Concepts of Algorithm, Programming Principle, ADT and class.	Lecture Discussion	Essay Type Answer MCQ
CLO1, CLO3	Algorithm Efficiency: Analysis Order of magnitude, Asymptotic Notations.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO1, CLO3	Sorting: Selection Sort, Insertion Sort, Bubble Sort, Advanced Sorting - Quick Sort, Merge Sort.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Exercise
CLO1, CLO3	Searching: Sequential Search, Binary Search.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Exercise
CLO1, CLO2, CLO4	Linked Lists: Pointer, Introduction to Linked List, Types of Linked Lists, Linked List Operation, Create Linked List, Linked List Declaration, Add node, Delete Node, Find Node, and Circular Linked list.	Lecture Assignment Demonstration	Essay Type Answer Report Exercise Presentation
CLO1, CLO2, CLO4	Stack: Introduction to Stack, Stack Operations–pop and push, Stack Implementation, stack application.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

CLO1, CLO2, CLO4	Queue: introduction, Queue Implementation (Array and link list), Queue operations, Circular queue.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO1, CLO3, CLO4	Tree: Introduction, Terms related to tree, Binary search tree, Tree Operation – create node, add node, delete node, print node, search node, Tree Traversal (pre-order, in-order and post-order).	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO1, CLO2	Heap: Introduction, Maintaining the heap property, Building a heap, The heap sort algorithm.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO2, CLO3	Graphs: Breadth first search (BFS), Depth first search Graphs (DFS), Minimum Spanning Tree, Dijkstra's Algorithm.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO2, CLO3	Hashing: Hash function, Need for a good hash function, Hash table, Collision resolution techniques, Linear probing, Implementation of hash table with linear probing, Quadratic Probing, Implementation of hash table with quadratic probing, Double hashing, Implementation of hash table with double hashing, Applications.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√			√
CLO2	√	√		√	√

CLO3	√	√		√	√
CLO4	√	√			√

Suggested Texts:

- Data Structure and Algorithm, 7th Edition by Schaum's Outline Series.
- Data Structure, 1st Edition by Edward M. Reinggold.
- Fundamentals of Data Structures, 1st Edition by Horowitz E. and Sahni, S Galgotia.
- C++ Plus Data Structures, 5th Edition by Nell Dale.
- Data Structures Using C, 1st Edition by Aaron M. Tenenbaum.
- Introduction to Algorithms, 3rd Edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest.
- Data Structure and Program Design in C, 2nd Edition by Kruse, Tondo, Leung, Publisher: Prentice-Hall.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of data structures.
 - Understand the fundamental steps of designing algorithms.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any problem and derive its solution.
 - Be able to design and develop algorithms using various data structures.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to identify the complex and real life problems to solve using the effective understanding of data structure and algorithms.

ICT 2103: Digital Logic Design

Course Objectives:	The objective of this course is to • Introduce the concept of digital and binary systems. • Design and analyze combinational logic circuits. • Design and analyze sequential logic circuits. • Understand the basic software tools for the design and implementation of digital circuits and systems.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To introduce the wave shaping circuits, Switching Characteristics of diode and transistor to the students.	PL01
	CLO2- To analyze different types of Multi vibrators and their design procedures.	PL02
	CLO3- To understand Sampling Gates and to Design NAND and NOR gates using various logic families.	PL03
	CLO4- To design digital circuits using Boolean Algebraic and state transition techniques.	PL03, PL05

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO3, CLO4	Number System: Review of number system, Binary, Octal, Hexadecimal, BCD, and ASCII. Boolean Algebra and Minimization: Introductory concept of number systems and codes. Boolean constants and variables, truth tables. Basic logic function. Boolean expressions, Implementing circuits from Boolean expressions, Boolean theorems, De Morgan's theorem. Sum-of-product and product -of-Sum forms. Simplifying logic circuits, algebraic simplification, the Karnaugh map method, Qnine McCluskey design method.	Lecture Assignment	MCQ Exercise
CLO3, CLO4	Logic Gates and Combination Circuits: Different types of logic gates. Circuit design using NAND or NOR gates only. Alternate logic-gate representations, Designing combinatorial logic circuits. Exclusive OR and NOR circuits. Logic circuits with multiple outputs designing without a truth table.	Lecture Assignment	Short Answer Discussion

CLO3, CLO4	Flip-flops: SR, JK, D and T flip flops. The D latch. Master slave FF. Flip flop application. FF synchronization. Data stores and transfer. Frequency division counting. One shot. Arithmetic circuits: Adder circuits. Carry propagation, carry look-ahead adder. IC parallel adder. The 2's complement addition and subtraction system. The BCD adder. Binary multiplier	Lecture Assignment	Short Answer Presentation
CLO3, CLO4	Counters and Register: Asynchronous Counter, Ripple counters, counters with mod numbers $< 2^n$, IC asynchronous counters, asynchronous down counter, propagation delay and ripple counters, Synchronous down and up/down counters. Decoding a counter, Decoding glitches. Cascading BCD counters, Shift-register.	Lecture Assignment	Short Answer presentation
CLO3, CLO4	Counter Application: frequency counter, digital clock. IC register. MSI Logic Circuits: Decoders, BCD-to-decimal decoders, BCD-to-7-segment decoder/drivers. Encoders. Multiplexes applications. Demultiplexer, Encoders. Multiplexes applications. Demultiplexer.	Lecture Assignment	Short Answer Presentation
CLO3, CLO4	Integrated-Circuit Logic Families: Digital IC terminologies, TTL series characteristics, open-collector TTL, ECL family, MOS digital ICs, MOSFET, CMOS tristate logic, TTL-CMOS-TTL interfacing, Memory Devices: Memory terminology, general memory operation, semiconductor memory technologies, different types of ROMs, semiconductor RAMs, static and dynamic RAMs Magnetic bubble memory, CCD memory, VHDL & FPGA Concept.	Lecture Assignment	Short Answer Presentation
CLO2	Pulse Operations, Analysis and design of Bistable Multivibrator: Pulse transformers, pulse transmission, pulse generation; Analysis and Design of Fixed bias transistor binary, Commutating capacitors, Triggering circuits, Non saturating Binary, Schmitt trigger circuit and its Applications, Analysis and design of Monostable, Astable Multivibrator: Analysis and design of Monostable multivibrators (Collector-coupled and Emitter-coupled) using transistors and Analysis and design of Astable multivibrator (Collector coupled and Emitter-coupled) using transistors.	Lecture Assignment	Short Answer Presentation

CLO1	Interfacing with analog world. Digital-to-analog conversion (D/A converter circuitry, DAC application), analog-to-digital conversion (Successive approximation ADC, flash ADC, digital-ramp ADC), Sample-and-hold circuits.	Lecture Assignment	Short Answer Presentation
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Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√	√	√
CLO2	√	√	√	√	√
CLO3	√	√	√		√
CLO4	√	√	√		√

Suggested Texts:

- Digital Systems, Principles and Applications, 11th Edition by Ronald J Tocci.
- Digital Computer Electronics, 3rd Edition by A P Malvino.
- Pulse and Digital Circuits, 1st Edition by Venkata Rao K, Rama Sudha K, Manmadha rao G, Publisher: Pearson, 2010.
- Pulse, Switching and Digital Circuits, 5th Edition by David A. Bell, Publisher: OXFORD University Press.
- Digital Electronics, 1st Edition by Taub & Schilling.

Intended Learning Outcomes:

- Knowledge and understanding
 - Demonstrate practical skills in the programming and testing of digital systems on FPGA and microcontroller development boards.
 - Understand the time and frequency domain aspects.
 - Understand how to analyze, build and troubleshoot digital circuits.
- Cognitive skills (thinking and analysis)
 - Be able to identify, formulate, and solve engineering problems.
 - Be able to analyze and synthesize combinational logic circuits.
 - Be able to apply Boolean algebra to solve logic functions.
 - Be able to interpret schematics to determine principal circuit functions.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves making effective presentations and giving and receiving clear instructions.

- Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to apply digital electronic circuit analysis skill to build efficient and constructive devices.
 - Be able to apply logic design circuits with Programmable Logic Devices.

ICT 2105: Numerical Analysis

Course Objectives:	The objective of this course is to ● Provide the student with numerical methods of solving the non-linear equations, interpolation and approximation, numerical differentiation and integration.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the nature and operations of Numerical Analysis.	PLO1
	CLO2- To formulate the usage of numerical methods in modern scientific computing.	PLO2, PLO3
	CLO3- To demonstrate familiarity with theories and concepts used in Numerical Analysis and identify the steps required to carry out a piece of research on a topic in Numerical Analysis.	PLO2, PLO7
	CLO4- To conduct derivation of the Numerical Methods, studying their convergence rate and performance, applicability of the methods on different test examples.	PLO4, PLO6

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
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CLO1, CLO3	Numerical calculation: Introduction and fundamental concepts of numerical methods for linear equations. Approximation in numerical computation, Truncation and rounding errors, Fixed and floating-point arithmetic, Propagation of errors.	Lecture Discussion Assignment	Short Answer Exercise
CLO2	Roots of Nonlinear Equations using bracketing and open ended methods: Bisection, False Position, Fixed Point, Newton-Raphson, Secant method. Nature of convergence of all methods should also be discussed.	Lecture Discussion Problem Based Learning (PBL) Assignment	Short Answer Exercise
CLO3, CLO4	Direct Solution of Linear Equations: Gaussian Elimination(Naïve Gaussian Elimination, Gaussian Elimination with pivoting.), Gauss-Jordan Method, LU Decomposition Iterative Solution of Linear Equations: Jacobi's Method, Gauss-Seidel Method.	Lecture Problem Based Learning (PBL) Assignment	Exercise
CLO3, CLO4	Curve Fitting- Interpolation and Approximation: Direct Method of Interpolation, Lagrange, Newton's Interpolation Polynomial, Interpolation with Equidistant Points, Spline Interpolation.	Lecture Group Assignment Demonstration	Exercise Practical Exam
CLO3, CLO4	Regression Analysis: Linear, Transcendental and Polynomial equation	Lecture Project Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam
CLO3, CLO4	Numerical Integration: Newton-Cotes Methods, Trapezoidal and Simpson rules ($\frac{1}{2}$ rule, $\frac{2}{3}$ rule), Romberg Integration.	Lecture Project Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam

CLO3, CLO4	Numerical Solution of Ordinary Differential Equation: Taylor series, Picard, Runge-Kutta, Heun's, Euler's method.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam
CLO3, CLO4	Solution of partial Differential Equations: Determination of characteristics equation of a matrix using Faddeev-Leverrier method; Eigenvalue and Eigenvector and matrix inversion.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√		√	√
CLO3	√				√
CLO4	√			√	√

Suggested Texts:

- Numerical Methods, 1st Edition by E. Balagurusamy.
- Numerical Methods for Engineers, 7th Edition by Steven C. Chapra, Raymond P. Canale.
- Numerical Mathematics and Computing, 7th Edition by Cheney & Kinkaid.
- Numerical Method, 3rd Edition by Jain & Iyenger.
- Computer Oriented Numerical Methods, 4th Edition by Rajaraman, V.

Intended Learning Outcomes:

- Knowledge and understanding

- ▢ Students will understand the basic numerical techniques with the underlying mathematical explanation to find out solutions to ordinary and partial differential equations, different linear and nonlinear equations etc.
- ▢ Students will also gain the knowledge of how to interpolate or extrapolate with the data available.
- Cognitive skills (thinking and analysis)
 - ▢ Students will be able to analyze large systems of linear equations, problems of a matrix which can be obtained numerically where analytical methods fail to give solutions.
 - ▢ Most of the engineering problems are characterized in the form of either nonlinear ordinary differential equations or partial differential equations. Students will be able to perceive and analyze the problems to solve the given ODE numerically, which will be useful in attempting any engineering problem.
- Communication skills (personal and academic)
 - ▢ Individual assignments at the end of every topic based session will help students to select a numerical method taking into account accuracy and cost requirements, interpret and validate the computed results, which may lead to further refinement of the mathematical model.
 - ▢ Consecutively students will grow an ability to design and conduct experiments as well as to analyze and interpret data.

ICT 2107: Statistics and Probability for Engineers

Course Objectives:	The objective of this course is to ● Provide the basic theorem of statistical and various statistical data representation methods. ● Understand the concept of data collection methods, measure of location, measure of dispersion, absolute and relative measures. ● Provide tools for supporting decisional processes through the management of databases and the use of statistic and mathematic models. ● Provide an introduction to probability theory, random variables, markov processes and queuing models. It is important because of its direct application in areas such as genetics, finance and telecommunications.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand different types of measures of location and dispersions.	PL01
	CLO2- To draw different types of graphs and diagrams and their uses.	PL02, PL04
	CLO3- To estimate index and intensity of relationship among different phenomenon.	PL02, PL05

	CLO4- To calculate probability of occurrence of any event using different probability distribution formula.	PLO4, PLO7
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Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	Elements of Statistics: Nature and scope of statistics, Nature & representation of statistical data; Attributes and variables; population and sample, discrete and continuous variables.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1, CLO2	Method of data collection: Interviews, Questionnaires and surveys, Observations, Documents and records, Focus groups, Oral histories.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Home Exercise
CLO1, CLO2	Sampling procedure: Simple random, stratified, systematic and cluster sampling technique.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1	Measures of location: Characteristics of an ideal measure; Arithmetic mean; Geometric mean; Harmonic mean; Median; Mode.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise
CLO1, CLO2	Measure of dispersion: Characteristics of an ideal measure: Absolute & Relative measures; Range; Standard deviation; Mean deviation; Quartile deviation; Quartiles; Deciles; Percentiles; Coefficient of dispersion; Coefficient of variation; Skewness and kurtosis.	Lecture Group Assignment Problem Based Learning (PBL)	Essay Type Answer Report Exercise

CLO4	Elements of Probability: Meaning and definition of probability; A priori and a posteriori probability; Basic terminology of probability; Random variables; Probability function; Expectation of sum and products.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO4	Probability Distribution: Binomial, Poisson, Normal & Exponential distribution.	Lecture Problem Based Learning (PBL)	Essay Type Answer Home Exercise
CLO3, CLO2	Regression and correlation: Relationship between variables; Fitting of regression lines; Simple correlation; Rank correlation; Multiple correlation and regression.	Lecture Group Assignment Problem Based Learning (PBL)	Essay Type Answer Report Exercise Presentation
CLO3	Tests of Significance: Tests of means, Variance, Correlation coefficient, regression coefficient and associations, model validation.	Lecture Individual Assignment Problem Based Learning (PBL)	Essay Type Answer Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√			√
CLO2	√	√			√
CLO3	√	√		√	√
CLO4	√	√			√

Suggested Texts:

- Fundamentals of Probability and Statistics for Engineers, 1st Edition by T.T. Soong.
- Fundamentals of Biostatistics, 7th Edition by Bernard Rosner.
- Probability & Statistics for Engineers, 9th Edition by Johnson R.A, Miller & Freud's.
- Biostatistics - A Foundation for Analysis in the Health Sciences, 11th Edition by Wayne W. Daniel, Chaad L. Cross.

Intended Learning Outcomes:

- Knowledge and understanding
 - ▢ Understand the basic concepts of statistics, probability and causal relationships.
 - ▢ Understand how to collect sample from a population.
 - ▢ Understand the means of presentation of data and results.
- Cognitive skills (thinking and analysis)
 - ▢ Be able to understand and analyze any sample data.
 - ▢ Be able to understand cause and effect among various phenomena.
- Communication skills (personal and academic)
 - ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - ▢ Be able to identify the complex and real life data using the effective understanding of the statistical parameters and analysis of data.

ICT 2109: Data Communication

Course Objectives:	The objective of this course is to ● Discuss about the analog and digital transmission, performance metrics of communication channels. ● Provide the details about synchronous and asynchronous communications, source and channel coding. ● Give an overview about the switching, radio system, optical fiber and satellite communications. ● Understand the basic concepts and structure of telecommunication networks for narrowband and broadband services. ● Understand the basic settings in the operation of telecommunication systems and devices.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the components of a data communications system.	PL01
	CLO2- To analyze the selection process of various transmission media in different networks.	PL02, PL07

	CLO3- To evaluate the data traffic in an Internet connected networks.	PLO5, PLO8
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Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	Introduction: Basic concept of information; Overview on data communication, Entropy, Information rate, Channel and channel capacity, Transmission modes, Transmission impairments, Guided and unguided media, Fiber optics communication, Satellite communications, frequency bands and characteristics.	Lecture Assignment	Short Answer Exercise
CLO2, CLO3	Data Encoding & Multiplexing: Sampling principal; Nyquist sampling rate; PAM, PWM, PPM, PCM, DPCM; Delta modulation; A-law & μ -law commanding; ASK, FSK, PSK & QPSK; NRZ, Bipolar AML, Manchester, B8ZS, HDB3 coding, constellation, bit error rate(BER); Noise; Echo cancellation; Inter-symbol Interference; Concepts of channel coding and capacity; Concepts of multiplexing, FDM, TDM.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO3	Data Link Control: Flow control; Error detection; Error control; HDLC; Other data link control protocols.	Lecture Assignment	Short Answer Discussion
CLO1	Circuit Switching & Packet Switching: Switching network; Circuit switching network; Circuit switching concepts; Routing in CS; Control signaling; Packet switching principles; Routing in PS; X.25.	Lecture Assignment	Short Answer Presentation
CLO1, CLO2	ATM and Frame Relay: Protocol architecture; ATM logical connection; ATM cells; Transmission of ATM cells; ATM service categories; ATM adaptation layer; Frame relay.	Lecture Assignment	Short Answer Assignment MCQ
CLO1	ISDN: Overview; ISDN channels; ISDN protocols; Broadband ISDN.	Lecture Assignment Project	Short Answer Group Exercise MCQ

CLO3	Error Control Codes: Overview, channel codes, linear block codes, cyclic codes, MLDC codes, convolution codes, Trellis code modulation.	Lecture Assignment Project	Short Answer Group Exercise MCQ
CLO1, CLO2	Satellite Communication: frequency bands and characteristics, types of satellites, transmission impairments, capacity allocation; Multiple access techniques.	Lecture Assignment	Short Answer Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√		√	√
CLO3	√	√		√	√

Suggested Texts:

- Data Communications and networking, 5th Edition by Behrouz A. Forouzan.
- Data and Computer Communication, 10th Edition by William Stallings.
- Telecommunication System Engineering: Analog and Digital Network Design, 4th Edition by Roger L. Freeman.
- Analog and Digital Communications, 3rd Edition by Hwei P Hsu.
- Communication Systems, 5th Edition by S. Haykin, M. Moher.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic of data communications.
 - Understand the reference models for data communications, their layers and corresponding functions, services and protocols.
 - Understand the protocol techniques, local area networks and how Internet is built.
 - Understanding error detection and correction techniques.
 - Understanding different communication techniques.
- Cognitive skills (thinking and analysis)
 - Be able to apply the skills within data communications and computer networks.

- ▮ Be able to apply data communication theory in practice.
- ▮ Be able to apply different coding techniques in practice.
- Communication skills (personal and academic)
 - ▮ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▮ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.

ICT 2102: Data Structures Lab

Course Objectives:	The objective of this course is to • Develop skills to design and analyze simple linear and nonlinear data structures. • Strengthen the ability to the students to identify and apply the suitable data structure for the given real world problem.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To design and apply appropriate data structures to solve real world problems.	PL02, PL03
	CLO2- To use programming tools to write and debug codes for abstract data types.	PL05, PL06

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	Introduction to Programming Principle, ADT and class.	Lecture Discussion	Short Answer Exercise
CLO1, CLO2	Lab practices on Selection Sort, Insertion Sort, Bubble Sort, Quick Sort, and Merge Sort.	Lecture Discussion Problem Based Learning (PBL)	Short Answer Exercise Practical Exam

CLO1, CLO2	Working with Sequential Search, Binary Search.	Lecture Problem Based Learning (PBL)	Exercise Practical Exam
CLO1, CLO2	Working with Pointer, Linked List, Create Linked List, Linked List Declaration, Add node, Delete Node, Find Node, and Circular Linked list.	Lecture Group Assignment Demonstration	Exercise Report Presentation
CLO1, CLO2	Lab practices on Stack, Stack Operations–pop and push, Stack applications.	Lecture Project Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam
CLO1, CLO2	Lab practices on Queue, Queue Implementation (Array and link list), Queue operations, and Circular queue.	Lecture, Project Problem Based Learning (PBL) Demonstration	Report, Exercise Presentation Practical Exam
CLO1, CLO2	Working with tree, Binary search tree, Tree Operation – create node, add node, delete node, print node, search node, Tree Traversal (pre-order, in-order and post-order).	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam
CLO1, CLO2	Lab practice on creating heap, maintaining the heap property, Building a heap, and heap sort algorithm	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

CLO1, CLO2	Working with Breadth first search (BFS), Depth first search Graphs (DFS), Minimum Spanning Tree, Dijkstra's Algorithm.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam
CLO1, CLO2	Lab practices on Hash function, Hash table, Collision resolution techniques, Implementation of hash table with linear probing, Implementation of hash table with quadratic probing, Implementation of hash table with double hashing, Applications.	Lecture Project Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√	√	√
CLO2	√	√	√	√	√

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of data structures and algorithms.
 - Understand the fundamental steps of designing algorithms.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any problem and derive its solution.
 - Be able to design and develop algorithms using various data structures.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to identify the complex and real life problems to solve using the effective understanding of data structure and algorithms.

ICT 2104: Digital Logic Design Lab

Course Objectives:	The objective of this course is to • Get familiar with basic logic gates-AND, OR, NOT, XOR, XNOR. • Enable students to design combinational logic circuits. • Enable students to design sequential logic circuits. • Enable students to understand the operation of various digital circuits.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To introduce the students the wave shaping circuits, Switching Characteristics of diode and transistor.	PLO1
	CLO2- To analyze different types of Multi vibrators and their design procedures.	PLO2, PLO5
	CLO3- To understand Sampling Gates and to Design NAND and NOR gates using various logic families.	PLO1, PLO3
	CLO4- To design digital circuits using Boolean Algebraic and state transition techniques.	PLO3, PLO4

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO3, CLO4	Verify the truth table of basic logic gates: AND, OR, NOT, NAND, NOR. Also truth table of Ex-OR, Ex-NOR, truth table of OR, AND, NOT, realized using NAND & NOR gates, SOP and POS expression.	Lecture Assignment Hands on Experiments	Short Answer Exercise Experiment

CLO3, CLO4	Half adder / subtractor & Full adder / subtractor using NAND and NOR gates and to verify their truth tables, 4-bit ripple adder / subtractor using basic Half adder / subtractor and Full Adder / Subtractor.	Lecture Assignment	MCQ Presentation Experiment
CLO3, CLO4	Truth table of 4-to-1 multiplexer and 1-to-4 demultiplexer. Realize the multiplexer using basic gates only. Also, construct 8-to-1 multiplexer and 1-to-8 demultiplexer using blocks of 4-to-1 multiplexer and 1-to-4 demultiplexer.	Lecture Assignment	Short Answer Discussion
CLO3	Binary to gray and gray to binary converter. Also, verify the truth table for all possible combinations, R-S, J-K and D Flip-flops with and without clock signal and verify their truth table.	Lecture Assignment	Short Answer Presentation
CLO3	Divide by 2, 4 & 8 asynchronous counter. Construct a 4-bit binary and ring counter for a particular output pattern using D Flip-flop, parallel in/ parallel out and serial in/ serial out registers using clock.	Lecture Assignment	Short Answer Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√				√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√

Intended Learning Outcomes:

- Knowledge and understanding
 - ▮ Demonstrate practical skills in the programming and testing of digital systems on FPGA and microcontroller development boards.
 - ▮ Understand the time and frequency domain aspects.
 - ▮ Understand how to analyze, build and troubleshoot digital circuits.
- Cognitive skills (thinking and analysis)

- Be able to identify, formulate, and solve engineering problems.
- Be able analyze and synthesize combinational logic circuits.
- Be able to apply Boolean algebra to solve logic functions.
- Be able to interpret schematics to determine principal circuit functions.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to apply digital electronic circuit analysis skill to build efficient and constructive devices.
 - Be able to apply logic design circuits with Programmable Logic Devices.

ICT 2106: Internet and Web Technology Lab

Course Objectives:	The objective of this course is to ● Develop an ability to design and implement static and dynamic website. ● Acquire knowledge and skills for creation of web site considering both client and server side programming. ● Explore different web extensions and web services standards.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand internetworking mechanism and web publishing.	PLO1
	CLO2- To analyze real time scenario to solve through web technology.	PLO2, PLO3
	CLO3- To develop website with web programming languages.	PLO5, PLO10
	CLO4- To design complex application.	PLO3, PLO6

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
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CLO1, CLO2	Introductions Syllabus Review Tools for developing client-side web applications Accessing Web Folders JavaScript – Introduction, Data Operations, Basic Input and Output, Document Objects, Browser Objects.	Problem Based Learning (PBL)	Exercise
CLO3	HTML basics	Problem Based Learning (PBL)	Exercise
CLO3	Style sheets	Problem Based Learning (PBL)	Exercise
CLO3	Dynamic page using DOM, CSS, and JS	Problem Based Learning (PBL)	Exercise
CLO3, CLO4	Form processing at the client side	Problem Based Learning (PBL)	Exercise
CLO3, CLO4	HTML and DHTML Common tags- Block Level and Inline Elements, Lists, Tables, Images, Forms, Frames; Cascading Style sheets, CSS Properties; Java Script: Introduction to Java Script, Objects in Java Script, Dynamic HTML with Java Script.	Problem Based Learning (PBL)	Exercise
CLO3, CLO4	JDBC: Data Base, Database Schema, A Brief Overview of The JDBC Process, JDBC Driver Types, JDBC Packages, Database Connection, Associating The JDBC-ODBC Bridge With Database, Creating, Inserting, Updating And Deleting Data In Database Tables, Result Set, Metadata.	Problem Based Learning (PBL)	Exercise
CLO3	Web server, server side processing, PHP basics.	Problem Based Learning (PBL)	Exercise
CLO3	Introduction to PHP: Basics of PHP, Functions, Error Handling, Interaction between PHP and MySQL, Database using Forms, Using PHP to manipulate and Retrieve Data in MySQL.	Problem Based Learning (PBL)	Exercise
CLO4	Web based real time Problem Solving project.		Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√				√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√

Intended Learning Outcomes:

1. Explain different components and technologies of World Wide Web as a platform.
2. Design and develop websites using fundamental web languages, technologies and tools.
3. Distinguish between server-side and client-side web technologies.
4. Describe various web technology and application development issues and trends.
5. Conduct independent research on a subject related to the course material.

ICT 2100: Course Viva**Course Content:**

The content of the viva includes the syllabus of all major courses.

2nd Year 2nd Semester**ICT 2201: Algorithm Analysis and Design**

Course Objectives:	The objective of this course is to • Analyze the asymptotic performance of algorithms. • Write rigorous correctness proofs for algorithms. • Demonstrate a familiarity with major algorithms and data structures. • Apply important algorithmic design paradigms and methods of analysis. • Synthesize efficient algorithms in common engineering design situations.
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Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the different types of computational complexities for optimizing the algorithms.	PLO1
	CLO2- To design and analyze different types of algorithms for solving the various complex problems.	PLO2, PLO3
	CLO3- To analyze and select the most suitable and effective algorithm for solving certain real life problem.	PLO4, PLO7

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO2	Introduction to Algorithm and Mathematical Induction: Introductory concepts and definitions related to algorithm formula, notation for describing algorithms, introduction to mathematical induction, examples, counting regions in the plane, simple coloring problem, simple inequality, Euler's formula, Finding edge-disjoint paths in a graph, Arithmetic versus geometric means, loop invariant etc.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1, CLO3	Analysis of Algorithms Complexity and Design of Algorithm by Induction: The O notation, time and space complexity, recurrence relations: intelligent guesses divide and conquer relations, recurrence relation with full history, design by induction: evaluation polynomials, maximal induced subgraph. Finding one-to-one mappings, the Celebrity problem, The skyline problem, finding the maximum consecutive subsequence, strengthening the induction hypothesis, dynamic programming: the Knapsack problem, etc.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
	Algorithms Involving Trees: Find the minimum and maximum value of a BST, Find successor and predecessor of a BST, Insert a data in a BST, Delete a data from a BST, Calculate the time complexity of various operation on a BST, Insert into and delete from AVL tree, Searching, insertion and deletion in a B-tree, Insertion and deletion in a red-black tree.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation

CLO2, CLO3	Algorithm Involving Sequences and Sets: Introduction, Binary search and variations, Interpolation search, sorting: Bucket sort, Radix sort, Insertion sort, Selection sort, Merge sort, Quick sort, Heap sort, order statistics, data compression, string matching, sequence comparisons, probabilistic algorithms, finding a majority, etc.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2, CLO3	Methods for the design of efficient algorithms: Divide and conquer, greedy method, dynamic programming, back tracking, branch and bound;	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation
CLO1, CLO2	Graph and Geometric Algorithms: Introduction, Eulerian graphs, graph traversals: Depth-first Search, Breadth-first Search, Topological Sorting, minimum-cost Spanning trees, network flows, Hamiltonian tours, decompositions of graphs, construction polynomials, convex hulls, closest pair, intersection of horizontal and vertical line segments, etc.	Lecture Discussion Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO2	Reductions and NP-Completeness: Introduction, examples of reductions, reductions involving linear programming reductions for lower bounds, polynomial time reductions, nondeterminism and Cook's Theorem, examples of NP-completeness Proofs, techniques for dealing with NP-complete problems, etc.	Lecture Group Assignment Demonstration	Essay Type Answer Report Exercise Presentation
CLO2, CLO3	Parallel Algorithms: Introduction, models of parallel computation, algorithms for shared-memory machines, algorithms for interconnected networks, systolic computation, etc.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√		√	√

CLO2	√	√		√	√
CLO3	√	√	√	√	√

Suggested Texts:

- Introduction to Algorithms, 3rd Edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest.
- Fundamentals of Computer Algorithms, 2nd Edition by Ellis Horowitz and Sartaj Sahni.
- Algorithms, 3rd Edition by Robert Sedgewick.
- Introduction to Algorithm, 1st Edition by Udi Manber.
- Introduction to the Design and Analysis of Algorithms, 3rd Edition by Anany V. Levitin.
- Algorithms + Data Structures= Programs, 1st Edition by Niklaus Wirth.
- Data Structures & Algorithms in C++, 4th Edition by Adam Drozdek.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of algorithm analysis and design.
 - Understand the fundamental steps of designing and optimizing algorithms.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any problem and derive its solution.
 - Be able to design and optimize algorithms to solve particular complex problems effectively.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to identify the complex and real life problems to solve using the effective understanding of the design and analysis of algorithms.

ICT 2203: Database Management Systems

Course Objectives:	The objective of this course is to • Understand the role of a database management system in an organization. • Understand basic database concepts, including the structure and operation of the relational data model. • Construct simple and moderately advanced database queries using Structured Query Language (SQL). • Understand and successfully apply logical database design principles, including E-R diagrams and database normalization. • Design and implement a small database project using Microsoft Access. • Understand the concept of a database transaction and related database facilities, including concurrency control, journaling, backup and recovery and data object locking and protocols. • Describe and discuss selected advanced database topics, such as distributed database systems and the data warehouse. • Understand the role of the database administrator.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To differentiate database systems from file systems by enumerating the features provided by database systems and describe each in both function and benefit.	PL01, PL02
	CLO2- To define the terminology, features, classifications, and characteristics embodied in database systems.	PL01, PL02
	CLO3- To analyze an information storage problem and derive an information model expressed in the form of an entity relation diagram.	PL02, PL04
	CLO4- To demonstrate an understanding of the relational data model.	PL02, PL05
	CLO5- To transform an information model into a relational database schema and to use a data definition language and/or utilities to implement the schema using a DBMS.	PL02,
	CLO6- To formulate , using relational algebra, solutions to a broad range of query problems.	PL03
	CLO7- To formulate , using SQL, solutions to a broad range of query and data update problems.	PL03

	CLO8- To demonstrate an understanding of normalization theory and apply such knowledge to the normalization of a database.	PLO3, PLO4
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Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2, CLO4, CLO5	Introduction: Data, Information, File Processing Concept, fundamental concept of Database Management System (DBMS), purpose of DBMS, advantages and implications of DBMS, view of data. Relational Databases, Database Architecture, Transaction Management.	Lecture Discussion	Essay Type Answer MCQ
CLO4	Data Models: The evolution of data models, importance of data models, degrees of data abstraction, business rules.	Lecture Discussion	Essay Type Answer Exercise
CLO3, CLO4	Data modeling using Entity Relationship Model: Overview of database design and ER model, Entity Types, Entity Sets, Attributes, Keys, Relationships, relationship Types, Roles, Structural Constraints.	Lecture Discussion Assignment	Exercise Report
CLO4, CLO5	Data modeling using Entity Relationship Model: Direct ERD Issues, weak entity sets, Codd's rules, Relational Database design: features of good relational database design, refining the ER Design for different databases, Naming Conventions, Extended ER features: Aggregation, Specialization and Generalization.	Lecture Discussion	Essay Type Answer Exercise
CLO4, CLO5	Relational Model: Relational Model Concepts, Structure of relational database, Relational Database Design Using ER-to-Relational Mapping, Relational Constraints and Relational Database Schemas.	Lecture Assignment Demonstration	Exercise Report
CLO5, CLO8	Relational database model: Decomposition, Functional Dependency, Data Anomalies, Basics Concept of Normalization, Normalization (1NF, 2NF, 3NF, BCNF).	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Exercise Report

CLO6	Relational algebra and calculus: Basics relational algebra operations: introduction, Selection and projection, Set operations, Renaming, Joins, syntax, semantics, Operators, grouping and ungrouping, relational Comparison, Calculus: Tuple relational calculus, Domain relational Calculus, calculus vs algebra, computational capabilities.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Exercise Report
CLO7	Constraints, Views and SQL: What is constraints, types of constraints, Integrity constraints, Views: Introduction to views, data independence, security, updates on views, comparison between tables and views. SQL: Data manipulation languages, Null values and set operations, Aggregate functions, Nested sub queries, derived relations, SQL Joins, Views, Query optimization, Assertion and Trigger, Procedures and Functions.	Lecture Assignment Problem Based Learning (PBL) Demonstration	Exercise Report
CLO1, CLO3	File organization and data storage: Indexing: primary and secondary indexes, B+ trees, hash tables.	Lecture Demonstration Assignment	Essay Type Answer Report
CLO1	Transaction management and Concurrency control: ACID properties, serializability and concurrency control, Lock based concurrency control, Deadlock, Wait for Graph, Recovery, Access control and security.	Lecture Demonstration	Essay Type Answer Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√		√
CLO2	√	√	√		√
CLO3	√	√	√		√
CLO4	√	√	√		√
CLO5	√	√	√		√
CLO6	√	√	√		√
CLO7	√	√	√	√	√
CLO8	√	√	√		√

Suggested Texts:

- Database Systems Concepts, 7th Edition by A. Silberschatz, H. Korth and S. Sudarshan, Publisher: McGraw Hill.
- Fundamentals of Database Systems, 7th Edition by Ramez Elmasri.
- An Introduction to Database Systems, 7th Edition by C. J. Date.
- An Introduction to Database Systems, 3rd Edition by R. Ramakrishnan and J. Gehrke.

Intended Learning Outcomes:

- Knowledge and understanding
 - ▢ Students will have a broad understanding of database concepts and database management system.
 - ▢ Students will be able to model an application's data requirements using conceptual modeling tools like ER diagrams and design database schemas based on the conceptual model.
- Cognitive skills (thinking and analysis)
 - ▢ Students will be able to analyze problems, identify and define the computing requirements appropriate to its solution.
 - ▢ Students will be able to conceptualize data using different data models.
- Communication skills (personal and academic)
 - Through different individual and group assignments provided to the students, they would be able to understand the technical requirements of a particular informational problem.
- Practical and subject specific skills (transferable skills)
 - Be able to identify the simple and complex database related problems to solve using the effective understanding of the database management systems.

ICT 2205: Analog and Digital Communication

Course Objectives:	The objective of this course is to • Introduce the concepts of analog communication systems and to equip students with various issues related to analog communication, such as radio wave propagation, modulation, demodulation, broadcasting transmitters & receivers. • Provide the fundamental knowledge about the building blocks of digital communication system. • Understanding and analyzing the signal flow in a digital communication system. • Analyze error performance of a digital communication system over various modulation techniques and waveform coding techniques in presence of noise and other interferences will also be covered to good extend.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To explain various concepts and principles of radio wave propagation, (de)-modulation and (de/en) coding.	PL02, PL06
	CLO2- To apply statistical model in the analog and digital communication system.	PL01
	CLO3- To explain the components of analog and digital communication system.	PL01
	CLO4- To determine critically and to use modulation and line coding in various system.	PL02, PL03
	CLO5- To evaluate the performance of the communication systems.	PL03, PL05

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Radio Wave Propagation: Surface and space wave propagation, Sky wave through Ionosphere. Pulse method for measuring height and electron concentration of Ionospheric region; Chapman theory of layer formation, Ionospheric storm.	Lecture Assignment	Short Answer Exercise

CLO1	Modulation and Demodulation: Linear modulation - AM, SSB, DSB, and SSB generation, PLL Circuit to generate linear modulated signals, low and high power modulators, Exponential modulation- FM and PM, demodulation of AM, FM. Broadcasting Transmitter: Transmitter classification, Elements of transmitter, AM and FM transmitters, SSB transmitter, stabilized master oscillator, Frequency multipliers, Mixer circuits, RF power amplifier, Pre-emphasis circuits, Transmitter performance-carrier frequency requirements, audio frequency response, distortion and signal to distortion ratio.	Lecture Assignment	Short Answer Exercise
CLO4	Radio Receiver: Receiver classification, Elements of receiver, AM and FM receivers, SSB receiver, Comparison of AM and FM receivers, Noise in receiver, AGC circuits, AFC circuits, Noise limiters, Receiver sensitivity, Cross modulation, Spurious responses.	Lecture Assignment	Short Answer Presentation
CLO2, CLO3, CLO5	Representation of Random Signals and Noise in Communication System: Signal Power and Spectral Representations, White noise, Thermal noise, PSDF of White Signals. Input and Output Relationship for Random Signals and Noise Passed Through a Linear Time Invariant System, Band Limited White Noise, ARC Filtering of White Noise. Noise performance of Analog Communication Systems: Signal-to-Noise Ratio in Linear Modulation, Synchronous Detection of DSB. Signal-to-Noise Ratio for AM and SSB, FM, Effect of Noise in Envelope and Square Law Detection of AM, Threshold Effects in Nonlinear Detectors.	Lecture Assignment	Short Answer Presentation
CLO2, CLO3, CLO5	Elements of Digital Communication Systems: Model of Digital Communication Systems, Digital Representation of Analog Signal, Certain issues in Digital Transmission, Advantages of Digital Communication Systems, Bandwidth-S/N tradeoff, Hartley Shannon Law.	Lecture Assignment	Short Answer Presentation
CLO2, CLO3, CLO5	Pulse Code Modulation: PCM Generation and Reconstruction, Quantization noise, Non uniform Quantization and Companding, DPCM, Adaptive DPCM, DM and Adaptive DM. Noise in PCM and DM. Line Coding: Techniques and Analysis	Lecture Assignment Problem Based Learning (PBL)	Short Answer Exercise

CLO2, CLO3, CLO5	Digital Modulation Techniques: Digital Modulation Techniques: Introduction, ASK, ASK Modulator, Coherent ASK Detector, Non-Coherent ASK Detector, FSK, Bandwidth and Frequency Spectrum of FSK. Non coherent FSK Detector, Coherent FSK Detector, FSK Detection Using PLL, BPSK, Coherent PSK Detection, QPSK, Differential PSK.	Lecture Assignment Problem Based Learning (PBL)	Short Answer Group Discussion Exercise
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Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√		√
CLO2	√	√	√		√
CLO3	√	√	√		√
CLO4	√	√	√		√
CLO5	√	√	√	√	√

Suggested Texts:

- Electronic communication systems, 4th Edition by George Kennedy, Barnard Davis.
- Principles of communication systems, 3rd Edition by Taub and Schilling.
- Analog and Digital Communication systems, 4th Edition by Martin S Roden.
- Analog & Digital Communication, 4th Edition by B P Lathi.
- Communication Systems Engineering, 2nd Edition by Proakis & Salehi.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the fundamental concept of wave propagation, basic communication system and its various components.
 - Understand the performance metric of communication systems.
- Cognitive skills (thinking and analysis)
 - Be able to find the performance of various modulation schemes.
 - Be able to apply various modulation and encoding schemes in real-life applications.
 - Be able to identify different types of radio transmission, its techniques and applications.

- ▢ Be able to formulate the power spectra, bandwidth efficiency, spread spectrum modulation.
- ▢ Be able to design and develop digital communication system through simulation.
- Communication skills (personal and academic)
 - ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, giving, and receiving clear instructions.
 - ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - ▢ Be able to use incorporating digital modulation and coding schemes in communication systems.

ICT 2207: Matrices, Vector, Fourier analysis and Laplace Transforms

Course Objectives:	The objective of this course is to • Construct, or give examples of, mathematical expressions that involve vectors, matrices, and linear systems of linear equations. • Evaluate mathematical expressions to compute quantities that deal with linear systems and eigenvalue problems. • Analyze mathematical statements and expressions (for example, to assess whether a particular statement is accurate, or to describe solutions of systems in terms of existence and uniqueness). • Write logical progressions of precise mathematical statements to justify and communicate your reasoning. • Apply linear algebra concepts to model, solve, and analyze real-world situations. • Construct and interpret linear transformations in $\mathbb{R}^2$ or $\mathbb{R}^3$ (for example, interpret a linear transform as a projection or as a shear).	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To identify a system of linear equations (or linear system) and describe its solution set. Write down the coefficient matrix and augmented matrix of a linear system.	PLO1

	CLO2- To use elementary row operations to reduce matrices to echelon forms. Make use of echelon forms in finding the solution sets of linear systems. Perform standard operations with vectors in Euclidean space. Understand the meaning of linear independence/dependence and span. Interpret linear systems as vector equations.	PLO2, PLO3
	CLO3- To define matrix-vector product and be able to interpret linear systems as matrix equations.	PLO1, PLO3
	CLO4- To determine the parametric vector form of solutions of linear systems.	PLO3, PLO4
	CLO5- To relate the solution set of a consistent inhomogeneous linear system to the solution set of its associated homogeneous equation.	PLO6

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2, CLO3, CLO5	Theory of Matrices: Concepts of matrices, types of matrices, transposes, determinant, Sarrus' rule, Laplace expansion, inverse matrix, elementary row operations of matrices (echelon form and row reduced echelon form of matrices), LU-Decomposition and rank. Systems of linear equations and its solution by Cramer's Rule, Gaussian elimination method and inverse matrix method, Kronecker-Capelli theorem. Diagonalization of matrices: Introduction, eigenvalues and eigenvectors, diagonalization of matrices and the Cayley-Hamilton theorem.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation

CLO2, CLO3	Vectors and Scalars, Algebra of vectors, Scalar triple product, Vector triple product, Vector differentiation and vector integration, Gradient, Divergence and Curl, Cartesian, Spherical, Polar and Cylindrical system.	Lecture Discussion Group Assignment Demonstration	Report Exercise Presentation
CLO1, CLO2, CLO3, CLO4, CLO5	Laplace Transform: Introduction, Laplace transform of elementary function, Properties of Laplace Transforms, Inverse Laplace transforms, and their properties, Convolution theorem, Heaviside's expansion formula and their Applications.	Lecture Discussion Group Assignment Demonstration	Report Exercise Presentation
CLO1, CLO2, CLO3, CLO4, CLO5	Fourier analysis: Real and complex form of Fourier series, Fourier Integral, Fourier transforms and their uses in solving boundary value problems.	Lecture Discussion Group Assignment Demonstration	Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√		√	√
CLO2	√	√		√	√
CLO3	√	√		√	√
CLO4	√	√		√	√
CLO5	√	√		√	√

Suggested Texts:

- Elementary Linear Algebra, Author: Howard Anton, Chris Rorres, 11th Edition by John Wiley & Sons, Inc.

- Elementary Linear Algebra. Author: Ron Larson, David C. Falvo, 6th Edition by Houghton Mifflin Harcourt Publishing Company, Boston, New York.
- Linear Algebra and Its Applications, 4th Edition by David C. Lay.
- Vector Analysis, 2nd Edition by Murray R Spiegel (Schaum's Outlines Series).
- A Text Book on Co-ordinate Geometry with Vector Analysis, 1st Edition by Rahman & Bhattacharjee.
- Fourier analysis with application to boundary value problems, 1st Edition by Murray R Spiegel (Schaum's Outlines Series).
- Laplace Transforms, 1st Edition by Murray R Spiegel (Schaum's Outlines Series).

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of matrices, vectors, Fourier analysis and Laplace transform.
 - Have preparation in successful continuation of their studies towards more advanced and specialized courses in their field.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis the different mathematical problems related to the matrix and vectors.
 - Be able to solve the Initial and Boundary value problems that arise frequently in any field of Engineering.
 - Be able to decompose a function into its constituent frequencies.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to understand and analysis the important theorem and ideas of matrices, vectors, Fourier analysis and Laplace transform to solve the complex problems, particularly related to the computer science and engineering fields.

ICT 2209: Financial and Managerial Accounting

Course Objectives:	The objective of this course is to • Prepare financial reports that provide information about a firm's performance to external parties such as investors, creditors and tax authorities. • Improve the effectiveness of both the management planning and control functions. • Provide the information that is needed for sound economic decision making in the perspective of accounting.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the importance of accounting and accounting as an information systems.	PL01
	CLO2- To learn how to prepare journal, ledger and trial balance.	PL04
	CLO3- To learn preparation of basic financial statements considering adjusting and closing entries.	PL04
	CLO4- To apply cost concepts and apply different costing techniques in decision making.	PL06, PL07
	CLO5- To analyze short term and long term investment prospects and making relevant decisions.	PL08, PL011

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Financial Accounting: objectives and importance of accounting; Accounting as an information system; computerized system and applications in Accounting; role of accountancy profession.	Lecture Assignment	Short Answer Exercise MCQ
CLO2, CLO3	Recording System: double entry mechanism; accounts and their classification; Accounting equation; Accounting cycle: journal, ledger & trial balance.	Lecture Assignment	Short Answer Exercise
CLO4	Preparation of Financial Statements: importance and limitations of statement of Comprehensive Income; preparation of statement of Comprehensive Income; usefulness and limitations of statement of Financial Position; preparation of statement of Financial Position; importance of statement of change in Equity; preparation of statement of change in Equity.	Lecture Assignment	Short Answer Case Study Discussion

CLO4	Financial Statement Analysis and Interpretation: different techniques of financial statement analysis; ratio analysis & interpretation, sensitivity analysis; cash flow analysis.	Lecture Assignment Presentation	Exercise Case Study Discussion
CLO4	Management Accounting: definition; Management Accounting; scope and functions of Management Accounting; evolution of Management Accounting; Financial Accounting Vs. Management Accounting; Cost Accounting Vs. Management Accounting; standard of ethical conduct for Management Accountants.	Lecture Assignment	Short Answer Exercise MCQ
CLO4	Cost Classification and Cost Behavior Analysis: cost concepts and classification; methods of segregating costs into fixed and variable costs (High-Low Method; Inspection of Accounts/Accounts Classification Method; Engineering Method; Ordinary Least Square Method); overhead costs; meaning and classifications of overhead costs; overhead recovery method/rate; Job order costing: preparation of job cost sheet and quotation price.	Lecture Assignment	Short Answer Assignment
CLO4	Inventory Valuation: absorption costing and marginal/variable costing technique; Cost-Volume-Profit (CVP) analysis; usefulness and limitations of CVP analysis; breakeven analysis; computations of breakeven point and shutdown point.	Lecture Assignment	Short Answer Exercise Case Study
CLO5	Investment Decisions: the concept of relevant cost; using relevant cost information in decision making; limiting factor Analysis; addition or deletion of products or departments; special sales orders; sell or process further decision; capital budgeting; various techniques of evaluation of capital investments.	Lecture Assignment Presentation	Exercise Case Study Discussion

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√		√	√

CLO2	√	√		√	√
CLO3	√	√		√	√
CLO4	√	√		√	√
CLO5	√	√		√	√

Suggested Texts:

- Intermediate accounting. Hoboken, 3rd Edition by Kieso, D. E., Weygandt, J. J., & Warfield, T. D., Publisher: John Wiley & Sons.
- Introduction to Management Accounting, 13th Edition by C. T. Horngren, G. Sundem and W. O. Stratton.
- Financial Accounting study manual, Professional Stage- Knowledge level and Application level, Publisher: ICAB.
- Management Accounting: Information for Decision-Making and Strategy Execution, 6th Edition by Atkinson, A.A., Kaplan, R.S., Matsumura, E.M. and Young, S.M.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the concept of Financial Accounting and Management Accounting.
 - Understand the use of financial statements for investment decisions.

Cognitive skills (thinking and analysis)

- Be able to explain basic concepts of Financial Accounting and Management Accounting.
- Be able to record economic transactions and prepare basic financial statements for the interested users.
- Be able to use different financial statement analysis techniques and interpret the results.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to use different financial statement analysis techniques to make proper short term and long term investment decisions.

ICT 2202: Algorithm Analysis Design and Lab

Course Objectives:	The objective of this course is to Learn how to analyze a problem & design the solution for the problem. In addition to that, solution must be optimum, i.e., time complexity & memory usage of the solution must be very low.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the different types of computational complexities for optimizing the algorithms.	PL01
	CLO2- To design and analyze different types of algorithms for solving the various complex problems.	PL02, PL03, PL05
	CLO3- To analyze and select the most suitable and effective algorithm for solving certain real life problem.	PL04, PL011

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Working with time and space complexities.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Short Answer Exercise Practical Exam Presentation
CLO2	Lab practices on Selection Sort, Insertion Sort, Bubble Sort, Sequential Search, Binary Search.	Lecture Discussion Problem Based Learning (PBL)	Short Answer Exercise Practical Exam
CLO1, CLO3	Working with Greedy algorithms: Activity Selection Problem, Job Sequencing Problem, Kruskal's and Prim's Minimum Spanning Tree, Dijkstra's Shortest Path Algorithm, Max Flow Problem, Fractional Knapsack Problem.	Lecture Group Assignment Problem Based Learning (PBL)	Exercise Report Practical Exam

CLO2, CLO3	Lab Practices on Divide and Conquer Algorithms: Binary Search, Quicksort, Merge Sort, Strassen's Algorithm, Cooley–Tukey Fast Fourier Transform (FFT) algorithm.	Lecture Problem Based Learning (PBL) Demonstration	Exercise Practical Exam Presentation
CLO2	Lab practice on building Binary Search Tree (BST), Find the minimum and maximum value of a BST, Find successor and predecessor of a BST, Insert a data in a BST, Delete a data from a BST, Insert into and delete from AVL tree, Searching, insertion and deletion in a B-tree, Insertion and deletion in a red-black tree.	Lecture Problem Based Learning (PBL) Demonstration	Exercise Practical Exam Presentation
CLO2, CLO3	Lab practices on Dynamic Programming: Longest Common Subsequence, Travelling Salesman Problem, Bellman–Ford Algorithm, Floyd Warshall Algorithm, Matrix Chain Multiplication.	Lecture Project Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam
CLO2, CLO3	Working with Backtracking: 8 Queen Problem, Combinational Sum, Word Break Problem, m Coloring Problem.	Lecture Project Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam
CLO2, CLO3	Working with Branch and Bound algorithms: Job Assignment Problem, 0/1 Knapsack, 8 puzzle Problem, N Queen Problem.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam
CLO2, CLO3	Lab practices on NP-Complete and NP-Hard Problems: Hamilton Path, Hamilton Cycle, Graph homomorphism problem, Minimum k-cut, Bandwidth problem	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation Practical Exam

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√	√	√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of the analysis and design of algorithms.
 - Understand the fundamental steps of designing and optimizing algorithms.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any problem and derive its solution.
 - Be able to design and optimize algorithms to solve complex and real life problems.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to identify the complex and real life problems to solve using effective understanding of the design and analysis of algorithms.

Course Objectives:	The objective of this course is to • Provide a sound introduction to the discipline of database management as a subject in its own right, rather than as a compendium of techniques and product-specific tools. • Familiarize the participant with the nuances of database environments towards an information-oriented data-processing oriented framework. • Give a good formal foundation on the relational model of data. • Present SQL and procedural interfaces to SQL comprehensively. • Give an introduction to systematic database design approaches covering conceptual design, logical design and an overview of physical design. • Motivate the participants to relate all these to one or more commercial product environments as they relate to the developer tasks. • Present the concepts and techniques relating to query processing by SQL engines. • Present the concepts and techniques relating to ODBC and its implementations.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the formal foundation of the relational model of data.	PLO1
	CLO2- To execute SQL and procedural interfaces to SQL comprehensively.	PLO5
	CLO3- To formulate systematic database design approaches covering conceptual design, logical design and an overview of physical design.	PLO2, PLO5
	CLO4- To present the concepts and techniques relating to query processing by SQL engines.	PLO3
	CLO5- To apply normalization techniques to normalize the database.	PLO6
	CLO6- To apply the concept of indexing, transaction and query processing to develop the database applications.	PLO6, PLO10

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
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CLO2, CLO3, CLO5	Creation of a database and writing SQL queries to retrieve information from the database.	Lecture Discussion Problem Based Learning (PBL) Assignment	Report Exercise Practical Exam
CLO1, CLO2	Apply the constraints like Primary Key, Foreign key, NOT NULL to the tables.	Lecture Discussion Problem Based Learning (PBL)	Report Exercise Practical Exam
CLO2, CLO4	Performing Insertion, Deletion, Modifying, Altering, Updating and Viewing records based on conditions.	Lecture Discussion Problem Based Learning (PBL) Assignment	Report Exercise Practical Exam
CLO2, CLO4	Nested queries, Sub queries.	Lecture Discussion Problem Based Learning (PBL) Assignment	Report Exercise Practical Exam
CLO2, CLO4	Implementation of JOINS.	Lecture Discussion Problem Based Learning (PBL) Assignment	Report Exercise Practical Exam
CLO4	Application of aggregate functions.	Lecture Discussion Problem Based Learning (PBL) Assignment	Report Exercise Practical Exam

CLO4	Queries using Conversion functions, string functions, date functions.	Lecture Discussion Problem Based Learning (PBL) Assignment	Report Exercise Practical Exam
CLO6	Views, Sequences, Synonyms	Lecture Discussion Problem Based Learning (PBL) Assignment	Report Exercise Practical Exam
CLO1, CLO3, CLO5	Database Design using ER modeling, normalization and Implementation for any application	Lecture Discussion Problem Based Learning (PBL) Group Assignment	Report Exercise Practical Exam
CLO1, CLO2, CLO3, CLO4, CLO5, CLO6	Case Study using real life database applications.	Lecture Project Problem Based Learning (PBL) Demonstration	Report Exercise Practical Exam Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√
CLO5	√	√	√	√	√
CLO6	√	√	√	√	√

Intended Learning Outcomes:

- Knowledge and understanding
 - ▯ Understand, appreciate and effectively explain the underlying concepts of database technologies.
 - ▯ Design and implement a database schema for a given problem-domain.
- Cognitive skills (thinking and analysis)
 - ▯ Students will also be able to write and analyze SQL commands to create tables and indexes, insert/update/delete data and query data in a relational DBMS.
 - ▯ Examine techniques pertaining to Database design practices.
- Communication skills (personal and academic)
 - ▯ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▯ Students will also be assigned the task of implementing a database based project which would refine their idea of design issues preserving data security.
- Practical and subject specific skills (transferable skills)
 - ▯ Students will be able to install, configure and interact with MySQL Relational Database Management System.
 - ▯ Students will also grow the ability to describe, define and apply the major components of the relational database model to database design.

Course Objectives:	The objective of this course is to • Implements different types of analog and digital communication technique with their different types of communication kits. • Train the students to analyze the modulation and demodulation techniques such as amplitude modulation, frequency modulation and understand their performance using both trainer kit and circuit implementation.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To explain various concepts and principles of radio wave propagation, (de)-modulation and (de/en)coding.	PLO2, PLO4
	CLO2- To apply statistical model in the analog and digital communication system.	PLO2, PLO5
	CLO3- To explain the components of analog and digital communication system.	PLO6, PLO9
	CLO4- To formulate the power spectra, bandwidth efficiency, spread spectrum modulation.	PLO3, PLO6
	CLO5- To determine critically and to use modulation and line coding in various system.	PLO3, PLO11

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2, CLO3, CLO5	Familiar with various analog and digital modulation scheme.	Demonstration Case studies Lecture	Practical Exam Report Viva-Voce
CLO1	BER analysis of the various modulation	Inquiry Based Learning (IBL) Lecture Exercise	Practical Exam Report Observation
CLO4	Design an optical fiber communication system	Demonstration Case studies Lecture	Practical Exam Viva-Voce Observation

CLO5	Effect of fading and noise on the signal	Group Assignment Lecture Problem Based Learning (PBL)	Observation Report Viva-Voce
CLO3	Performance of SISO, MISO and MIMO	Inquiry Based Learning (IBL) Lecture Exercise	Practical Exam Viva-Voce

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1					
CLO2	√	√			√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√
CLO5	√	√	√	√	√
CLO6	√	√			√

Intended Learning Outcomes:

- Knowledge and understanding
 - Acknowledge the principles of operation and the main features of analog and digital transmission.
 - Understand the performance metric of communication systems.
- Cognitive skills (Thinking and analysis)
 - Be able to find the performance of various modulation schemes.
 - Be able to apply various modulation and encoding schemes in the real-life applications.
 - Be able to identify the different types of radio transmission, its techniques and applications.
 - Be able to formulate the power spectra, bandwidth efficiency, spread spectrum modulation.
- Communication skills (Personal and academic)

- ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
- ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - ▢ Be able to apply electrical circuit analysis skill to build efficient and constructive devices.

ICT 2200: Project Work II and Course Viva

Course Content:

The content of the viva includes the syllabus of all major courses. Each group of student must have to do a project work based on the courses taught in the 2nd Year's 1st and 2nd Semester. A faculty member or representative from other dept./institute/industry will supervise the project.

3rd Year 1st Semester

ICT 3101: Operating System

Course Objectives:	The objective of this course is to • Understand the basic components of a computer operating system, and the interactions among the various components. • Provide an introduction on the policies for scheduling. • Covers deadlocks, memory management, synchronization, system calls, and file systems. • Describe the knowledge of main memory and virtual memory.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the history of modern computers.	PLO1
	CLO2- To analyze each of the major components of an operating system.	PLO2
	CLO3- To explore more advanced topics in the field, including memory management and file input/output.	PLO4

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Operating System: OS role in computer systems, Operating system concepts, Operating system structure.	Lecture Assignment	Short Answer Discussion
CLO1, CLO2	Process: Process model and implementation, Inter-Process Communication (IPC), classical IPC problems, process scheduling, multiprocessing and time-sharing, CPU management.	Lecture Assignment	Short Answer Discussion Assignment
CLO3	Memory management: Swapping, paging, segmentation, virtual memory.	Lecture Assignment	Short Answer Discussion
CLO3	Input/Output: Hardware, software, disk, terminals, clocks.	Lecture Assignment	Short Answer Discussion
CLO2, CLO3	Deadlock: Resource allocation and deadlock, deadlock detection, prevention and recovery.	Lecture Assignment	Short Answer Assignment MCQ
CLO2, CLO3	File Systems: Files, directories, security, protection, and Case study of some operating systems.	Lecture Assignment	Short Answer Discussion

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1				√	√
CLO2	√				√
CLO3	√	√			√

Suggested Texts:

- Operating System Concepts. 10th Edition by Avi Silberschatz, Peter Baer Galvin and Greg Gagne.
- Operating System: Design and Implementation. 3rd Edition by Andrew S. Tanenbaum.
- Operating system Concepts. 9th Edition by Kernigham & Ritchie.
- Operating system Concept and Design. 2nd Edition by M. Milenkovic.
- Operating system, 1st Edition by Madnick & Donovan.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the goals of standardization of OS (and other) interfaces.
 - Understand the execution of a program as the trace of consecutive machine language instructions residing in the computer memory and how typical programming structures manifest themselves in machine language.
 - Understand the layered structure of input/output (I/O) software and give a broad overview of I/O interrupt handling.
- Cognitive skills (thinking and analysis)
 - Be able to understand the typical (physical) computer memory hierarchy and the compromises involved in using such a hierarchy.
 - Be able identify the most common data structures required in an OS implementation.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.

ICT 3103: Computer Network

Course Objectives:	The objective of this course is to • Emphasize on the organization and management of local area networks (LANs). • Provide detail about computer network organization and implementation, gaining practical experience in installation, monitoring, and troubleshooting of current LAN systems. • Introduce computer communication network design and its operations. • Include Open Systems Interconnection (OSI) communication model; error detection and recovery; local area networks; bridges, routers and gateways; network naming and addressing; and local and remote procedures.
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Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the principles and concepts on computer networks.	PLO1
	CLO2- To analyze soundness or potential flaws in proposed protocols.	PLO2, PLO4
	CLO3- To explain and identify security and ethical issues in computer networking.	PLO7, PLO8, PLO11

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction: Introduction to Computer Networks, Network Goals, Applications of Networks, Network Structure, Network Architecture, The OSI Reference Model, Data Transmission in the OSI Terminology, Connection-Oriented and connectionless Service Primitives, Public Networks, The ARPANET, SNA.	Lecture Assignment	Short Answer Exercise
CLO1	Physical layer: Transmission media, PSTN, Mobile telephone system, ADSL.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO1, CLO2	Data Link Layer: Data link layer design issues, Error detection and correction, Elementary data link protocols, Sliding window protocols, HDLC, The data link layer in the internet.	Lecture Assignment	Short Answer Discussion
CLO1, CLO2	Medium access sub-layer: Multiple access protocols, ALOHA, CSMA/CD Protocol; Collision-free protocols, Wireless LAN protocols, Ethernet, Bluetooth, and Data link layer switching.	Lecture Assignment	Short Answer Presentation
CLO1, CLO2	Network Layer: Network layer design issues, Routing algorithms, Congestion control algorithms, Inter networking, Network layer in the internet, Network layer in ATM networks.	Lecture Assignment	Short Answer Assignment MCQ

CLO1, CLO2	Transport Layer: The transport service, Elements of transport protocols, The internet transport protocols.	Lecture Assignment Project	Short Answer Group Exercise MCQ
CLO1, CLO2	Application Layer: DNS-domain name system, Electronic mail, The world wide web, multimedia.	Lecture Assignment	Short Answer Group Exercise
CLO3	Network Security: Secrecy, Authentication, Digital signature.	Lecture Assignment Project	Short Answer Group Exercise MCQ

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√			√
CLO3	√			√	√

Suggested Texts:

- Computer Networks. 5th Edition by Tanenbaum.
- Computer Networking: A top down approach, 6th Edition by Kurose. Ross.
- Data Communications and Computer Networks, 5th Edition by William Stallings.
- Data Communications and networking, 4th Edition by Behrouz A. Forouzan.
- Computer Network: A System Approach, 4th Edition by Peterson and Davie.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the knowledge on designing and building a complete system.
 - Understand different layer of computer network.
- Cognitive skills (thinking and analysis)
 - Be able to analyze and capture network traffic.

- ▢ Be able to analyze soundness or potential flaws in proposed protocols.
- ▢ Be able to apply the theory of basic network performance analysis.
- Communication skills (personal and academic)
 - ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.

ICT 3105: ICT Business Analytics and Data Visualization

Course Objectives:	The objective of this course is to ● Introduce students to key information technology concepts that will allow them to understand how they provide the necessary data and information for business analytics. ● Enable students to use tools for conducting data analysis, Predictive Analytics and data visualization to assist in the delivery of delivering real-time actionable intelligence. ● Familiarize with the operational & technological best practices of business analytics.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To demonstrate the strategic value of information, data management, and Business Analytics to ICT based organizational decision-making activities.	PLO10
	CLO2- To analyze various datasets using complementary data analytics/business analytics tools to generate insights and/or foresights to approach the organization's complex problems.	PLO2, PLO5, PLO7
	CLO3- To analyze and review in more depth the current algorithms, methodologies and modelling for ICT business analytics based on the outcome of the student group assignment deliverable.	PLO10, PLO11, PLO12

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy

CLO1, CLO2	Introduction: Overview of ICT business analytics, concept of business value from corporate data, the exploitation of information for advantage, types and sources of information value.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1, CLO2	Business intelligence: Nature and value of business intelligence, business analytics, and how different types of analytics can add value to corporate ICT data sources, Enterprise data life cycle and data governance.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO2, CLO3	Combining and importing data: Knowledge discovery, data mining, data warehousing, data lake, ICT Business analytics, Online Analytical Processing (OLAP) analysis, metadata.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2, CLO3	Data exploration and visualizing trends: Data preparation, Data visualization, visualization techniques, dashboard, uses of data visualization tools (R studio, Python, Tableau)	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO3	Data management: The relationship between corporate strategy and Information Systems (IS) strategy pertaining data management, Privacy, ethical, legal issues associated with organizational data, anomaly detection	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation
CLO3	Cloud based data management and analytics: Building analytic models using cloud based data management	Lecture Discussion Group Assignment	Essay Type Answer Report, Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	

CLO1	√				√
CLO2	√	√			√
CLO3	√			√	√

Suggested Texts:

- Business Analytics: Data Analysis & Decision Making, 5th Edition by S. Christian Albright, Wayne L. Winston.
- The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios Paperback, 1st Edition by Steve Wexler, Jeffrey Shaffer, Andy Cotgreave.
- Good Charts: The HBR Guide to Making Smarter, More Persuasive Data Visualizations Paperback, 1st Edition by Scott Berinato.
- Essentials of Business Analytics, 2nd Edition by Camm, Cochran, Fry, Ohlmann, Anderson, Sweeney, Williams.

Intended Learning Outcomes:

- Knowledge and understanding
 - Enable all participants to recognize, understand and apply the language, theory and models of the field of business analytics.
 - Understand the core and necessary data mining techniques so that students understand how to work with large data sets and apply the appropriate data mining technique to answer business questions.
- Cognitive skills (thinking and analysis)
 - Foster an ability to critically analyze, synthesize and solve complex unstructured business problems.
 - Evaluate various ICT business analytic tools and techniques for the business interesting points finding from the student group work deliverable
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Propose a business analytics report to solve practical problems identified in an ICT business project.

Course Objectives:	The objective of this course is to • Provide an understanding of principal concepts, major issues, technologies and basic approaches in information and data security. • Develop an understanding of security policies (such as authentication, and confidentiality). • Understand of database security model. • Understand of multilevel secure relational model and poly-instantiation integrity. • Analyze of various types of security incidents and attacks, and learn methods to prevent, detect and react incidents and attacks.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the fundamentals of managing information and data security systems and personnel.	PLO1
	CLO2- To learn how security and management are interrelated.	PLO2, PLO4
	CLO3- To analyze common threats and recognize vulnerabilities of information systems.	PLO4, PLO7
	CLO4- To understand the laws and regulations surrounding information and data security.	PLO7, PLO8
	CLO5- To learn how to plan for disaster recovery.	PLO7
	CLO6- To learn how to conduct security audit.	PLO6, PLO11
	CLO7- To understand and apply security policies.	PLO3

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Defining Database and Information Security, Concept of Information Security, The three concepts of Information and data Security.	Lecture Assignment	Short Answer Exercise
CLO1, CLO2	Access Control in Current System, Multilevel Security, Different Techniques for Inference Control, Integrity Principle and Mechanism.	Lecture Assignment Demonstrate	Short Answer Identification Exercise

CLO2	Three Components of Trust Management, Trust Management using PGP Certificates, Concept of Trust Negotiation.	Lecture Assignment	Short Answer Discussion
CLO2	Data As a Service (DAS) Model, Encrypted Data Management, Query processing architecture for DAS.	Lecture Assignment	Short Answer presentation
CLO2	Introduction to Geospatial Database, Geospatial Data Models, Security of Geospatial data Through Access Control Models.	Lecture Assignment	Short Answer MCQ
CLO2, CLO3	Introduction to Data Warehouses and OLAP Systems, Security Problem Description, Classify Security Threats & Identify Security Requirements, Three-tier Security Architecture	Lecture Assignment Project	Short Answer Group Exercise MCQ
CLO2, CLO3	Hadoop Distributed File System, (HDFS), Map Reduce Algorithm.	Lecture Assignment	Short Answer Group Exercise MCQ
CLO3, CLO4	Steganography versus Watermarking, Watermarking Solution for Relational data.	Lecture Assignment Project	Short Answer Group Exercise MCQ
CLO3, CLO4	What is statistical database? Understanding of Data Compromise, How to Protect Data from Snooper, Query set size control mechanism..	Lecture Assignment	Short Answer Assignment MCQ
CLO3, CLO5	Attacks to Server Systems, Denial of Service Attack, Attacks to Network Systems.	Lecture Assignment Project	Short Answer Group Exercise MCQ
CLO5, CLO6	What is Risk Management process, Identifying Information Assets, Identifying Security Risk and evaluation, Risk Treatment?	Lecture Assignment	Short Answer Group Exercise MCQ
CLO4, CLO7	Information Security Governance, Information Security Management System (ISMS), Information Security Policy, Standards and Procedures, Information Security Evaluation.	Lecture Assignment Project	Short Answer Group Exercise MCQ

Assessment Pattern:

CLO	Continuous Assessment	Term Final
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	Class Test	Assignment	Project	Presentation	
CLO1	√				
CLO2	√				√
CLO3		√			√
CLO4	√			√	√
CLO5	√				√
CLO6	√				√
CLO7				√	√

Suggested Texts:

- Database Security and Auditing: Protecting Data Integrity and Accessibility, 1st Edition by Hasan A. Afyouni.
- Database Security, 1st Edition by Silvano Castano, Maria Fugini, Giancarlo Martella, and Pierangela Samarati.
- Implementing Database Security and Auditing, 1st Edition by Ron Ben Ratan, Publisher: Elsevier Digital Press, 2005.
- Database Security, 1st Edition by Alfred Basta, Melissa Zgola.
- Principles of Information Security, 6th Edition by Michael E. Whitman and Herbert J. Mattord, Publisher: Cengage Learning.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the principal concepts, major issues, technologies and basic approaches in information and data security.
 - Gain the basic principles of security and to know how it affects our daily life.
 - Gain an understanding of terms and terminology commonly used in security management.
 - Know how vulnerabilities occur and how to limit them.
 - Gain a fundamental understanding of what an attack/threats is, and how to identify and prevent them from occurring.
 - Gain a fundamental knowledge of security audit and security policies.
- Cognitive skills (thinking and analysis)
 - Be able to implement security plan and monitor solutions.
 - Be able to monitor and evaluate audit logs and set administrator alerts.
 - Be able to respond to any breach of security and adjust organizational security plan accordingly.

- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to demonstrate and apply knowledge of current trends in ICT security, particularly those that relate to security threats, security policies, dissenter management and security audit.

ICT 3109: Principles of Economics

Course Objectives:	The objective of this course is to • Provide an introduction to a broad range of economic concepts, theories and analytical techniques. • Cover both microeconomics - the analysis of choices made by individual decision-making units (households and firms) - and macroeconomics - the analysis of the economy as a whole. • Discuss the analysis of market, supply and demand, model in which trade-offs and choices will be considered through comparison of costs and benefits of actions. • Analyzed production and market structure at the firm level. Macroeconomic issues regarding the interaction of goods and services markets, labor and money at an aggregate level will be modelled. • Examine microeconomic market failures and macroeconomic objectives.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand basic concept of scarcity, choice and opportunity cost.	PLO1
	CLO2- To learn demand, supply, market equilibrium and comparative static analysis.	PLO2, PLO3
	CLO3- To examine the economic agent behavior and apply the micro and macroeconomic policies in science and engineering field.	PLO4, PLO6
	CLO4- To use the economic tools in choosing techniques and analyze the IT related issues to address the real-life problems.	PLO11, PLO12

Course contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	Fundamentals of Economics: Definition of economics and basic concepts of macroeconomics, Evolution of economic thought, Macroeconomic agents and their interactions, circular flow diagram, production possibility frontier, Law of demand, Determinants, demand curve, exception, Law of supply, Determinants, supply curve, exception, Intersection between demand and supply curve, Determination of equilibrium price and quantity, Changes of the market equilibrium.	Lecture Assignment	Short Answer Exercise Case Studies from the Real World Problems
CLO3, CLO4	Price Elasticity of Demand: Concepts, Calculations and interpretation, Determinants, Graphical Illustration, Income and cross elasticity of demand, Revenue implications of price elasticity of demand.	Lecture Assignment Discussion	Short Answer Identification of the Nature of the Commodities by Elasticity Exercise
CLO3	Utility and Consumer's Choice: Cardinal and ordinal approaches of utility, Law of diminishing marginal utility, Indifference curve, Budget line and Consumer's Utility maximization.	Lecture Assignment	Short Answer Discussion
CLO3	Production: Resources / Factors of production, Short run vs long run production behavior of a firm (special reference to IT firm), Cost structure of a firm and determination of profit maximizing output level.	Lecture Assignment	Short Answer Problem solving
CLO3, CLO4	Market and Government Policy: Types of market based on competition, Consumers and producer's reaction under different policy (Fiscal, monetary and growth policies) setting in different market. GNP, GDP and National income measurement.	Lecture Assignment Case studies	Short Answer Assignment Mathematical problems

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√				√
CLO3	√			√	√
CLO4		√			√

Suggested Texts:

- Economics, 11th Edition by Arnold, Roger, Publisher: South-Western, Cengage Learning - International Edition.
- Principles of Economics, 8th Edition by Mankiw, N. Gregory.
- Microeconomics, 11th Edition by Michel Parkin, Publisher: Addison Wesley Longman.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of scarcity, choice and opportunity cost.
 - Understand the use of economic tools to explain how unlimited wants are to be met with limited resources.
- Cognitive skills (thinking and analysis)
 - Be able to explain the real life problems of demand, supply and market equilibrium and its change under different government's policy setting.
 - Be able to analyze how the economic agents are driven by their optimization goals and what is the time path to reach such optimum levels.
- Communication skills (personal and academic)
 - An individual assignment (mathematical depiction, graphical illustration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to use the concepts and analytical tools to comprehend and construct economic arguments.

ICT 3102: Operating System Lab

Course Objectives:	The objective of this course is to • Familiarize with basic Linux commands and executing those commands on OS files. • Learn and develop programs using Shell Scripting. • Implementing OOP in Linux environment.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the history of modern computers.	PL01
	CLO2- To analyze each of the major components of an operating system.	PL02, PL05
	CLO3- To explore more advanced topics in the field, including memory management and file input/output.	PL04

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction to operating systems concepts, process management, memory management, file systems, virtualization, and distributed operating systems.	Demonstration	Practical Exam Viva-Voce
CLO2	Familiarization with UNIX system calls for process management and inter-process communication.	Demonstration Case studies, Lecture	Practical Exam Viva-Voce
CLO2	Experiments on process scheduling and other operating system tasks through simulation/implementation.	Demonstration	Practical Exam Viva-Voce
CLO2	Resource allocation and deadlock, deadlock detection, prevention and recovery.	Demonstration Assignment	Practical Exam Viva-Voce
CLO3	Files, directories, security, protection, and Case study of some operating systems.	Demonstration Assignment	Demonstration Assignment
CLO2, CLO3	Apply the operating system concepts by experimenting on either Unix operating systems.	Exercise	Completion

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1				√	√
CLO2	√	√			√
CLO3	√				√

Intended Learning Outcomes:

- Knowledge and understanding
 - ▢ Understand the goals of standardization of OS (and other) interfaces.
 - ▢ Understand the execution of a program as the trace of consecutive machine language instructions residing in the computer memory, and how typical programming structures manifest themselves in machine language.
 - ▢ Understand the layered structure of input/output (I/O) software and give a broad overview of I/O interrupt handling.
- Cognitive skills (thinking and analysis)
 - ▢ Be able to understand the typical (physical) computer memory hierarchy and the compromises involved in using such a hierarchy.
 - ▢ Be able identify the most common data structures required in an OS implementation.
- Communication skills (personal and academic)
 - ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
 - ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - ▢ Be able to read and understand simple bash scripts, modify them in a controlled way, and produce simple scripts.
 - ▢ Possess the courage and basic skills to use a unix-like OS, e.g., Linux, by means of a command line shell and an SSH terminal connection.

ICT 3104: Computer Network Lab

Course Objectives:	The objective of this course is to • Provide information about computer network organization and implementation. • Gain practical experience in installation, monitoring, and troubleshooting of current LAN systems. • Describe various types of routing techniques, VLAN operation and its management.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the principles and concepts on computer networks.	PLO1
	CLO2- To analyze soundness or potential flaws in proposed protocols.	PLO2, PLO5
	CLO3- To explain and identify security and ethical issues in computer networking.	PLO7

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Applications of Networks, Network Structure, Network Architecture, Connection-Oriented and connectionless Service Primitives, Public Networks.	Demonstration Case studies Lecture	Practical Exam Report Viva-Voce
CLO1	Familiarities with network physical layer, Mobile telephone system.	Demonstration	Practical Exam Report Viva-Voce
CLO2, CLO3	Working with data link layer design issues, Error detection and correction.	Problem Based Learning (PBL) Project	Practical Exam Viva-Voce

CLO2	Working with multiple access protocols, ALOHA, CSMA/CD Protocol; Collision-free protocols, Wireless LAN protocols, Ethernet, Bluetooth	Demonstration	Practical Exam Report Viva-Voce
CLO2	Lab practice on Network layer design, Routing algorithms, Congestion control algorithms, Inter networking, Network layer in the internet, Network layer in ATM networks.	Problem Based Learning (PBL) Project Lecture	Practical Exam Report Viva-Voce
CLO2	Introduction to the transport service, Elements of transport protocols, The internet transport protocols.	Group assignment Lecture	Practical Exam, Report Viva-Voce
CLO2	Lab practice on DNS-domain name system, Electronic mail, The world wide web, multimedia.	Problem Based Learning (PBL) Projects	Practical Exam Report Viva-Voce
CLO3	Lab practice on Secrecy, Authentication, Digital signature.	Exercise	Completion

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1		√			√
CLO2	√		√		√
CLO3	√			√	√

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the knowledge on designing and building a complete system.
 - Understand different layer of computer network.
- Cognitive skills (thinking and analysis)
 - Be able to analyze and capture network traffic.

- Be able to analyze soundness or potential flaws in proposed protocols.
- Be able to apply the theory of basic network performance analysis.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to design logical sub-address blocks with a given address block.

IT 3100: Special Study/Industrial Attachment and Viva

Students require to pursue Special Study or they may send to Industrial for industrial experience.

3rd Year 2nd Semester

ICT 3201: Software Engineering

Course Objectives:	The objective of this course is to ● Learn basic SW engineering methods and practices, selected models. ● Understand Role of project management including planning, scheduling and, risk management, verification and validation techniques ● Analyze quality control that will enable the students to apply these in subsequent projects and work experiences.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the current theories, models, and techniques that provide a basis for the software lifecycle	PLO1, PLO2
	CLO2- To analyze the software engineering lifecycle by demonstrating competence in communication, planning, analysis, design, construction, and deployment.	PLO2, PLO4

	CLO3- To demonstrate an ability to use the techniques and tools necessary for engineering practice.	PLO5, PLO6, PLO10, PLO12
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Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	Software Engineering Paradigms: Definition of software engineering, The classical life cycle, prototyping fourth generation technique, The product and the process, measurement matrices.	Lecture Assignment	Short Answer Exercise
CLO1, CLO2	Software Project Planning: Project planning objectives, S/W slope, Resources, Metrics for S/W productivity and quality, S/W project estimation, Decomposition techniques, Empirical Estimation Models, Automated Estimation tools, S/W project scheduling.	Lecture Assignment	Short Answer Identification Exercise
CLO2	Requirement Analysis Fundamentals: Analysis principle, Software Prototyping Specification, Requirement analysis Methodologies, Structured and Object oriented analysis, Data Flow-oriented Analysis methods.	Lecture Assignment	Short Answer Discussion
CLO2, CLO3	Software Design fundamentals: Design process, Design Fundamentals, S/W architecture, Program structure, Data structure, S/W procedure, Modularity, abstraction, Effective modular design, Procedural design, Data flow-oriented design, Top-down and bottom-up design, Design Process Considerations, Transform analysis, Transaction analysis, Data structure-oriented design, Logical construction of programs and systems, Data structured systems development, object-oriented design, Design concepts, methods & strategy. Real-time Design. Coding style, Code documentation, Data declaration, statement construction, Input/output. Software reliability and availability models: Software quality factors, software review, software quality metrics, Software reliability, Software quality assurance approach.	Lecture Assignment	Short Answer presentation

CLO2, CLO2	Software Testing Techniques: Testing fundamentals, White box testing, Basis path testing, Loop testing, Black Box testing.	Lecture Assignment	Short Answer Assignment MCQ
CLO2, CLO3	Software Testing Strategies: Verification and validation, Organization for software testing, Unit testing, Integration testing, Validation testing, System testing, The art of debugging.	Lecture Assignment	Short Answer Group Exercise MCQ
CLO2, CLO3	Software Maintenance and Configuration Management: Definition, Maintenance Characteristics, Maintainability, Maintenance tasks, Software configuration Management.	Lecture Assignment	Short Answer Group Exercise MCQ

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√			√	√
CLO2	√	√			√
CLO3	√		√		√

Suggested Texts:

- Software Engineering, 7th Edition by Ian Sommerville.
- Software Engineering: A Practitioners Approach, 7th Edition by R.S. Pressman.
- Software Architecture: Prospective on an Emerging Discipline, 1st Edition by Mary Shaw.
- Fundamentals of Software Engineering, 2nd Edition by Ghezzi, M. Jazayeri.
- Designing Object-oriented Software, 1st Edition by R. Wirfs-Brock, Brian Wilkerson, Lauren Wiener.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the software engineering basics.
 - Understand the professional and ethical responsibility.
 - Understand the software development life cycle.

- Cognitive skills (thinking and analysis)
 - ▢ Be able to design and conduct experiments, as well as to analyze and interpret data.
 - ▢ Be able to identify, formulate, and solve engineering problems.
 - ▢ Be able to use the techniques, skills, and modern engineering tools necessary for engineering practice.
 - ▢ Be able to analyze, design, verify, validate, implement, apply, and maintain software systems.
- Communication skills (personal and academic)
 - ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.

ICT-3203: Computer Architecture and Microprocessor

Course Objectives:	The objective of this course is to ● Gain the fundamental concepts of underlying modern computer organization and architecture. The student will familiarize about hardware design including logic design, basic structure and behavior of the various functional modules of the computer and how they interact to provide the processing needs of the user. The students will also become familiar with the architecture and the instruction set of an Intel microprocessor. Assembly language programming will be studied as well as the design of various types of digital and analog interfaces. Understand the architecture of 8085, 8086, 80286 and 80386.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To describe the basic design of computer logic, through simple combinational and sequential logic circuits	PLO1, PLO2
	CLO2- To understand the architecture of widely used microprocessor model like 8086.	PLO2
	CLO3- To analyze the architecture and instruction set of 80286 and 80386 microcontrollers.	PLO2, PLO5
	CLO4- To design I/O circuits and memory interfacing circuits.	PLO3, PLO6

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Design Methodology: Introduction, Combinational circuits, Sequential circuits, the register level, Register-level components, Design method, the processor-level, Processor-level components, Design techniques.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO1	Arithmetic Logic Unit: Fixed-point arithmetic, Addition, Subtraction, Multiplication and division, Processor Organization, Arithmetic Logic Unit, Design of Arithmetic Circuit, Design of Logic Circuit, Design of Arithmetic Logic Unit.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO1	Control Design: Basic concepts of control unit, Hardwired control, GCD processor control unit, Multiplier control unit, CPU control unit, Micro-programmed control; Microinstruction.	Lecture Assignment	Short Answer Discussion
CLO1	Memory Organization: Memory devices and characteristics, RAM organization, Serial access memory; Virtual memory, Main-memory allocation, Segments and pages, High speed memories, Cache memory.	Lecture Assignment	Short Answer Discussion
CLO1	Microprocessors: Concept of microprocessor, Evolution of microprocessors.	Lecture Assignment	Short Answer Discussion
CLO2	Internal architecture of Intel 8085,8086/8088 microprocessors: Instruction set and format, Programming in machine assembly language, Interrupt structure, DMA, I/O operation, Microprocessor interface ICs, peripheral interfacing, Microprocessor based system design, Coprocessor, Multiprocessor system.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO3	Intel 80286, 80386 microprocessors: memory management scheme, Protection mechanism, 80386 modes; Pentium microprocessor; Advanced microprocessors.	Lecture Assignment	Short Answer Discussion
CLO4	Interfacing with analog world: A/D conversion, digital ramp ADC, successive approximation ADC, flash ADC, tristate ADC, D/A converter, DAC specifications, DAC applications, Data acquisition, sample-and hold circuits, multiplexing.	Lecture Assignment	Short Answer Discussion

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√				√
CLO3				√	√
CLO4	√	√			√

Suggested Texts:

- Computer Architecture and Organization, 5th Edition by John P. Hayes.
- Microprocessors and Microcomputer based system Design, 2nd Edition by Md. Rafiquzzaman.
- Microprocessors and System Design, 1st Edition by Gibson & Cheu.
- Microprocessors and Interfacing: Hardware and Software, 1st Edition by D. V. Hall.
- Language Programming Technique in IBM PC, 1st Edition by Alan R. Miller.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the architecture and operation of typical microprocessors and microcontrollers.
 - Understand the programming and interfacing of microprocessors and microcontrollers.
- Cognitive skills (thinking and analysis)
 - Be able to demonstrate programming proficiency using the various addressing modes and data transfer instructions of the target microprocessor and microcontroller.
 - Be able to compare accepted standards and guidelines to select appropriate Microprocessor (8085 & 8086) and Microcontroller to meet specified performance requirements.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.

- Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.

ICT 3205: Signals and Systems

Course Objectives:	The objective of this course is to • Understand the continuous and discrete-time signals and systems, their properties, representations and methods that are necessary for the analysis of continuous and discrete-time signals and systems. • Knowledge of time-domain representation and analysis concepts as they relate to difference equations, impulse response and convolution, etc. • Knowledge of frequency-domain representation and analysis concepts using Fourier analysis tools, Laplace-transform • Mathematical and computational skills needed in application areas like communication, signal processing and control, which will be taught in other courses.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand various types of signals and systems.	PLO1
	CLO2- To learn the different representation of signals and systems.	PLO2
	CLO3- To calculate the output of various systems.	PLO4
	CLO4- To solve real-life problems using signal representations.	PLO3, PLO6

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Basic Signals and systems: Classification, operations and properties of signals and systems.	Lecture Assignment	Exercise Tutorial Assignment

CLO1	Linear Time-Invariant System: Discrete-Time LTI, Continuous-time LTI, convolutional sum, Impulse response, Properties and integral representation, Casual LTI, singularity functions.	Lecture Assignment	Assignment Exercise Tutorial
CLO2	Fourier Series and Transform: History, The Response of LTI Systems to Complex Exponentials, Fourier Series Representation of Continuous-Time Periodic Signals, Convergence of the Fourier Series, Properties of Continuous-Time Fourier Series, Fourier Series Representation of Discrete-Time Periodic Signal, Properties of Discrete-Time Fourier Series, Fourier Transform for aperiodic and periodic signals, Properties of the Continuous-Time and discrete time Fourier Transform.	Lecture Assignment	Exercise Assignment Simulation Tutorial
CLO1, CLO2	Digital Filters: Basic concept, classification, Design, Applications.	Lecture Assignment	Exercise Interview Assignment
CLO2, CLO3	Sampling: Introduction, Representation of a Continuous-Time Signal by Its Samples, The Sampling Theorem, Reconstruction of a Signal from Its Samples Using Interpolation, The Effect of Undersampling: Aliasing, Discrete-Time Processing of Continuous-Time Signals, Sampling of Discrete-Time Signals,	Lecture Assignment	Short Answer Assignment MCQ
CLO2, CLO3	The Laplace Transform: Basic concept, ROC, The Inverse Laplace Transform, Geometric Evaluation of the Fourier Transform from the Pole-Zero Plot, Properties of the Laplace Transform, Some Laplace Transform Pairs, Analysis and Characterization of LTI Systems Using the Laplace Transform, The Unilateral Laplace Transform, System represent using LT.	Lecture Assignment	Exercise Interview
CLO2, CLO3	The Z-Transform: Introduction to z-Transform, The Region of Convergence for the z-Transform, The Inverse z-Transform, Properties of the z-Transform, Geometric Evaluation of the Fourier Transform from the Pole-Zero Plot, Some Common z-Transform Pairs, Analysis and Characterization of LTI Systems Using z-Transforms, The Unilateral z-Transform, System represent using ZT.	Lecture Assignment	Exercise Assignment Tutorial MCQ

CLO2, CLO3, CLO4	Communication Systems and Linear Feedback Systems: Complex Exponential and Sinusoidal Amplitude Modulation and Demodulation, Frequency-Division Multiplexing, Single-Sideband Sinusoidal Amplitude Modulation, Amplitude Modulation with a Pulse-Train Carrier, Pulse-Amplitude Modulation, Sinusoidal Frequency Modulation, Discrete-Time Modulation, Introduction to Linear Feedback Systems, Some Applications and Consequences of Feedback, Root-Locus Analysis of Linear Feedback Systems, The Nyquist Stability Criterion, Gain and Phase Margins.	Lecture Assignment	Exercise Assignment Tutorial MCQ
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Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√				√
CLO3	√			√	√
CLO4		√			√

Suggested Texts:

- Signals and Systems, 2nd Edition by Simon Haykin, Barry Van Veen.
- Signals and Systems, 2nd Edition by Alan V. Oppenheim, Alan S. Willsky with S. Hamid.
- Signals and Systems using MATLAB, 3rd Edition by Luis Chaparro.
- Transforms in Signals and Systems (Modern Applications of Mathematics), 1st Edition by Peter Kraniavskas.
- Signals and Systems, Schaum's Outline Series, 4th Edition by Hwei P. Hsu.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the fundamental operations of signals and systems.
 - Understand the representation, properties, and response of different types of signals and systems.
- Cognitive skills (thinking and analysis)
 - Be able to compute the output of various types of systems.

- Be able to develop the continuous-time Fourier transform from the Fourier series and understand related topics such as time scaling, convolution theorem, Parseval's relation, and uncertainty principle and Eigen functions of the Fourier operator.
- Be able to identify linear time-invariant systems.
- Be able to calculate Laplace transform and Region of convergence.
- Be able to calculate Z-Transform and Region of convergence.
- Be able to analyze and reconstruct signals using sampling.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to use demonstrate signals and systems operations using simulation software.

ICT 3207: Server Administration and Management

Course Objectives:	The objective of this course is to ● Give students a comprehensive overview of the tools and techniques needed to successfully administer servers. ● Understand the importance of automation, and be able to create scripts and use other tools to automate server management procedures. ● Familiarize with the operational & technological best practices for improving resource management.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the computer system fundamentals necessary for Server Management.	PLO1
	CLO2- To develop the basic concept of server administration and maintenance skills to effectively manage a stable, small-scale networked environment.	PLO2, PLO11
	CLO3- To analyze and select the use of proven techniques to improve project outcomes through resource management	PLO4, PLO5, PLO12

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Computer System Fundamentals: Operating System Fundamentals, Networking Fundamentals, Security concepts Introduction about server administration: Introduction to the role of server, Illustration about the major tasks performed by a server administrator.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1	Role of VMware in basic server administration: Overview of VMware, Features of VMware, Technical support resources, VMware network and service configuration.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO1, CLO2	Linux server management: Introduction to Linux, Installation of Linux, Managing User and Group Accounts, Managing Partitions and the Linux File system, Managing Files in Linux, Backup and Disaster Recovery, Working with Linux Permissions and Ownership, Implementing DNS Server, DHCP Server, FTP & SAMBA Server, Proxy Server, Implementing Mail Server, Implementing Web Server.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2	Cloud Management: Introduction to Cloud Management, Familiarizing with AWS, Basic Linux deployment and Cloud Security Concepts.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2, CLO3	Introduction to IT resource management: Overview of IT resource management, Need for an automated resource management	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation

CLO2, CLO3	Building an IT resource management system: Role of a manager, Resource and role identification, Resource scheduling, Resource Optimization, Detailed analytics, Cloud based solutions.	Lecture Discussion Group Assignment	Essay Type Answer Report Exercise Presentation
CLO3	IT Resource Management Best Practices: 5 step framework formation based on industry best practices, Resource management life cycle.	Lecture Group Assignment Demonstration	Essay Type Answer Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√			√	√
CLO2	√				√
CLO3	√	√			√

Suggested Texts:

- The Practice of System and Network Administration, 2nd Edition by Thomas A. Limoncelli, Christine Hogan, Strata R. Chalup.
- UNIX and Linux System Administration, 4th Edition by Evi Nemeth, Garth Snyder, Trent R. Hein, Ben Whaley.
- Project Management: A Systems Approach to Planning, Scheduling, and Controlling, 10th Edition by Harold Kerzner, PhD.
- Online materials for IT resource management.
- Principles of Network and System Administration, 2nd Edition by Mark Burgess.
- Essential System Administration, 3rd Edition by Aeleen Frisch.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of Network Topology, Design, Monitoring, etc.
 - Understand Linux Server Installation & Configuration.
 - Understand the broader scope of IT resource management.

- Cognitive skills (thinking and analysis)
 - ▯ Be able to administer, secure and troubleshoot Server.
 - ▯ Be able to describe how to plan major and minor tasks and time so that services are stable and effective, and meet a Service Level Agreement.
 - ▯ Be able to explain the importance of IT resource management.
- Communication skills (personal and academic)
 - ▯ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▯ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - ▯ Be able to acquire the communication, leadership and teamwork skills necessary to work in teams, or in charge of teams, that are responsible for operating server environments.

ICT 3209: Smart Sensors Communication

Course Objectives:	The objective of this course is to ● Understand of various physical phenomenon of different types of sensors and microsystems. ● Design of sensors with appropriate electronic interface as a complete system. ● Discuss about various types of sensors like magnetic, optical, bio, chemical, radiation, electrical and mechanical etc. ● Emphasis on the integration of electronics with sensors to provide a smart transducer or a system on a chip with multiple integrated devices.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To select the right sensor for a given application.	PL01, PL02
	CLO2- To design basic circuit building blocks.	PL03
	CLO3- To simulate, synthesize, and layout a complete sensor or sensor system, MEMS device or Microsystems ready for fabrication tools.	PL04, PL05

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Sensors: Working Principles; Different types, Selection of Sensors for Practical Applications; Introduction of Different Types of Sensors such as Capacitive, Resistive, Surface Acoustic Wave for Temperature, Pressure, Humidity, Toxic Gas etc. Important Characteristics of Sensors: Determination of the Characteristics Impedance Spectroscopy: Equivalent circuit of Sensors and Modelling of Sensors.	Lecture Assignment	Short Answer Exercise
CLO1, CLO2	Importance and Adoption of Smart Sensors Architecture of Smart Sensors: Important components, their features; Interface Electronic Circuit for Smart Sensors and Challenges for Interfacing the Smart Sensor.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO1, CLO2	Wireless sensor network (WSN): Introduction to Wireless Sensor Networks; Network Architecture; Hardware Platforms; Medium Access Control Protocol design; Routing protocols for WSN; 8 Energy Harvesting WSNs.	Lecture Assignment	Short Answer Discussion
CLO2, CLO3	Internet of Things (IoT): Introduction to IoT; Internet of Things Architecture; Vision and Challenges for realizing the Internet of Things.	Lecture Assignment	Short Answer Presentation
CLO2, CLO3	Internet of Things Application Domains: Design and Implementation of IoT for Environmental Condition Monitoring; Development of WSN Based Smart Bed for Health Care Application (Case study); Study of Smart City, smart village and their Design (Case study); Application of IoT in agriculture (Case study).	Lecture Assignment Case study Demonstrate	Short Answer Presentation
CLO2, CLO3	Design and development of Security and Privacy Technologies related to IoT.	Lecture Assignment	Short Answer Presentation
CLO2, CLO3	Relevance of smart sensors and IoT to Developing World and Research Scope for the Internet of things.	Lecture Assignment	Short Answer Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√			√	√
CLO3	√	√			√

Suggested Texts:

- Handbook of Modern Sensors, 2nd Edition by Jacob Fraden.
- Semiconductor Sensors, 1st Edition by S. M. Sze.
- Fundamentals of Wireless Sensor Networks: Theory and Practice, 1st Edition by Willy, Waltenegus Dargie and Christian Poellabauer.
- 6LoWPAN: The Wireless Embedded Internet, 1st Edition by Zach Shelby, Carsten Bormann, Publisher: Wiley.
- Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, 1st Edition by Dr. Ovidiu Vermesan, Dr. Peter Friess, River Publishers.
- Interconnecting Smart Objects with IP: The Next Internet, 1st Edition by Jean-Philippe Vasseur, Adam Dunkels, Morgan Kuffmann.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the general concepts of Smart Sensors and Internet of Things (IoT).
 - Recognize various devices, sensors and applications.
- Cognitive skills (thinking and analysis)
 - Identify and compare approaches to automatically monitoring the environmental data.
 - Analyze network problems to determine the root cause and relevant of internet of things network.
 - Describe how network problem regarding data decimation and data gathering can be discover and mitigate.
 - Identify and describe routing protocols for WSN and IoT.
 - Explore and analyze recent approaches to solving low energy consumption and quality of services related problem in WSN and IoT.
- Communication skills (personal and academic)
 - Develop research skills related to design an IoT network to address a daily life problem.
 - Write a review of current literature on an aspect of Internet of Things.

- ▯ Select a technical paper and summarize it for the class in a brief presentation.
- Practical and subject specific skills (Transferable Skills)
 - ▯ Create IoT solutions using sensors, actuators and Devices.

ICT 3202: Software Engineering Lab

Course Objectives:	The objective of this course is to ● Learn basic SW engineering methods and practices, selected models. ● Identify role of project management including planning, scheduling and, risk management, verification and validation techniques and quality control that will enable the students to apply these in subsequent projects and work experiences.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the history of Software Engineering.	PLO1
	CLO2- To analyze each of the major components of Software Engineering.	PLO2, PLO5
	CLO3- To explore more advanced topics in the field, including software requirements, design, implementation, testing, and maintenance.	PLO3, PLO4 PLO10

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems	Demonstration	Practical Exam Viva-Voce
CLO2	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences	Demonstration Case Studies Lecture	Practical Exam Viva-Voce

CLO2, CLO3	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations	Demonstration	Practical Exam Viva-Voce
CLO3	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions	Demonstration Assignment	Practical Exam Viva-Voce

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√			√	√
CLO3	√	√	√		√

Intended Learning Outcomes:

- Knowledge and understanding
 - ▢ Software Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals in the aspect of Software.
 - ▢ The ability to apply standard practices and strategies in software service management using open-ended programming environments with agility to deliver a quality service for business success.
- Cognitive skills (thinking and analysis)
 - ▢ Be able to understand the typical design of model and implementation for the product of Software.
 - ▢ Be able to identify the most critical reasoning of cost estimation of software Engineering.
- Communication skills (personal and academic)
 - ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
 - ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
 - ▢ The ability to research, understand and implement computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics, and networking for efficient analysis and design of computer-based systems of varying complexity.

- Practical and subject specific skills (Transferable Skills)
 - Be able to read and understand simple software design in the aspect of business enterprise.
 - Be able to design software requirement specifications in the aspect of business solution.

ICT-3204: Computer Architecture and Microprocessor Lab

Course Objectives:	The objective of this course is to • Provide understanding of microprocessor, interfaces, interrupts. • Understand details about microprocessor with memory and peripheral chips involving system design. • Describe the concept of 8259A Programmable Interrupt Controller (PIC) and DMA.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the architecture of widely used microprocessor model like 8086.	PL01
	CLO2- To analyze the architecture and instruction set of 80286 and 80386 microcontroller.	PL02, PL05
	CLO3- To design I/O circuits and memory interfacing circuits.	PL03

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Code demonstrating on how 8, 16, 32 and 64-bit values can be used with software code	Demonstration Case studies Lecture	Practical Exam Viva-Voce
CLO2	Lab practice on the use of branching, flags, stacks, procedures, macros, and interrupts.	Problem Based Learning (PBL) Lecture	Practical Exam Viva-Voce
CLO2	Lab practice for the basic arithmetic and logic operations available in assembly language.	Problem Based Learning (PBL) Lecture	Practical Exam Viva-Voce

CLO3	Discuss string operations and demonstrating a variety of string operations including search and search & replace.	Demonstration Case studies Lecture	Practical Exam Viva-Voce
CLO3	Lab practice on using inline assembly language programming within a C++ program and with a standalone assembler, such as MASM or TASM.	Exercise	Completion

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√			√
CLO3	√		√		√

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the architecture and operation of typical microprocessors and microcontrollers.
 - Understand the programming and interfacing of microprocessors and microcontrollers.
- Cognitive skills (thinking and analysis)
 - Be able to demonstrate programming proficiency using the various addressing modes and data transfer instructions of the target microprocessor and microcontroller.
 - Be able to compare accepted standards and guidelines to select appropriate Microprocessor (8085 & 8086) and Microcontroller to meet specified performance requirements.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to analyze assembly language programs, select appropriate assemble into machine a cross assembler utility of a microprocessor and microcontroller.

- Be able to design electrical circuitry to the Microprocessor I/O ports in order to interface the processor to external devices.
- Be able to evaluate assembly language programs and download the machine code that will provide solutions real-world control problems.

IT3200: Project Work - III and Course Viva

Course Contents:

The content of the viva includes the syllabus of all major courses. Each group of student must have to do a project work based on the courses taught in the 3rd Year's 1st and 2nd Semester. A faculty member or representative from other dept./institute/industry will supervise the project.

4th Year 1st Semester

ICT 4100: Internship

Outline: The student will work full-time as an intern to particular company for a period of six months. S/he will be evaluated based on the marks provided by the company along with the marks of at least two presentations given at IIT.

Option - I

ICT 4101: Digital Signal Processing

Course Objectives:	The objective of this course is to • Develop the knowledge on signals used in digital signal processing. • Introduce signals, systems, time and frequency domain concepts and the associated mathematical tools those are fundamental to all DSP techniques. • Provide a thorough understanding and working knowledge of design, implementation, analysis and comparison of digital filters for processing of discrete time signals. • Study various sampling techniques and different types of filters and will also understand basic principles of Estimation Theory.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To analyze the digital systems using z-Transform and Discrete Time Fourier Transform.	PLO2
	CLO2- To identify the difference between fixed point and floating-point digital signal processor.	PLO1
	CLO3- To analyze the discrete time signals and systems in the frequency domain using Discrete Fourier Transform and Fast Fourier Transform.	PLO4, PLO5
	CLO4- To design FIR and IIR filters using a variety of techniques.	PLO3

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2, CLO3	Introduction to DSP, classifications of signals, continuous time and discrete time (DT) sinusoids, concept of frequency, advantages and limitations of DSP, applications of DSP, steps of ADC, sampling theorem, abasing, quantization, coding.	Lecture Assignment	Short Answer Exercise

CLO1, CLO4	Classification of DT signals, classification of DT systems, impulse response, FIR and IIR, block diagram of DT systems, analysis of LTI systems, convolution, properties of convolution, causality and stability of LTI systems, recursive and non-recursive systems, correlation, properties and applications of correlations. Z-transform, ROC, Inverse z-transform, properties of ztransform, concept of pole-zero, one-sided z-T.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO1, CLO3	Frequency analysis, Fourier series and Fourier transform for continuous time and discrete time signals, power density and energy density spectrums, DFT, properties of FT and DFT, invertibility of LTI systems, DFT as linear transformation, FFT, divide and conquer approach, radix-2 FFT. Structures of DT systems: Direct form, lattice structure, transposed structure.	Lecture Assignment	Short Answer Discussion
CLO2, CLO4	State-space system analysis. Digital filter: advantages and limitations of digital filters, adaptive filters, applications: inverse modeling, system identification, noise cancellation etc., characteristics of ideal and practical filters. Filter design: designing steps, window method, optimal method, IIR & FIR filter design methods.	Lecture Assignment	Short Answer Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√			√
CLO2	√	√			√
CLO3	√	√			√
CLO4	√	√			√

Suggested Texts:

- A Textbook of Digital Signal Processing, 1st Edition by Dr. Sanjay Sharma.
- Introduction to Digital Signal Processing, 1st Edition by Robert Meddins.
- Digital Signal Processing, 4th Edition by S. Salivahanan.
- Digital Signal Processing in Communication Systems, 1st Edition by Marvin E Frerking.

- Detection Theory: Applications and Digital, 1st Edition by Ralph D Hippenstiel.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand discrete time signal processing and characterization of random signals, filter design techniques and imperfections caused by finite word length.
 - Learn how to estimate the spectra of random signals that are to be processed by a discrete time filter and to appreciate the performance of a variety of modern and classical spectrum estimation techniques.
 - Learn the theory of modern digital signal processing and digital filter design.
- Cognitive skills (thinking and analysis)
 - Be able to apply filtering technique in complex digital filter design and digital simulation experience.
 - Be able to implement simple low pass, band pass, high pass IIR and FIR digital filters for typical signal processing applications using MATLAB.
 - Be able to explain frequency response of discrete time signals and systems.
- Communication skills (personal and academic)
 - Individual assignments including demonstration and presentation involve receiving clear instructions, designing and making an effective presentation.
 - A term project (group work) - where students will be assessed in terms of their effectiveness as an individual or leader in the team.
- Practical and subject specific skills (transferable skills)
 - Be able to apply various methods for designing new digital signal processing systems and for continued learning.

ICT 4103: Mobile Application Development

Course Objectives:	The objective of this course is to • Know the components and structure of mobile application development frameworks for Android and windows OS based mobiles. • Understand how to work with various mobile application development frameworks. • Learn the basic and important design concepts and issues of development of mobile applications. • Understand the capabilities and limitations of mobile devices.
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Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To design and implement various mobile applications using emulators.	PLO3, PLO5
	CLO2- To deploy applications to hand-held devices.	PLO3, PLO10

Course Contents:

CLO	Course Contents	Teaching Strategy	Learning	Assessment Strategy
CLO1	Android Basics: Introduction, Architecture, Environment Setup, Emulator or Create AVD, Hello World App, Application Components, Activity Lifecycle, Content Providers, Broadcast Receivers, Services, Fragments, App Folder	Lecture Discussion Group Assignment Demonstration		Exercise Report Presentation
CLO1	Intents and Service: Android Intents (Implicit, Explicit), Intent Filters, Implicit Intents Vs Explicit Intents	Lecture Discussion Group Assignment Demonstration		Exercise Report Presentation
CLO1	UI Layouts: View and ViewGroup, <i>UI Layouts, LinearLayout, RelativeLayout, TableLayout, FrameLayout, ListView, GridView, WebView</i>	Lecture Discussion Group Assignment Demonstration		Exercise Report Presentation
CLO1	UI Controls: TextView, EditText, AutoCompleteTextView, Button, ImageButton, Toggle Button, RadioButton, CheckBox, RadioGroup, ProgressBar, RatingBar, AlertDialog, Spinner, <i>DatePicker, TimePicker, TextClock</i>	Lecture Discussion Group Assignment Demonstration		Exercise Report Presentation
CLO1	Menus: Options Menu, Context Menu, Popup Menu.	Lecture Discussion Group Assignment Demonstration		Exercise Report Presentation

CLO1	Notifications and Alarms: Toast, Custom Toast, Progress Notification, Push Notifications using Firebase Cloud Messaging.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1	Animations: Rotate Animations, Slide Up Down Animations, Fade In Out Animations, Zoom In Out Animations.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1	Connectivity: Bluetooth, Bluetooth Turn ON or OFF, Bluetooth Device Discoverable, Bluetooth List Paired Devices, WiFi.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1	Multimedia: Camera App, Audio, Audio Recorder, Audio Manager, Video Player.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1	Parsing: Android XML Parsing using DOM Parser, SAX Parser, XMLPullParser. JSON Parsing etc.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1	Storage: Android Shared, Preferences, Session Management, Internal Storage, External Storage, Android SQLite Database, SQLite ListView.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO2	Events and Themes: Input Events, Styles and Themes, Custom Views, Drag and Drop, Google Maps API, Send Email, Send SMS, Phone Calls, Gestures, Sensors, Test App on Real Device, Publish App on Google Play Store.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation

CLO2	AdMob Ads: Integrate AdMob Ads, Banner Ads in App, Interstitial Ads in App, Rewarded Video Ads in App.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO2	Graphics: Android Clipboard, ImageSwitcher, Loading Spinner, Navigation Drawer Sliding Menu, Internet Connection, TextToSpeech, Login and Registration Screen Design, Tabs with Fragments and ViewPager, ScrollView, Facebook Integration, Firebase Cloud Messaging.	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√	√	√
CLO2	√	√	√	√	√

Suggested Texts:

- Head First Android Development: A Brain-Friendly Guide, 2nd Edition by Dawn Griffiths, David Griffiths.
- Android Programming, The Big Nerd Ranch Guide, 2nd Edition by Bill Phillips, Chris Stewart, Brian Hardy & Kristin Marsicano.
- Android Cookbook: Problems and Solutions for Android Developers, 2nd Edition by Ian F. Darwin.
- Android Programming for Beginners, 2nd Edition by John Horton.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of Mobile Application Development.
 - Understand the fundamental issues of Mobile Apps, software architecture, operating platforms, user interfaces and data storage.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis the different Mobile Applications related to User Interface, Layout, Widgets, Fragment, Animation, Clipboard, Facebook, Cloud, etc.
- Communication skills (personal and academic)

- An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
- Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to understand and analysis the important theorem and ideas of Mobile Application Development to solve the realistic problems, particularly related to the computer science and engineering fields.

ICT 4105: Cyber Security

Course Objectives:	The objective of this course is to ● Provide students with knowledge of cyber security principles along with industry-based vendor-neutral IT security knowledge and skills. Students will be introduced to a broad range of cyber security related topics such as infrastructure security, communications network security, cryptography, access control, authentication, external threats and operational and organizational security.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the basic and fundamental risk management principles as it relates to Cyber Security and gather skills and understanding the need to manage risks and recover disasters.	PL01, PL02
	CLO2- To develop an understanding of cyber security policies as well as protocols to implement such policies.	PL02, PL06, PL08
	CLO3- To demonstrate and apply knowledge of current trends in ICT security, particularly those that relate to security protocols and policy, cryptography, malware, digital forensics and legal evidence.	PL03, PL07
	CLO4- To apply skills in the identification of security threats, implementation of secure system properties, security testing, and incident response know basic and fundamental risk management principles as it relates to Cyber Security and gather skills and understanding the need to manage risks and recover disasters.	PL06, PL011, PL012

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	1. Significance of Information in Our Life 2. Information System and Security 3. Importance of Information Protection 4. The Evolution of Information Security 5. Security Services and Mechanisms 6. How to Build a Security Program	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1, CLO2, CLO4	1. Cyber World Vs. Real World 2. What is Cyber Security? 3. The State of Security Today 4. Why Security Matters Today? 5. Information Security Vs. Cyber Security 6. Hacker Vs. Cracker	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1, CLO2, CLO3, CLO4	1. The Interrelated Components of the Computing Environment 2. Cyber Security Models: a) the CIA Triad b) the Star Model c) the Parkerian Hexad 3. Variations on a Theme: a) Computer Security b) Information Security c) Network Security d) Information Assurance	Lectures Independent Research Class Discussions	Exercise Report Presentation
CLO1, CLO4	1. What is Cyber Crime? 2. Risks of Conducting a Cyber Crime 3. Categories of Cyber Crime 4. Types of Malware: Virus, Worm, Spyware, Trojan 5. Phishing and Identity Theft 6. Cyber Harassment and Cyberstalking	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation

CLO1, CLO4	1. Analogy for Understanding Vulnerability, Attack and Threat 2. Cyber Attack and its Classification 3. Motivation of Cyber Attacker 4. Penetration Testing 5. Cyber Threats 6. Consequences of Attacks, Threats and Vulnerabilities	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO4	1. What is Risk? 2. Risks Involved in Cyberspace 3. Types of Risk 4. Risk Strategies 5. The Risk Management Framework (RMF)	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1	1. What does Disaster Recovery and Business Continuity Mean? 2. Creating Disaster Recovery Plan and Procedures 3. Challenges of Disaster Recovery Plan 4. Traditional Disaster Recovery 5. Business Continuity Planning 6. Back Plan and Policy	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1, CLO3	1. Cyber Surveillance 2. Issues Making Cyber Security Difficult 3. Security Policies 4. Security Standards 5. Security Procedures 6. Security Guidelines	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1, CLO2, CLO3, CLO4	1. State of Security Today 2. Issues Making Cyber Security Difficult 3. Policy Development: National ICT Act and ICT Policy 4. Role of Government In Securing Cyberspace 5. Role of Private Sectors In Securing Cyberspace 6. International Laws in Securing Cyberspace.	Lectures Independent Research Class Presentation	Exercise Report Presentation

CLO1, CLO4	1. Prevention Tips 2. Mobile Protection 3. Social Network Security 4. Preventing Cybercrime Using Software 5. Defense Against Hackers 6. Ways to Prevent Cyber Crime Targeted at Individual, Family, Personal Property, Business, Organization, Government Agency	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1, CLO2, CLO3, CLO4	1. Identification, Verification and Authentication 2. Types of Authentication 3. Authentication Factors 4. Biometric Authentication 5. Authorization 6. Access Control Lists	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1, CLO2, CLO3, CLO4	1. Demystifying Computer/Cyber Crime 2. Understanding IP Addresses 3. The Explosion of Networking 4. The Explosion of Wireless Networks 5. Interpersonal Communication	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1, CLO2, CLO3, CLO4	1. Compliance with Standards and Controls 2. Information Security Standards 3. Security Standard Example 4. Regulations Affecting Information Security Professionals 5. Laws Affecting Information Security Professionals 6. Certification and Accreditation (C&A)	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO3	1. Security Education, Training, and Awareness 2. Human Factors in Security 3. Developing and Implementing a Security Training Plan 4. Cross-Domain Training (IT and Other Security Domains)	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation

CLO1, CLO2, CLO4	1. Ethics Vs. Law 2. Ethical Issues in Cyberspace: Netiquette 3. Cyber Law: Bangladesh Perspective 4. Digital Security Act 2018 of Bangladesh 5. Intellectual Property Right 6. Copyright and Patent Law	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1, CLO2, CLO3, CLO4	1. Past, Present, and Future Cyber Environment 2. Key Future Uncertainties 3. Possible Future Scenarios 4. A New Lexicon for Cyber Security Terms 5. Current and Future Cyber Security Research 6. Keeping Secure in the Long Term	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√			√
CLO3	√	√			√
CLO4	√	√			√

Suggested Texts:

- Information Security: The Complete Reference, 2nd Edition by Mark Rhodes, Ousley.
- Handbook of Applied Cryptography, 1st Edition by Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone.
- Information Security Management: Concepts and Practice, 1st Edition by Bel G. Raggad.
- Cyber Security and Cyber War: What Everyone Needs to Know, 1st Edition by P.W. Singer, Allan Friedman.
- Cyber Security Basics: Protect Your Organization by Applying the Fundamentals, 1st Edition by Don Franke.

Intended Learning Outcomes:

- Knowledge and understanding

- ▢ Understand the principal concepts, major issues, technologies and basic approaches in information security.
- ▢ Gain the basic principles of Cyber Security and know how it affects our daily life.
- ▢ Gain an understanding of terms and terminology commonly used in Cyber Security.
- ▢ Know how vulnerabilities occur and how to limit them.
- ▢ Gain a fundamental understanding of what an attack/threat is and how to identify and prevent them from occurring.
- ▢ Gather skills and understanding needed to manage risks and recover disasters.
- Cognitive skills (thinking and analysis)
 - ▢ Be able to identify differences in regulatory and legal environments as it relates to cyber security.
 - ▢ Be able to develop an understanding of cyber security policies as well as protocols to implement such policies.
- Communication skills (personal and academic)
 - ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - ▢ Be able to demonstrate and apply knowledge of current trends in ICT security, particularly those that relate to security protocols and policy, cryptography, malware, digital forensics and legal evidence.

IT 4107: Cryptography and Network Security

Course Objectives:	The objective of this course is to • Acquire the fundamentals idea of Cryptography. • Acquire knowledge on standard algorithms used to provide confidentiality, integrity and authenticity. • Understand the various key distribution and management schemes. • Understand how to deploy encryption techniques to secure data in transit across data networks. • Design security applications in the field of Information technology.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the terms and terminology commonly used in Cryptography and Network Security and possess a fundamental knowledge of Network Security and know how to provide the security of a network.	PLO1, PLO5, PLO7

	CLO2- To understand what a threat, attack and vulnerability are and know how to identify and prevent them from occurring.	PLO2, PLO3
	CLO3- To demonstrate and apply knowledge of current trends in ICT security, particularly those that relate to security protocols and policy, cryptography, malware, digital forensics, and legal evidence.	PLO3, PLO5, PLO10
	CLO4- To apply skills in the identification of security threats, implementation of secure system properties, security testing, and incident response and investigate emerging security trends and their application to professional practice.	PLO3, PLO4, PLO5, PLO7

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	1) Significance of Information and its Protection in Our Life 2) OSI Security Architecture 3) Security Services, Goals and Mechanisms 4) Attacks, Threats, Vulnerability, and Control	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO1	1) Why Mathematics is Necessary in Cryptography and Network Security? 2) Integer and Binary Operations 3) Modular Arithmetic 4) Additive Inverse, Multiplicative Inverse and Extended Euclidean Algorithm 5) Euler's Totient Function 6) Prime Factorization and Primality Testing 7) Matrix and its Multiplicative Inverse	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	1) General Idea Behind Cryptography and Steganography 2) Basic Terminologies Related to Them 3) Various Cryptographic and Steganographic Techniques 4) Symmetric-key Vs. Asymmetric-key Cryptography 5) Traditional Vs. Modern Ciphers: a) Additive, Multiplicative, Affine, Auto-key, Playfair, Vigenere, Key-less and Keyed-ciphers b) DES, AES, RSA, Rabin, and Elliptic Curve Cryptosystems 6) Three Pass Protocol	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO3	1) Hash Function and its Importance in Network Security 2) Popular Hash Functions and their Functions 3) Desirable Properties of Hash Function 4) Hash Function Vs. MAC 5) Digital Signature Vs. Conventional Signature Vs. Cryptosystem 6) Processes of Signing and Verification 7) Services Provided by Digital Signature	Lectures Independent Research Class Discussions	Exercise Report Presentation
CLO3, CLO4	1) Identification, Verification, Authentication, and Authorization 2) Message Authentication Vs. Entity Authentication 3) Authentication Factors 4) Providing Authentication by Hash and MAC 5) Various Schemes of Knowledge-based and Challenge-response Authentication 6) Biometric Authentication	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO2, CLO3	1) Challenges in Symmetric-key and Public-key Distribution 2) Need for KDC 3) Various Protocols for Creating Session Key 4) Kerberos as a KDC and an authentication protocol. 5) Symmetric-key agreement protocols	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
	6) Need for digital certificates and certification authorities for public key distribution 7) Idea of a Public-Key Infrastructure (PKI)		
CLO2, CLO3	1) Need for Web Security 2) Available Tools to Achieve Website Security 3) Providing Web Security by SSL and SET Protocols 4) Creating Dual Signature 5) Virtual Private Network and Tunneling	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO3, CLO4	1) What does Access Control Mean? 2) Access Control Mechanisms: 3) Need for a Firewall, its Characteristics and Benefits, Firewall Techniques and Types, Implementing a Firewall 4) IDS Vs. IDPS, Requirement Fulfilled by an IDS, Architecture of an IDS	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation
CLO3, CLO4	1) General Structure of an E-mail Application Program 2) Providing E-mail Security by PGP 3) MIME and S/MIME and their Applications 4) Structure of Messages Exchanged in PGP and S/MIME 5) Architecture of IP Security and its Application	Lecture Discussion Group Assignment Demonstration	Exercise Report Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√			√	√
CLO3	√			√	√
CLO4	√	√			√

Suggested Texts:

- Cryptography and Network Security, 2nd Edition by Behrouz A Forouzan, Publisher: Tata McGraw Hill Education Pvt. Ltd., New Delhi, 2007.
- Cryptography and Network Security, 4th Edition by William Stallings., Publisher: Prentice Hall, New Delhi, 2006.
- Cryptography and Network Security, 2nd Edition by Atul Kahate, Publisher: Tata McGraw Hill Education Pvt. Ltd., New Delhi, 2009.
- Applied Cryptography, 2nd Edition by Bruce Schneier, Publisher: John Wiley & Sons, New York, 1996.
- Mastering Network Security, Bk & Cd-Rom Edition by Chris Brenton, Publisher: BPB Publication, New Delhi, 2002.
- Network Security, 2nd Edition by Steven L Shaffer, Alan R Simon, Publisher: AP Professional, New York, 2001.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the principal concepts, major issues, technologies and basic approaches in cryptography and Network security.
 - Gain the basic principles of Network Security and know how it affects our daily life.
 - Gain an understanding of terms and terminology commonly used in Network Security.
 - Know how vulnerabilities occur and how to limit them.
 - Gain a fundamental understanding of what an attack / threat is, and how to identify and prevent them from occurring.
 - Evaluate tools and technologies for use in protecting the network and individual network systems
- Cognitive skills (thinking and analysis)
 - Be able to implement security plan and monitor solutions.
 - Be able to monitor and evaluate audit logs and set administrator alerts.
 - Be able to respond to any breach of security and adjust organizational security plan accordingly.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - Be able to demonstrate and apply knowledge of current trends in ICT security, particularly those that relate to security protocols and policy, cryptography, malware, digital forensics, and legal evidence.

ICT 4109: Big Data Analytics and Application

Course Objectives:	The objective of this course is to • Understand the Big Data Platform and its Use cases. • Provide an overview of Apache Hadoop. • Provide HDFS Concepts and Interfacing with HDFS. • Understand Map Reduce Jobs. • Provide hands on Hadoop Eco System. • Apply analytics on Structured, Unstructured Data. • Exposure to Data Analytics with R.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To identify Big Data and its Business Implications.	PL01
	CLO2- To list the components of Hadoop and Hadoop Eco-System.	PL01, PL02
	CLO3- To access and Process Data on Distributed File System.	PL02, PL05
	CLO4- To manage Job Execution in Hadoop Environment.	PL04, PL05
	CLO5- To develop Big Data Solutions using Hadoop Eco System.	PL03
	CLO6- To analyze Infosphere Big Insights Big Data Recommendations.	PL02
	CLO7- To apply Machine Learning Techniques using R.	PL03, PL05

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2, CLO4, CLO5	Types of Digital Data, Introduction to Big Data, Big Data Analytics, History of Hadoop, Apache Hadoop, Analyzing Data with Unix tools, Analyzing Data with Hadoop, Hadoop Streaming, Hadoop Echo System, IBM Big Data Strategy, Introduction to Infosphere Big Insights and Big Sheets.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise

CLO2, CLO3, CLO4, CLO5	The Design of HDFS, HDFS Concepts, Command Line Interface, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, Hadoop I/O: Compression, Serialization, Avro and File-Based Data structures.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO2, CLO3, CLO4, CLO5	Anatomy of a Map Reduce Job Run, Failures, Job Scheduling, Shuffle and Sort, Task Execution, Map Reduce Types and Formats, Map Reduce Features.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO5, CLO6	Pig: Introduction to PIG, Execution Modes of Pig, Comparison of Pig with Databases, Grunt, Pig Latin, User Defined Functions, Data Processing operators. Hive: Hive Shell, Hive Services, Hive Metastore, Comparison with Traditional Databases, HiveQL, Tables, Querying Data and User Defined Functions. Hbase: HBasics, Concepts, Clients, Example, Hbase Versus RDBMS. Big SQL : Introduction	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO7	Machine Learning: Introduction, Supervised Learning, Unsupervised Learning, Collaborative Filtering. Big Data Analytics with BigR.	Lecture Group Assignment Problem Based Learning (PBL)	Report Exercise Presentation
CLO7	Basic Neural Network and Tensor Flow. Neural Networks and Deep Learning are emerging as the highest precision tools for many large scale classification and pattern recognition problems. We will learn how to use Tensor Flow both on GPU and CPU machines.	Lecture Discussion Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment	Term Final
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	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√
CLO5	√	√	√	√	√
CLO6	√	√	√	√	√
CLO7	√	√	√	√	√

Suggested Texts:

- Big Data Analytics: Systems, Algorithms, Applications, 1st Edition by Prabhu, C.S.R., Sreevallabh Chivukula, A., Mogadala, A., Ghosh, R., Livingston, L.M.J.
- Big Data for Beginners: Understanding SMART Big Data, Data Mining & Data Analytics For improved Business Performance, Life Decisions & More!, 1st Edition by Vince Reynolds.
- A Hands-On Introduction to Data Science, 1st Edition by Chirag Shah.
- Guide to Big data Hadoop Distributed File System A book for beginners/intermediate, 1st Edition by Kartikeya Mishra.
- The Ultimate Guide to Data Analytics, Data Mining, Data Warehousing, Data Visualization, Regression Analysis, Database Querying, Big Data for Business and Machine Learning for Beginners, 1st Edition by Herbert Jones.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of algorithm analysis and design.
 - Understand the fundamental steps of designing and optimizing algorithms.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any problem and derive its solution.
 - Be able to design and optimize algorithms to solve particular complex problems effectively.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)

- Be able to identify the complex and real life problems to solve using the effective understanding of the design and analysis of algorithms.

ICT 4111: Optical Fiber Communication

Course Objectives:	The objective of this course is to - Analyze the operation of LEDs, laser diodes, and PIN photo detectors (spectral properties, bandwidth, and circuits) and apply in optical systems. - Explain the principles of compare and contrast single- and multi-mode optical fiber characteristics. - Analyze and design optical communication and fiber optic sensor systems. - Locate, read and discuss current technical literature dealing with optical fiber systems.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the components of fiber optic communication.	PLO1
	CLO2- To analyze signal degradation and loss measurement.	PLO2, PLO4
	CLO3- To evaluate the performance of optical network.	PLO4, PLO5

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction to Optical Fibers: Evolution of fiber optic system, Element of an Optical Fiber Transmission link, Total internal reflection, Acceptance angle, Numerical aperture, Skew rays, Ray Optics, Optical Fiber Modes and Configurations , Mode theory of Circular Wave guides, Overview of Modes-Key Modal concepts, Linearly Polarized Modes, Single Mode Fibers-Graded Index fiber structure.	Lecture Assignment	Short Answer Exercise

CLO2	Signal Degradation of Optical Fibers: Attenuation – Absorption losses, Scattering losses, Bending Losses, Core and Cladding losses, Signal Distortion in Optical Wave guides-Information Capacity determination -Group Delay-Material Dispersion, Wave guide Dispersion, Signal distortion in SM fibers-Polarization, Mode dispersion, Intermodal dispersion, Pulse Broadening in GI fibers-Mode Coupling -Design Optimization of SM fibers-RI profile and cut-off wavelength.	Lecture Assignment Demonstrate	Short Answer Problem Solving Exercise
CLO1, CLO3	Fiber Optical Sources and Coupling: Direct and indirect Band gap materials-LED structures -Light source materials -Quantum efficiency and LED power, Modulation of a LED, lasers Diodes-Modes and Threshold condition -Rate equations -External Quantum efficiency -Resonant frequencies -Laser Diodes, Temperature effects, Introduction to Quantum laser, Fiber amplifiers- Power Launching and coupling, Lencing schemes, Fiber -to- Fiber joints, Fiber splicing-Signal to Noise ratio , Detector response time.	Lecture Assignment	Short Answer Case study Discussion
CLO1	Principles of Fiber Optics Communications Analog & Digital Transmission, Digital Coding, Electrical & Optical Bandwidth, Dispersion Effects , Bandwidth and Data Rate Dynamic Range , Noise and Bit Error Rate, optical modulation	Lecture Assignment	Short Answer Presentation
CLO1, CLO3	Fiber Optic Receiver and Measurement: Fundamental receiver operation, Pre amplifiers, Error source, Receiver Configuration, Probability of Error, Quantum limit. Fiber Attenuation measurements Dispersion measurements, Fiber Refractive index profile measurements, Fiber cut- off Wave length Measurements, Fiber Numerical Aperture Measurements, Fiber diameter measurements.	Lecture Assignment	Short Answer Assignment MCQ

CLO1, CLO3	Optical Networks and System Transmission Basic Networks – SONET / SDH – Broadcast and select WDM Networks, Wavelength Routed Networks, Nonlinear effects on Network performance, Link Power budget, Rise time budget, Noise Effects on System Performance-Operational Principles of WDM Performance of WDM and EDFA, system Solutions, Optical CDMA, Ultra High Capacity Networks.	Lecture Assignment Project	Short Answer Group Exercise MCQ
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Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√			√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√

Suggested Texts:

- Understanding Fiber Optics, 4th Edition by Jeff Hecht, Publisher: Prentice Hall, 2003.
- Introduction to Optical Fiber Communication Systems, 1st Edition by William B. Jones. Jr., Holt, Publisher: Rinheart and Winston, Inc. 1988.
- Optical Fiber Communication, 2nd Edition by Gerd Keiser, Publisher: McGraw Hill, 1991.
- Optical Fiber Communications – Principles and Practice, 2nd Edition by John M. Senior, Publisher: Prentice Hall, 1992.
- Fiber Optic Communication, 4th Edition by Palaise.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic of optical communications.
 - Understand signal degradation, optical fiber modes, receiver operation and measurement.
 - Understand the protocol techniques and operation principle of optical network.
- Cognitive skills (thinking and analysis)
 - Be able to apply the skills within fiber optic communications and optical networks.
 - Be able to apply network algorithm in practice.
- Communication skills (personal and academic)

- ▮ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
- ▮ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.

ICT 4113: IT Professional and Ethics

Course Objectives:	The objective of this course is to ● Provide necessary theoretical knowledge and skills to critically analyze ethical dilemmas, evaluate alternatives and communicate to all stakeholders with the decision taken in a convincing manner. ● Understand ethical issues as the students pertain to professional and personal identity. ● Learn to consider thyself and the world around from these basic ethical positions.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To demonstrate an understanding of codes of ethics in Information and Communication Technology.	PLO8
	CLO2- To identify social responsibilities of computer professionals and information technologists.	PLO6
	CLO3- To evaluate solutions to ethical problems and make a recommendation for the best course of action.	PLO4, PLO7, PLO8
	CLO4- To design structured program for complex computational applications	PLO3, PLO11

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction: Understanding how computers impact on society, professional ethics in ICT, and practical codes of conduct as an expression of these professional ethics.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise

CLO2	Professionalism in computing: Professional bodies in IT and related fields, Professional bodies trade unions, and other organizations, characteristics and functions of a professional body, social responsibilities of IT professionals to the public at large, fellow members, clients; Trust, honesty and integrity.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO1, CLO3	Ethics in computing: Philosophy, ethics, and applied ethics, Factors affecting in making ethical decisions, Theories in ethics, Common computer ethics fallacies such as Computer game fallacy, law-abiding citizen fallacy, shatterproof fallacy, candy-from-a –baby fallacy, hacker's fallacy, free information fallacy, Code of conducts in IT professional bodies	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO1, CLO3, CLO4	Legal issues in ethical Analysis of Information Systems: Intellectual property, Software licensing issues, Computer evidence, Online transactions, Electronic signatures, Online communities, Computer misuse and frauds.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO1, CLO3, CLO4	Social issues of privacy: Personal information, Data protection principles, Knowledge and consent in the Internet, Privacy enhancing and invasive tools/methods	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation
CLO1, CLO3, CLO4	Conflicts of Interest: Contracts (MOU, NDA, User agreements), Liabilities, Warranty and maintenance, Offline dispute resolution methods, Issues in resolving dispute relating online transactions.	Lecture Discussion Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	

CLO1	√	√		√	√
CLO2	√	√		√	√
CLO3	√	√		√	√
CLO4	√	√		√	√

Suggested Texts:

- George Reynolds, Ethics in Information Technology, 6th Edition by Thomson Course Technology.
- Ethics for the Professions, 1st Edition by John Rowan & Samuel Zinaich, Jnr. Wadsworth.
- A Modern Legal Ethics: Adversary Advocacy in a Democratic Age, 1st Edition by Daniel Markovits.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand, identify and apply different ethical philosophies, frameworks, and methodologies for professionals
 - Analyze the local and global impact of computing on individuals, organizations and society.
- Cognitive skills (thinking and analysis)
 - Be able to identify and relate appropriate privacy measures and their management for the computing environment.
 - Be able to reflect a professional behavior particularly in the fields of IS and software development.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - An awareness of ethical, social and cultural issues within a global context and their importance in the exercise of professional skills and responsibilities.

ICT 4115: IT Project and Service Management

Course Objectives:	The objective of this course is to • Understand the concepts of Project Management for planning in execution of projects. • Understand the feasibility analysis in Project Management and network analysis tools for cost and time estimation. • Analyze, apply and appreciate contemporary project management tools and methodologies. • Understand the service management processes. • Apply service management frameworks for enhancing service firms' competitiveness.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand project characteristics and various stages of a project.	PLO1
	CLO2- To analyze the learning and understand techniques for Project planning, scheduling and execution Control.	PLO2, PLO6
	CLO3- To apply the risk management plan and analyze the role of stakeholders.	PLO4, PLO10
	CLO4- To analyze and select the most suitable methods and framework for project and service management respectively.	PLO11, PLO12

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Stages of a project: Feasibility studies and the establishment of a business case for a project, Requirements elicitation, analysis and verification: purpose and methods, establishing project objectives, goals and measures of success, Stages of a development project. Adapting the development life cycle to projects where off-the-shelf packages to be installed.	Lecture Discussion	Short Questions

CLO2	Project planning and estimating: Use of product and work breakdown structures (PBS and WBS), Use of (activity on node) precedence plans and network analysis; Critical path analysis, Gantt charts, resource allocation, including the identification of resource types and the resolution of resource clashes, Agile approaches to planning: the use of time-boxing; product and sprint backlogs; prioritization of increments (e.g. using MoSCoW rules), methods, advantages and disadvantages and relative accuracy of different estimating techniques.	Lecture Discussion Case Study	Essay Type Answer MCQ Exercise
CLO2	Progress monitoring, project control, and reporting: What to monitor and why: key project metrics related to time/progress (e.g. planned and actual activity duration) costs (e.g. planned and actual effort and other costs) scope/size of functionality and quality (e.g. Earned value analysis: planned and earned value, actual costs; cost and schedule performance indicators, including their graphical representation, Assessment of implications and impact on the project of deviations and changes to project plan.	Lecture Discussion Case Study	Essay Type Answer Exercise Presentation
CLO3	Risk management: Risk identification: types of risk, risk checklists, Risk prioritization: assessment of likelihood and impact of risk; qualitative and quantitative methods of assessing risk exposure, Risk management tactics, including risk avoidance, risk transfer, risk reduction, risk mitigation and contingency planning, Cost benefit analysis of planned risk reduction actions, risk reduction leverage, Risk registers.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO4.	Service management concepts and frameworks: Services and service management. Customer-facing services. Resource-facing services. Identifying IT services. Service quality SERVQUAL model The service lifecycle. Processes. Functions. Roles Service strategy. Service portfolio. Service catalog management. Service level agreements. Introduction to ITIL v3, ISO 20000 and FitSM.	Lecture Discussion	Short Questions

CLO4	Specification of services and business models: Business models with a CANVAS template. Market understanding. The empathy map. Service definition and service portfolio specification. Understanding the organization of the entity that will deliver the services.	Lecture Discussion Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO2	Planning and delivery processes: Service portfolio management, Service level management, Service reporting, Service availability and continuity management, Capacity management, Information security management, Customer relationship management and Supplier relationship management.	Lecture Group Assignment Demonstration	Essay Type Answer Report Project Presentation
CLO4	Service management tools: Tool assessment framework, analysis of specific ITSM tools Selection of the tool that best fits a given SMS.	Demonstration	Project Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√		√	√
CLO2	√	√		√	√
CLO3	√	√		√	√
CLO4	√	√		√	√

Suggested Texts:

- Software Project Management, 5th Edition by Bob Hughes and Mike Cotterell.
- The Practical Guide to World-Class IT Service Management, 1st Edition by Kevin J Smith.
- Project Management for IT-related Projects, 3rd Edition by Bob Hughes.
- Effective Project Management: Traditional, Agile, Extreme, 7th Edition by Robert K. Wysocki.
- Service Management: Operations, Strategy, and Information, 7th Edition by James A Fitzsimmons and Mona J. Fitzsimmons.

Intended Learning Outcomes:

- Knowledge and understanding
 - ▯ Understand the basic concepts of project management.
 - ▯ Understand the fundamental steps of IT service management.
- Cognitive skills (thinking and analysis)
 - ▯ Be able to understand and analyze various stages of a project.
 - ▯ Be able to identify IT services as a means to provide functionality and value to customers in the context of specific case studies.
- Communication skills (personal and academic)
 - ▯ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▯ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - ▯ Be able to apply techniques for Project planning, scheduling and execution control.
 - ▯ Be able to select the appropriate tools to support a given designed service management solution.

ICT 4117: IT Risk Management

Course Objectives:	The objective of this course is to ● To describe a management view of cyber security. ● To illustrate the general principles of risk analysis. ● To explain the fundamentals of business continuity management and how it can help address security incidents. ● To analyze security components within organizational context: identity and access management, data protection, security operations, etc. ● To understand third party risk assessments.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To explain various Information security threat and controls for it.	PL01, PL02
	CLO2- To analyze security incidents and design countermeasures.	PL03, PL04

	CLO3- To apply cloud and mobile security models.	PLO3, PLO5
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Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	Business priorities and secure development: Business priorities and information security risks, Integrating security into software development process, Security training and awareness.	Lecture Discussion	Short Question MCQ
CLO1, CLO2	Cyber threats and information security risks: Cyber Threats, including motivation, trends, and threat monitoring, general principles of risk analysis.	Lecture Discussion Case Study	Essay Type Answer MCQ Exercise
CLO1, CLO2	IT risk analysis, risk management, and security risk metrics: Major steps of risk analysis (probability, impact, prioritization, etc.), Approaches to managing risks (reduction, mitigation transfer, and acceptance), Managing risk with metrics.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Assignment
CLO1, CLO2	Identity and access management (IAM) and Data protection: What is it? Identity and access management approach, IAM service components (manage access, enforce access, report access), IAM reference architecture and strategy, Identifying critical assets / data classification (data elements, PII, process), Data loss prevention (data in motion, data at rest, and data at endpoint), Data privacy (privacy laws, data flow, data inventory, integrated framework).	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO1, CLO2	Incident response planning and business continuity: Security incident response planning (prepare, identify, assess, contain, investigate, resolve, learn), Business Continuity Planning: making sure the organization can continue functioning after a security incident.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Case study Exercise Presentation

CLO1, CLO2	Third party risk management: Third party risk management, Third party risk assessment.	Lecture Discussion	Essay Type Answer
CLO1, CLO2	Security monitoring: Security monitoring overview and how it can help manage risk, Overview of security monitoring data, logging requirements and monitoring tools.	Lecture Group Assignment Demonstration	Essay Type Answer Exercise Presentation
CLO3	Cloud and Mobile security models: Cloud security concerns and risks, which applications and data can be migrated to the Cloud? Impact to traditional security management, Approaches to securing mobile devices, Emerging challenges, including Bring your own device (BYOD), Overview of managed security services, Evaluation and assurance and information security audits.	Lecture Group Assignment Project	Essay Type Answer Report

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√	√	√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√

Suggested Texts:

- Measuring and managing information risk: A FAIR approach, 1st Edition by Jack Freund, Jack Jones.
- The Failure of Risk Management: Why It's Broken and How to Fix It, 2nd Edition by Douglas W. Hubbard.
- How to Measure Anything in Cyber Security Risk, 1st Edition by Douglas W. Hubbard and Richard Seiersen.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of security components and risk analysis.
 - Understand the fundamental steps of security threats and risk management.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analyze risk incident and prepare response.
 - Be able to design and optimize mobile and cloud security models.

- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to detect cyber threats and information security risks and mitigate it through proper response planning and method.

ICT 4102: Digital Signal Processing Lab

Course Objectives:	The objective of this course is to ● Practically implement the convolution, correlation, DFT, IDFT, Block convolution, Signal smoothing, filtering of long duration signals, and Spectral analysis of signals.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To analyze the digital systems using z-Transform and Discrete Time Fourier Transform.	PL02, PL05
	CLO2- To identify the difference between fixed point and floating-point digital signal processor.	PL01
	CLO3- To analyze the discrete time signals and systems in the frequency domain using Discrete Fourier Transform and Fast Fourier Transform.	PL04, PL05
	CLO4- To design FIR and IIR filters using a variety of techniques.	PL03

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2	Lab practice on Generate and plot sequences over the indicated interval, complex valued signal and its magnitude, phase, the real part, and the imaginary part in four separate subplots, Decompose into even and odd components, convolution, cross correlation from noise-corrupted-and-shifted version	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report

CLO2, CLO3, CLO4	Lab practice on impulse response from difference equation, system stability checking, DFT evaluation at 501 equispaced points between $[0, \pi]$ and its magnitude, angle, real and imaginary part, Numerically the discrete-time Fourier transform computation at 501 equispaced frequencies between $[0, \pi]$, DFT and its periodicity, conjugate-symmetry property of discrete-time Fourier transform	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report
CLO1, CLO3	Lab practice on the steady-state response of an LTI system, magnitude and the phase response and verification of a 3 rd order lowpass filter, z-transform properties and the z-transform table, z-transform, inverse z-transform, causal sequence and its pole-zero plot, DFS representation of periodic sequence, the discrete -time Fourier transform with its magnitude and phase, parallel form, direct, linear-phase, and cascade form structures, frequency sampling.	Problem Based Learning (PBL) Lecture Group Assignment	Practical Exam Exercise Report

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√	√	√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand discrete time signal processing and characterization of random signals, filter design techniques, and imperfections caused by finite word length.
 - Learn how to estimate the spectra of random signals that are to be processed by a discrete time filter, and to appreciate the performance of a variety of modern and classical spectrum estimation techniques.
 - Learn the theory of modern digital signal processing and digital filter design.
- Cognitive skills (thinking and analysis)
 - Be able to apply filtering technique in complex digital filter design and digital simulation experience.
 - Be able to implement simple low pass, band pass and high pass IIR and FIR digital filters for typical signal processing applications using MATLAB;

- Be able to explain frequency response of discrete time signals and systems.
- Communication skills (personal and academic)
 - Individual assignments after each lab work and group assignments including demonstration and presentation involve receiving clear instructions, designing and writing an effective report and making an effective presentation.
 - A term project (group work) - where students will be assessed in terms of their effectiveness as an individual or leader in the team.
- Practical and subject specific skills (transferable skills)
 - Be able to apply various methods for designing new digital signal processing systems and for continued learning.

ICT 4104: Mobile Application Development Lab

Course Objectives:	The objective of this course is to ● To understand the components and structure of mobile application development frameworks for Android and windows OS based mobiles. ● To understand how to work with various mobile application development frameworks. ● To learn the basic and important design concepts and issues of development of mobile applications. ● To understand the capabilities and limitations of mobile devices. ● Construct real time embedded circuit.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To learn and develop mobile applications using GUI and Layouts.	PLO-1
	CLO2- To explain mobile applications using Event Listener and Databases (SQLite).	PLO-2
	CLO3- To apply RSS Feed, Internal/External Storage, SMS, Multithreading, GPS and Analyze and discover own mobile app for simple needs.	PLO3, PLO5

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy

CLO1	Familiar with an application that uses GUI components, Font and Colors.	Demonstration Case studies Lecture	Practical Exam Report Viva-Voce
CLO1	Application that uses Layout Managers and event listeners. Develop a native calculator application.	Inquiry Based Learning (IBL) Lecture	Practical Exam Report Observation
CLO2, CLO3	Basic graphical primitives on the screen and Databases.	Demonstration Problem Based Learning (PBL)	Practical Exam Viva-Voce Observation
CLO2, CLO3	Use of mobile applications using RSS Feed, Internal/External Storage, SMS, Multithreading and GPS.	Group assignment Lecture Problem Based Learning (PBL)	Observation Report Viva-Voce
CLO2, CLO3	Data to the SD card, creates an alert upon receiving a message, creates alarm clock.	Inquiry Based Learning (IBL) Lecture Exercise	Practical Exam Viva-Voce

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				
CLO2	√	√		√	√
CLO3	√	√	√	√	√

Intended Learning Outcomes:

- Knowledge and understanding
 - Identify the basic knowledge on mobile application environment and technology
 - Explain the concepts and processes of mobile application development
 - Discuss design and development issues specific to mobile applications

- ▯ Design and develop mobile applications, using development tools and environments.
- Cognitive skills (Thinking and analysis)
 - ▯ Be able to identify, formulate, and solve engineering problems.
 - ▯ Be able to development of frameworks for Android and windows OS based mobiles.
 - ▯ Be able learn the basic and important design concepts and issues of development of mobile applications.
 - ▯ Be able to understand the capabilities and limitations of mobile devices.
 - ▯ Be able to analyze and discover own mobile app for simple needs.
- Communication skills (Personal and academic)
 - ▯ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▯ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable skills)
 - ▯ Be able to build and apply necessary aps to build real life program.
 - ▯ Be able to Cross-platform and networking events or websites development skills

4th Year 2nd Semester

ICT 4201: Artificial Intelligence

Course Objectives:	The objective of this course is to • Present an overview of artificial intelligence (AI) principles and approaches. • Develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents: Search, Knowledge representation, inference, logic and learning.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To analyze the various Knowledge organizations and management, Agent, rule base, inference engine, search methods, Matching Techniques.	PL01, PL02

	CLO2- To identify the difference between Natural Language processing, Pattern recognition, expert systems and neural network.	PLO1, PLO2
	CLO3- To analyze the various algorithm related to NLP, pattern recognition, rule base, Genetic algorithm and ANN.	PLO4, PLO5
	CLO4- To apply Fuzzy logic, Generic Algorithm and ANN in many real-life applications and also NLP, perceptron and Robotics in sensor data.	PLO3, PLO6, PLO8

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Fundamentals of AI: What is AI? Its foundation and history Intelligent Agents: Agents and Environments, The Structure of Agents, Nature of the Environment, Solving Problems by Searching, Problem-Solving Agent, various types of searching technique.	Lecture Assignment	Short Answer Exercise
CLO1	Informed Search and Exploration: Informed (Heuristic) Search Strategies, Heuristic Functions, Local Search Algorithms and Optimization Problems, Local Search in Continuous Spaces, Online Search Agents and Unknown Environments. Constraint Satisfaction Problems: Constraint Satisfaction Problems, Backtracking Search for CSPs, Local Search for Constraint Satisfaction Problems. Adversarial Search: Optimal Decisions in Games, Alpha-Beta Pruning, Imperfect, Real-Time Decisions, Games That Include an Element of Chance.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO1	Knowledge and Reasoning: Knowledge-Based Agents, The Wumpus World, Propositional Logic and Predicate Logic, Knowledge Representation.	Lecture Assignment Demonstrate	Short Answer Exercise
CLO1, CLO4	Planning: The Planning Problem, Planning with State-Space Search, Partial-Order Planning, Planning Graphs, Planning with Propositional Logic, Planning and Acting in the Real World.	Lecture Assignment	Short Answer Discussion

CLO1	Learning: Learning from Observations, Knowledge in Learning, Statistical Learning Methods, Reinforcement Learning.	Lecture Assignment	Short Answer Discussion
CLO3, CLO4	Fuzzy Logic Systems: Fuzzy Logic Systems Architecture, Example of a Fuzzy Logic System, Application Areas of Fuzzy Logic Artificial Neural Network: Basic Structure of ANNs, Types of Artificial Neural Networks, Machine Learning in ANNs, Back Propagation Algorithm, Bayesian Networks (BN), Application.	Lecture Assignment Demonstrate	Short Answer Presentation
CLO2, CLO3, CLO4	Natural Language Processing: Components of NLP, Steps in NLP, Implementation Aspects of Syntactic Analysis. Perception: Image Formation, Early Image Processing Operations, Extracting Three-Dimensional Information, Object Recognition Robotics: Robotic Perception, Planning to Move, planning uncertain movements, Moving.	Lecture Assignment Demonstrate	Short Answer Discussion

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√			√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√

Suggested Texts:

- Artificial Intelligence: A Modern Approach, 3rd Edition by Stuart Russell, Peter Norvig.
- Neural Networks and Fuzzy Systems: A Dynamical Systems Approach to Machine Intelligence, 1st Edition by Kosko Bart.
- Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis and Applications, Eastern Economy Edition by G. A. Vijayalakshmi Pai and Sanguthevar Rajasekaran.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand solving problems by searching problem solving agent and various types of searching technique.
 - Learn knowledge-based system, FL and ANN.
 - Learn how to apply NLP and Perception in data.
- Cognitive skills (thinking and analysis)
 - Be able to apply inference engine, fuzzy-genetic algorithm and artificial neural network in designing intelligent agent.
 - Be able implement simple searching method, NLP, rule base systems, genetic algorithm and artificial neural network for various real-life applications, such as robotic application, image analysis, character recognition, pattern analysis, etc. using MATLAB.
 - Be able to incorporate machine intelligence.
- Communication skills (personal and academic)
 - Individual assignments including demonstration and presentation involve receiving clear instructions, designing and making an effective presentation.
 - A term project (group work) - where students will be assessed in terms of their effectiveness as an individual or leader in the team.
- Practical and subject specific skills (transferable skills)
 - Be able to apply various methods for designing new digital signal processing systems and for continued learning.

ICT 4203: Parallel and Distributed System

Course Objectives:	The objective of this course is to ● Provide an understanding of the principles of parallel and distributed system along with their applications in real world. ● Understand their architecture both in software and hardware platform. ● Learn challenges that the system faces while implementation. ● Learn security and failure handling and finally the importance of consistency and synchronization in distributed system.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To distinguish the features of parallel and distributed system.	PLO1, PLO2

	CLO2- To demonstrate knowledge of the core architectural aspects of distributed systems, detail learning of the main underlying components of distributed systems (such as RPC, RMC, DCE RMC file systems).	PLO4, PLO5
	CLO3- To use and apply important methods in distributed systems to support scalability and fault tolerance.	PLO3, PLO7

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction: Why use parallel and distributed systems? Why not use them?, Speedup and Amdahl's Law.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO2	Hardware and Software architectures: multiprocessors (shared memory), networks of workstations (distributed memory), clusters, threads and shared memory, processes and message passing, distributed shared memory (DSM), distributed shared data (DSD).	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO2	Parallel Algorithms: Concurrency and synchronization, Data and work partitioning, Common parallelization strategies, Granularity, Load balancing, Examples: parallel search, parallel sorting, etc.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2	Distributed Systems: System Architecture, Communication, Mid-session Recess, Replication & Consistency, Distributed Shared Memory, Synchronization & Coordination, Middleware, Fault Tolerance, Security, Naming, Distributed File Systems.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation

CLO2, CLO3	Shared-Memory Programming: Threads, Pthreads, Locks and semaphores. Distributed-Memory Programming: Message Passing, MPI, PVM, Other Parallel Programming Systems: Trademarks: Distributed shared memory.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2, CLO3	System Configurations: Aurora: Scoped behavior and abstract data types, Enterprise: Process templates, Protocols for DSM systems, Impact of network protocols (TCP/IP, UDP/IP, bulk-data transfer, etc.), System area networks (SAN) (e.g., Myrinet).	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√			√
CLO2	√	√			√
CLO3	√	√			√

Suggested Texts:

- Parallel Programming: Techniques and Applications Using Networked Workstations and Parallel Computers, 2nd Edition by B. Wilkinson and M. Allen, Publisher: Prentice Hall.
- Distributed Systems: Principles and Paradigms, 2nd Edition by Andrew S. Tanenbaum & Maarten van Steen, Publisher: Pearson Prentice Hall.
- Distributed Systems: Concepts and Design, 3rd Edition by George Coulouris, Jean Dollimore & Tim Kindberg, Publisher: Addison-Wesley.
- Distributed Operating Systems, Eastern Economy Edition by Pradeep K. Sinha, Publisher: IEEE Press.
- Introduction to Parallel Computing, 2nd Edition by Kumar, Grama, Gupta and Karypis, Publisher: Benjamin Cummings Publishing Co.
- Using MPI: Portable Parallel Programming with the Message-Passing Interface, 3rd Edition by William Gropp, Ewing Lusk, and Anthony Skjellum.

Intended Learning Outcomes:

- Knowledge and understanding

- Understand the basic concepts of distributed and parallel systems.
- Understand the fundamental steps of designing and optimizing system.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any problem and derive its solution.
 - Be able to design and optimize algorithms to solve particular complex problems effectively.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to identify the complex and real life problems to solve using effective understanding of the design and analysis of algorithms.

ICT 4205: Wireless and Cellular Networks

Course Objectives:	The objective of this course is to ● Build an understanding of the fundamental concepts of cellular network. ● Familiarize the student with the basic taxonomy and terminology of the wireless networking. ● Introduce the student to advanced networking concepts, preparing the student for entry Advanced courses in computer networking. ● Gain expertise in some specific areas of networking such as the design and maintenance of individual networks.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To design and analyze WLAN, home RF and WSN for an educational/research institute or office.	PL02, PL03
	CLO2- To implement MANET or WSN in any emergency situation (lack of infrastructure) and learn features of 5G mobile communications.	PL03, PL09
	CLO3- To understand protocol and access technique 4G network and able to analyze the built in functions of MATLAB-18 for LTE to simulate, such network to measure the performance under fading channel.	PL04, PL05, PL06

	CLO4- To understand designing and optimization of cellular network.	PLO7, PLO9
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Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Classification of wireless network: Home RF and Bluetooth, IEEE 802.11 family, protocol stack of IEEE 802.11, frame format of Wi-Fi, CSMA/CA of MAC sublayer, binary exponential backoff algorithm and flow chart	Lecture Assignment	Short Answer Exercise
CLO1, CLO2	Mobile ad-hoc network (MANET): Destination-Sequenced Distance Vector (DSDV) protocol, Cluster-Head Gateway Switch Routing Protocol, Wireless Sensor Network (WSN), Sensor Node Structure, LEACH and DEEP Clustering Protocol	Lecture Assignment Demonstrate	Essay Type Answer Short Answer Identification Exercise
CLO1	Wide Area Network: concept of cell and cell cluster, co-site, adjacent channel and co-channel interferences of WAN, handover and roaming, channel allocation scheme	Lecture Assignment	Essay Type Answer Short Answer Discussion
CLO4	Advanced Mobile Communication: Vision of IMT 2000, principle of CDMA/WCDMA, architecture of 3G mobile (UMTS) communication, satellite based mobile communications	Lecture Assignment	Essay Type Answer Short Answer Exercise Report Presentation
CLO1, CLO4	WiMAX: Development of IEEE 802.16, adaptive modulation and channel coding of WiMAX, BW allocation algorithms, Wi-Fi and WiMAX integrated network, 802.16 Protocol Stack, a security sublayer, MAC common part sublayer and Service Specific Convergence Sub-layer, TDD and FDD operation	Lecture Assignment	Essay Type Answer Short Answer Exercise Report Presentation

CLO3, CLO4	4G Technology: Development of 4G long-term evolution (LTE), femtocell deployment , OFDMA-based physical layer access and MIMO of LTE, architecture of LTE, LTE frame structure and RB	Lecture Assignment	Essay Type Answer Short Answer Exercise Report Presentation
CLO1, CLO4	5G Wireless Systems: Cognitive radio network, cooperative spectrum sensing, objectives of 5G mobile communication, activities of METIS, 5G Challenges, Massive MIMO, D2D and M2M communications, Moving Networks and Ultra-dense Networks of 5G.	Lecture Assignment	Short Answer Exercise Report Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√			√
CLO2	√	√			√
CLO3	√	√			√
CLO4	√	√			√

Suggested Texts:

- Wireless communication, 2nd Edition by Rappaport.
- Wireless & Mobile Network Architectures, 1st Edition by Yi bing Lin.
- An Introduction to LTE: LTE, LTE-Advanced, SAE and 4G Mobile Communications, 2nd Edition by Christopher Cox.
- Computer and Communication Networks, 2nd Edition by Nader F. Mir.
- 5G Wireless Systems: Simulation and Evaluation Techniques (Wireless Networks), 1st Edition by Yang Yang, Jing Xu, Guang Shi, Cheng-Xiang Wang.
- Cognitive Radio Communications and Networks, 1st Edition by Alexander M. Wyglinski, Maziar Nekovee, Y. Thomas Hou.
- Wireless Communications, 1st Edition by Andrea Goldsmith
- Wireless and Mobile Network Architecture, Wiley Student Edition by Yi-Bing Lin and Imrich chlamtac.
- Wireless Communication and Networks, 2nd Edition by William Stallings.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand and apply knowledge of mathematics, science and engineering.
 - Understand and design a system, component or process to meet desired needs within realistic constraints.
- Cognitive skills (thinking and analysis)
 - Keep himself updated on latest wireless technologies and trends in the communication field.
 - Be able to describe and explain radio standards and communication protocols on the link and networking layers for wireless personal area networks and inter-working with wireless local area networks and cellular networks.
 - Understand the transmission of voice and data through various networks.
 - Be familiar with architectures, functions and performance of wireless sensor networks systems and platforms.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.

ICT 4102: Artificial Intelligence Lab

Course Objectives:	The objective of this course is to ● Provide understanding of some fundamental issues and algorithms in artificial intelligence (AI). ● Provide some fundamental tools and algorithms required to produce ● Exhibit limited human-like abilities, particularly in the form of Problem Solving by search, representing and reasoning with knowledge, planning, and learning.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To analyze the various knowledge organizations and management, agent, rule base, inference engine, search methods and matching techniques.	PLO2, PLO5
	CLO2- To identify the difference between natural language processing, pattern recognition, expert systems and neural network.	PLO2, PLO4

	CLO3- To analyze the various algorithm related to NLP, pattern recognition, rule base, Genetic algorithm and ANN.	PLO2, PLO4
	CLO4- To apply Fuzzy logic, Generic Algorithm and ANN in many real-life applications and also Apply NLP, perceptron and Robotics in sensor data.	PLO3, PLO6, PLO8

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Getting started with python and Create environment for AI and machine learning. Preprocessing and labeling data.	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report
CLO1	Understand supervised learning – Classification. Building classifier in Python using LR/DT/RF. Understand performance of the classifier.	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report
CLO3, CLO4	Understand clustering. Algorithm for clustering the data. Measuring the clustering performance. Design KNN based classifier for a real time problem.	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report
CLO1	Extracting statistics from time series data. Analyzing sequence data by HMM. Analysis of stock market.	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report
CLO1	Concept of searching in AI. Understand various types of searching algorithm. Using searching algorithm design for games.	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report

CLO3	Reading, writing and displaying an image. Color space conversion. Edge, face and eye detection.	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report
CLO4	Design and construct a FLC for solving real life problem. Design and construct a NF system for solving real life problem.	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report
CLO3, CLO4	Build an ANN based classifier. Apply ANN classifier in designing logic gates.	Lecture Problem Based Learning (PBL) Group Assignment	Short Answer Practical Exam Report
CLO3, CLO4	Understand the difference between shallow and deep machine learning. Design and construct a CNN model in python and use it as image classifier.	Problem Based Learning (PBL) Lecture Group Assignment	Practical Exam Exercise Report

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√			√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand solving problems by Searching Problem-Solving Agent, various types of searching technique.
 - Learn knowledge-based system, FL and ANN.
 - Learn how to apply NLP and Perception in data.

- Cognitive skills (thinking and analysis)
 - Be able to apply inference engine, fuzzy-genetic algorithm and artificial neural network in designing intelligent agent.
 - Be able to implement simple searching method, NLP, rule base systems, genetic algorithm and artificial neural network for various real-life applications, such as robotic application, image analysis, character recognition, pattern analysis, etc. using MATLAB/Python.
 - Be able to incorporate machine intelligence.
- Communication skills (personal and academic)
 - Individual assignments after each lab work and group assignments including demonstration and presentation involve receiving clear instructions, designing and writing an effective report and making an effective presentation.
 - A term project (group work) - where students will be assessed in terms of their effectiveness as an individual or leader in the team.
- Practical and subject specific skills (transferable skills)
 - Be able to apply various methods for designing new digital signal processing systems and for continued learning.

ICT 4200: Research Project

Study of the problems in the field of Information and Communication Technology.

Option - II

ICT 4211: Data Mining and Knowledge Discovery

Course Objectives:	The objective of this course is to ● Understand about data mining and its importance as well as application of data mining in real life situation. ● Understand types of data mining techniques. ● How to apply the techniques of clustering, classification, association finding, feature selection and visualization on real world data. ● How to set up a data mining process for an application, including data preparation, modelling and evaluation.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To demonstrate advanced knowledge of data mining concepts and techniques.	PL01, PL02

	CLO2- To apply the techniques of clustering, classification, association finding, feature selection and visualization on real world data and data mining software and toolkits in a range of applications.	PLO3, PLO5
	CLO3- To determine whether a real world problem has a data mining solution.	PLO4, PLO6
	CLO4- To set up a data mining process for an application, including data preparation, modelling and evaluation.	PLO2, PLO3, PLO4

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO3	Introduction: Definition of data mining, its application and challenges; Area and types of data mining.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1	Related technologies - Machine Learning, DBMS, Statistics, Data Mining Goals, Stages of the Data Mining Process, Data Mining Techniques, Knowledge, and Representation Methods. Data Warehouse and OLAP: Data Warehouse and DBMS.	Lecture Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO2, CLO4	Data preprocessing: Types of data and data set; Data cleaning, selection, Data transformation, Data reduction, Discretization and linearization.	Lecture Problem Based Learning (PBL) Home work	Essay Type Answer Group Assignment Presentation
CLO2, CLO3	Exploratory analysis: Descriptive statistics; Visualization using Plots and diagrams and OLAP.	Lecture Problem Based Learning (PBL) Assignment	Essay Type Answer Individual Assignment Presentation
CLO2, CLO4	Classification: Types of classification, Hunts algorithms for Decision trees classification. Validating a model.	Lecture Problem Based Learning (PBL)	Essay Type Answer Individual Assignment

CLO2	Clustering: K-means, hierarchical and density-based clustering and pitfalls of these clustering.	Lecture Problem Based Learning (PBL)	Essay Type Answer Individual Assignment
CLO2	Text mining: extracting attributes (keywords), structural approaches (parsing, soft parsing), Bayesian approach in classifying text.	Lecture Problem Based Learning (PBL)	Essay Type Answer
CLO2, CLO3, CLO4	Web mining: classifying web pages, extracting knowledge from the web, Data Mining software and applications.	Lecture Problem Based Learning (PBL)	Essay Type Answer

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√

Suggested Texts:

- Introduction to Data Mining, 2nd Edition by Vipin Kumar, Anil K. Patek, and Arun K. Jain,
- Data Mining: Practical Machine Learning Tools and Techniques, 2nd Edition by Morgan Kaufmann, 2005.
- Data Mining and Data Warehousing, 2nd Edition by Bharat Bhushan Agarwal, Sumit Prakash Tayal, Publisher: Mcgraw Hill.
- Data Warehousing, Data Mining and OLAP, 1st Edition by Alex Berson and Stephen J. Smith, Publisher: Mcgraw Hill.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the data mining and different types of data mining along with their application.
 - Gain knowledge on data processing techniques.
 - Gain clear knowledge on how to validate any mining model.

- Cognitive skills (thinking and analysis)
 - ▯ Be able to understand processing and selection of data for any other task.
 - ▯ Be able to understand hidden information that exists beneath the descriptive output of data.
- Communication skills (personal and academic)
 - ▯ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - ▯ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - ▯ Be able to identify the complex and real life data using effective understanding of the data mining.

ICT 4213: Digital Image Processing

Course Objectives:	The objective of this course is to ● Study the image fundamentals and mathematical transforms necessary for image processing. ● Study the image enhancement techniques. ● Study image restoration procedures. ● Study the image segmentation and compression procedures. ● Gain understanding of algorithm, analytical tools and practical implementations of various digital image applications.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To analyze images in the frequency domain using various transforms and review the fundamental concepts of a digital image processing system.	PL01, PL02
	CLO2- To evaluate the techniques for image enhancement and image restoration and categorize various compression techniques.	PL04, PL05
	CLO3- To interpret image compression standards with image segmentation and representation techniques.	PL04, PL05

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction: Elements of digital image processing systems, Elements of visual perception, brightness, contrast, hue, saturation, Color image fundamentals - RGB, HSI models, Image sampling, Quantization, dither, Two-dimensional mathematical preliminaries, 2D transforms - DFT, DCT, KLT, SVD.	Lecture Discussion	Essay Type Answer Exercise
CLO2	Digitization of images and its properties: Sampling, quantization, metric properties, topological properties, histogram, noise.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO2, CLO3	Data structures for image analysis and Image processing: Computer program = data + algorithm, Levels of Image Data Representation, Relational models, Traditional Image Data Structures, Topological Data Structures, Relational Structures, Hierarchical Data Structures, Pyramids.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO3	Segmentation: detection of discontinuities, edge linking and boundary detection, thresholding, region-oriented segmentation, use of motion in segmentation.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2, CLO3	Image transforms: Z-transform, 2D Fourier transform, discrete cosine transform, Hadamard transform, Walsh transform, Slant transform.	Lecture Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation

CLO2	Image compression: run-length coding, transform coding, standards.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
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Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√	√	√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√

Suggested Texts:

- Digital Image Processing and Analysis: Application with MATLAB and CVIP tools, 3rd Edition by SE Umbaugh, Taylor & Francis, Publisher: CRC Press, 2018.
- Digital Image Processing, 4th Edition by Rafael C. Gonzalez.
- Principles of Digital Image Processing: Advanced Methods, 2013th Edition by Wilhelm Burger, Mark J. Burge.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of image analysis.
 - Students will learn basic image processing theories and their real-world applications.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any image.
 - Be able to design and conduct digital imaging experiments.
 - Be able to analyze and interpret image and video data.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.

- Practical and subject specific skills (transferable skills)
 - Be able to identify the complex image patterns to be used for machine learning.

ICT 4215: Bio-informatics

Course Objectives:	The objective of this course is to • Introduce students to the fundamentals of molecular biology and recent advance in genomics technology. • Introduce basic principles of modern bioinformatics. • Teach the students how to apply many of the basic bioinformatics methods that are of common use in the field. • Provide students with a practical and hands-on experience with common bioinformatics tools and databases. • Learn basic theories and programs used for database searching, protein and DNA sequence analysis, and prediction of protein function.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To explain the basic principles of Molecular Biology that underpin Bioinformatics analyses and apply these principles when analyzing biological data.	PL01, PL02
	CLO2- To collect data from Data Bank and Survey a selected field within Bioinformatics, synthesize information from primary literature and coherently report the findings in a written document.	PL02, PL04, PL05
	CLO3- To analyze biological data using a variety of Bioinformatics tools.	PL04, PL05
	CLO4- To interpret correctly the outputs from tools used to analyze biological data and make meaningful predictions from these outputs.	PL03, PL04, PL06

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy

CLO1	Introduction: Introduction to bioinformatics and importance and application. Area of Bioinformatics	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1	Molecular Biological terms: Structure of Chromosome, Gene, Protein, amino acid, nucleotides, Introduction to single letter code of amino acids, symbols used in nucleotides.	Lecture Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO2, CLO3, CLO4	Motif: Regulatory Motifs in DNA Sequences; Profiles; The Motif Finding Problem; Search Trees; Finding Motifs; Finding a Median String; Motif in protein.	Lecture Problem Based Learning (PBL) Home work	Essay Type Answer Group Assignment Presentation
CLO1, CLO3	Mapping Algorithm: Molecular Biology Primer; Copying DNA; Cutting and Pasting DNA ; Measuring DNA Length; Probing DNA; How Do Individuals of a Species Differ?; How Do Different Species Differ?; Why Bioinformatics?; Restriction Mapping; Impractical Restriction Mapping Algorithms; Practical Restriction Mapping Algorithm.	Lecture Problem Based Learning (PBL)	Essay Type Answer
CLO2	Bioinformatics databases: Classification of biological databases, Introduction, Nucleotide sequence databases, Protein sequence databases. Protein information resources: Biological data basics – primary secondary data basics – protein pattern data basics data retrieval – Entrez and PDB.	Lecture Problem Based Learning (PBL)	Essay Type Answer Individual Assignment
CLO3, CLO4	Alignment techniques: Database searching; comparison of two sequences; identity and similarity – local and global similarity; global and local alignment; Alignment with Gap Penalties; multiple sequence alignment – data basis of multiple alignment – secondary data base.	Lecture Problem Based Learning (PBL)	Essay Type Answer Individual Assignment
CLO3, CLO4	Genome rearrangements and phylogeny: Definition of genome rearrangements and its implications; Types of genome rearrangements, 4-approximation Algorithm; Concept of phylogeny.	Lecture Problem Based Learning (PBL)	Essay Type Answer

Assessment Pattern:

CLO	Continuous Assessment	Term Final
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	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√

Suggested Texts:

- Bioinformatics for Dummies, 2nd Edition by Jean-Michel Claverie, Cedric Notredame.
- Bioinformatics: a concept-based Introduction, 1st Edition by Venkatarajan Subramanian Mathura.
- An Introduction to Bioinformatics Algorithms, 2nd Edition by Neil C. Jones and Pavel A. Pevzner.
- Principles of Bioinformatics by Shanmughavel.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand bioinformatics and its data collection and analyzing techniques.
 - Gain knowledge on sequence matching.
 - Gain clear knowledge on how species are different within and between their groups.
- Cognitive skills (thinking and analysis)
 - Be able to understand human genome structure and its effect on species.
 - Be able to understand hidden information that exists in genetic level.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to identify the complex and real life data using the effective understanding of the genetic and proteomic analysis.

Course Objectives:	The objective of this course is to • Provide Insights About the Roles of a Data Scientist. • Enable to Analyze of Big Data. • Train the Techniques and Tools for Transformation of Data. • Make Understand Data Mining. • Learn Data Visualization and Optimization.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To learn the necessary skills to manage visualize and analyze data.	PLO1, PLO2
	CLO2- To learn concepts such as exploratory data analysis, statistical inference and modeling, machine learning and visualization.	PLO4, PLO5
	CLO3- To learn the necessary skills to develop data products including R programming, data wrangling, reproducible research and communicating results.	PLO9, PLO10, PLO11

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Data Science Introduction, Introduction to Python, Conditionals and Loops and Patterns.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1	Fraud and Risk Detection, Healthcare, Internet Search, Targeted Advertising Website Recommendations, Advanced Image Recognition, Speech Recognition, Airline Route Planning, Gaming, Augmented Reality.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO1	Data Collection and Data Blending, Business Intelligence and Data Warehousing.	Lecture Discussion	Essay Type Answer Exercise Presentation

		Problem Based Learning (PBL) Demonstration	
CLO2	Data Mining, Machine learning process steps.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2	Classifier, Classification model, Feature Extraction, Binary Classification, Multi-class classification, Multi-label classification.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation
CLO2	K-Means Clustering, Mean-Shift Clustering, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), Expectation–Maximization (EM) Clustering using Gaussian Mixture Models (GMM), Agglomerative Hierarchical Clustering.	Lecture Discussion Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO2	Different types of methods: Word frequency, Collocation, Concordance, Text Classification, Text Extraction.	Lecture Group Assignment Demonstration	Essay Type Answer Report Exercise Presentation
CLO3	Applications of Time series analysis, clustering, statistical tests, linear and non-linear modeling and many statistical computing and analysis options will be conducted by R.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√			√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√

Suggested Texts:

- Data Science Concepts and Techniques with Applications, 1st Edition by Usman Qamar Muhammad Sumair Raza.
- Practical Statistics for Data Scientists, 2nd Edition by Peter C. Bruce, Andrew Bruce, O'REILLY.
- Doing Data Science: Straight Talk from the Frontline, 1st Edition by Cathy O'Neil and Rachel Schutt, O'REILLY.
- The Art of Data Science, Null Edition by Roger D. Peng and Elizabeth Matsui.
- Big Data: A Revolution That Will Transform How We Live, Work and Think, 1st Edition by Viktor Mayer-Schönberger and Kenneth Cukier.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of algorithm analysis and design.
 - Understand the fundamental steps of designing and optimizing algorithms.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any problem and derive its solution.
 - Be able to design and optimize algorithms to solve particular complex problems efficiently.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to identify the complex and real life problems to solve using effective understanding of the design and analysis of algorithms.

ICT 4219: Simulation and Modeling

Course Objectives:	The objective of this course is to • Provide fundamental and practical concepts of computer simulation and modeling techniques and its role in engineering management problem solving. • Cover models and model building applications of simulation in engineering, engineering management issues and introduction to ARENA system modeling framework. • Design both broad in scope and practical in its applications covering areas, such as management, manufacturing, maintenance and service.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand the fundamental concepts of computer simulation and its role in engineering problem solving.	PLO1, PLO2
	CLO2- To develop and model engineering problems and apply procedures for modeling systems using ARENA framework. Appreciate the advantages of using simulation and modeling for taking decision in engineering problems.	PLO3, PLO5, PLO6
	CLO3- To understand the need to incorporate simulation and modeling considerations throughout the design and execution of a project aiming at understanding its limitations and ways of improvement.	PLO4, PLO7

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
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CLO1	Simulation modeling basics: systems, models and simulation with examples.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1	Classification of simulation models: Classification, Steps in a simulation study; Concepts in discrete-event simulation: event-scheduling vs. process-interaction approaches, time-advance mechanism, and continuous simulation models; Combined discrete-continuous models.	Lecture Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO2	Application of Simulation: Analysis and modeling of some practical systems, Organization of a discrete-event simulation model and estimate different parameters.	Lecture Problem Based Learning (PBL) Home Work	Essay Type Answer Group Assignment Presentation
CLO2, CLO3	Building a model: Monte Carlo simulation; Simulation of queuing systems. Building valid and credible simulation models.	Lecture Problem Based Learning (PBL) Assignment	Essay Type Answer Individual Assignment Presentation
CLO2, CLO3	Validation: Validation principles and techniques, statistical procedures for comparing real-world observations and simulated outputs, input modeling. Random number: Generating random numbers and random varieties; Output analysis. Simulation languages.	Lecture Problem Based Learning (PBL)	Essay Type Answer Individual Assignment

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√

Suggested Texts:

- Simulation Modeling and Analysis, 2nd Edition by Law M, Kelton W.

- Discrete-event modeling and simulation: a practitioner's approach by Gabriel A. Weiner.
- Excel Data Analysis Modeling and Simulation, 1st Edition by Hector Guerrero.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand different types of simulation and its application.
 - Gain knowledge on the process of creating a simulation model.
 - Gain clear knowledge on how to validate any simulation model.
- Cognitive skills (thinking and analysis)
 - Be able to understand any system, its input, output, and all types of data that can be analyzed.
 - Be able to create an artificial system using simulation.
- Communication skills (personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - Be able to identify the complex and real life data using effective understanding of simulation modeling processes.

IT 4221: Machine Learning

Course Objectives:	The objective of this course is to • Learn supervised and unsupervised learning models. • Learn various types of machine learning algorithm. • Understand the tuning parameters of the algorithms.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To develop an appreciation for what is involved in learning models from data.	PL03
	CLO2- To understand a wide variety of learning algorithms.	PL01, PL02

	CLO3- To understand how to evaluate models generated from data.	PLO4, PLO5
	CLO4- To apply the algorithms to a real-world problem, optimize the models learned and report on the expected accuracy that can be achieved by applying the models.	PLO6, PLO9, PLO10

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO3	Introduction: Definition of learning systems. Goals and applications of machine learning. Aspects of developing a learning system: training data, concept representation, function approximation.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1	Inductive Classification: The concept learning task. Concept learning as search through a hypothesis space. General-to-specific ordering of hypotheses. Finding maximally specific hypotheses. Version spaces and the candidate elimination algorithm. Learning conjunctive concepts. The importance of inductive bias.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO1	Decision Tree Learning: Representing concepts as decision trees. Recursive induction of decision trees. Picking the best splitting attribute: entropy and information gain. Searching for simple trees and computational complexity. Occam's razor. Overfitting, noisy data, and pruning.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2, CLO4	Ensemble Learning: Using committees of multiple hypotheses. Bagging, boosting and Decorate. Active learning with ensembles. Experimental Evaluation of Learning Algorithms: Measuring the accuracy of learned hypotheses. Comparing learning algorithms: cross-validation, learning curves, and statistical hypothesis testing.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation

CLO2, CLO3, CLO4	Computational Learning Theory: Models of learnability: learning in the limit; probably approximately correct (PAC) learning. Sample complexity: quantifying the number of examples needed to PAC learn. Rule Learning: Propositional and First-Order: Translating decision trees into rules. Heuristic rule induction using separate and conquer and information gain. First-order Horn-clause induction (Inductive Logic Programming) and Foil. Learning recursive rules. Inverse resolution, Golem, and Progol.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation
CLO2, CLO4	Artificial Neural Networks: Neurons and biological motivation. Linear threshold units. Perceptron: representational limitation and gradient descent training. Multilayer networks and backpropagation. Hidden layers and constructing intermediate, distributed representations. Overfitting, learning network structure, recurrent networks. Support Vector Machines: Maximum margin linear separators. Quadratic programming solution to finding maximum margin separators. Kernels for learning non-linear functions. Bayesian Learning: Probability theory and Bayes rule. Naive Bayes learning algorithm. Parameter smoothing. Generative vs. discriminative training. Logistic regression. Bayes nets and Markov nets for representing dependencies.	Lecture Discussion Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO2, CLO4	Instance-Based Learning: Constructing explicit generalizations versus comparing to past specific examples. k-Nearest-neighbor algorithm. Case-based learning. Text Classification: Bag of words representation. Vector space model and cosine similarity. Relevance feedback and Rocchio algorithm. Versions of nearest neighbor and Naive Bayes for text. Clustering and Unsupervised Learning: Learning from unclassified data. Clustering. Hierarchical Agglomerative Clustering. k-means partitional clustering. Expectation maximization (EM) for soft clustering. Semi-supervised learning with EM using labeled and unlabeled data.	Lecture Group Assignment Demonstration	Essay Type Answer Report Exercise Presentation

CLO2, CLO4	Language Learning: word-sense disambiguation, sequence labeling. Hidden Markov models (HMM's). Viterbi algorithm for determining most-probable state sequences. Forward-backward EM algorithm for training the parameters of HMM's. Use of HMM's for speech recognition, part-of-speech tagging, and information extraction. Conditional random fields (CRF's). Probabilistic context-free grammars (PCFG). Parsing and learning with PCFGs. Lexicalized PCFGs.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
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Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√

Suggested Texts:

- Neural Networks and Fuzzy Logic by Ling S, UTS.
- Introduction to Artificial Neural Systems, 1st Edition by Zurada J.M., Publisher: West Publishing Company.
- Neural Networks for Pattern Recognition, 1st Edition by Bishop C., Publisher: Oxford Univ. Press, 2004.
- Artificial Intelligence, 4th Edition by Stuart Russell, Peter Norvig.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the basic concepts of machine learning algorithm analysis and design.
 - Understand the fundamental steps of designing and optimizing machine learning algorithms.
- Cognitive skills (thinking and analysis)
 - Be able to understand and analysis any problem and derive its solution.
 - Be able to design and optimize machine learning algorithms to solve particular complex problems effectively.
- Communication skills (personal and academic)

- ▢ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
- ▢ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (transferable skills)
 - ▢ Be able to identify the complex and real life problems to solve using effective understanding of design and analysis of machine learning algorithms.

ICT 4223: Embedded System Design

Course Objectives:	The objective of this course is to ● Discuss the major components that constitute an embedded system. ● Implement small programs to solve well-defined problems on an embedded platform. ● Develop familiarity with tools used to develop in an embedded environment.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand what is a microcontroller, microcomputer and embedded system.	PL01
	CLO2- To understand different components of a micro-controller and their interactions.	PL01, PL05
	CLO3- To become familiar with programming environment used to develop embedded systems.	PL03, PL05
	CLO4- To understand key concepts of embedded systems, like IO, timers, interrupts, interaction with peripheral devices.	PL01, PL05
	CLO5- To learn debugging techniques for an embedded system.	PL04, PL07

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO3, CLO5	Introduction: Embedded Systems Overview, Design Challenge, Processor Technology, IC Technology, Design Technology, Trade-Offs.	Lecture Assignment	Short Answer Exercise

CLO2	Custom Single Purpose Processors: Combinational Logic; Sequential Logic; Custom Single Purpose Processor Design; Rt-Level Custom Single Purpose Processor Design; Optimizing Custom Single Purpose Processors.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO1, CLO2	General Purpose Processors: Basic Architecture; Operation; Programmer's View; Development Environment; ASIPs; Selecting a Microprocessor; General Purpose Processor Design.	Lecture Assignment	Short Answer Discussion
CLO1, CLO2	Standard Single-Purpose Processors: Timers, counters And Watchdog Timer; UART; Pulse Width Modulators; LCD Controllers; Keypad Controllers; Stepper Motor Controllers; Analog to Digital Converters; Real Time Clock.	Lecture Assignment	Short Answer Presentation
CLO2, CLO3, CLO4	Memory: Memory Write Ability and Storage Permanence; Common Memory Types; Composing Memory; Memory Hierarchy and Cache; Advanced RAM.	Lecture Assignment	Short Answer Presentation
CLO2, CLO3, CLO4	Interfacing: Communication Basics; Microprocessor Interfacing: I/O Addressing; Microprocessor Interfacing: Interrupts; Microprocessor Interfacing: Direct Memory Access; Arbitration; Multilevel Bus Architecture; Advance Communication Principles; Serial Protocols; Parallel Protocols; Wireless Protocols.	Lecture Assignment	Short Answer Presentation
CLO2, CLO3, CLO4	Introduction To Real Time Operating Systems: Tasks and Task States; Tasks and Data; Semaphores and Shared Data.	Lecture Assignment	Short Answer Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√	√	√	√
CLO2	√	√	√	√	√
CLO3	√	√	√	√	√
CLO4	√	√	√	√	√
CLO5	√	√	√	√	√

Suggested Texts:

- Embedded Systems: A Contemporary Design Tool, 2nd Edition by Peckol, James, Publisher: John Wiley and Sons, 2008.
- Analog Interfacing to Embedded Microprocessor Systems, 2nd Edition by Ball, Stuart, Newnes, 2004.
- Embedded System Design, A Unified Hardware/Software Introduction, Student Edition by Frank Vahid, Tony Givargis, Publisher: John Wiley, 2006 reprint.
- An Embedded Software Primer, 1st Edition by David .E. Simon, Fourth Impression 2007, Publisher: Pearson Education.
- Computers as Components: Principles of Embedded Computing System Design, 3rd Edition by Morgan Kaufmann, Marilyn Wolf, 2012.

Intended Learning Outcomes:

- *Knowledge and understanding*
 - Understand fundamental of a microcontroller, microcomputer and embedded system.
 - Become familiar with different components of a micro-controller and their interactions.
 - Become familiar with programming environment used to develop embedded systems.
 - Learn how to design an embedded system using microelectronics devices.
 - Learn the debugging techniques for an embedded system.
 - Understand key concepts of embedded systems like IO, timers, interrupts, interaction with peripheral devices.
- *Cognitive skills (thinking and analysis)*
 - Be able to understand importance of embedded systems, differences between microcontrollers and microprocessor, hardware platforms and peripherals.
 - Students will be able to use different software for embedded system design.
 - Students will be able to explain the architecture of various controllers, the pin diagram and internal structure of input/output devices, RAM organization and working registers.
 - Students will be able to complete small project on embedded system.
- *Communication skills (personal and academic)*
 - Individual assignments including demonstration and presentation involve receiving clear instructions, designing and making an effective presentation.
 - A term project (group work) - where students will be assessed in terms of their effectiveness as an individual or leader in the team.
- *Practical and subject specific skills (transferable skills)*
 - Be able to apply various methods for designing new digital signal processing systems and for continued learning.

ICT 4225: Research Methodology

Course Objectives:	The objective of this course is to • Understand research terminology. • Be aware of the ethical principles of research, ethical challenges and approval processes. • Describe quantitative, qualitative and mixed methods approaches to research. • Identify the components of a literature review process. • Critically analyze published research.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To understand a general definition of research design.	PLO1
	CLO2- To know why educational research is undertaken and the audiences that profit from research studies.	PLO2, PLO9
	CLO3- To identify the overall process of designing a research study from its inception to its report.	PLO10, PLO11
	CLO4- To get familiar with ethical issues in educational research, including those issues that arise in using quantitative and qualitative research and know the primary characteristics of quantitative research and qualitative research and more.	PLO8, PLO11

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1, CLO2, CLO3	Study design: cross-sectional, case-control, cohort, and intervention studies. Measures of disease frequency and risk and alternative sources of epidemiological data. Interpretation of epidemiological and statistical concepts as causality, random errors, bias, confounding.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

CLO1, CLO2, CLO3	Describing univariate and bivariate data: tables and graphs; proportions; measures of central tendency (mean, median) and variability (range, standard deviation, percentiles); correlation coefficients, differences and ratios.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration Demonstrate	Essay Type Answer Report Exercise Presentation
CLO1, CLO2, CLO3	Statistical inference: confidence intervals and p-values, hypotheses tests. Simple and multiple linear and logistic regression analysis. Statistical analyses using software SPSS.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
CLO4	Qualitative research methods: observational method, interviews, focus group discussions, participatory methods. Measures of illness perceptions and experiences; participant accounts of everyday life. Describing the data collection process; sampling principles; the role of gatekeepers; the interactions between researcher and researched; the links between theory and method Qualitative data analysis, validity and triangulation in qualitative research.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√	√		√	√
CLO2	√	√		√	√
CLO3	√	√		√	√
CLO4	√	√		√	√

Suggested Texts:

- Basic epidemiology, 2nd Edition by Bonita, Ruth; Beaglehole, Robert, Kjellström, Tord, Geneva, WHO.

- Health research methodology: a guide for training in research methods, 2nd Edition by Manila, WHO.
- Principles of biostatistics, 2nd Edition by Pagano, Marcello, Gauvreau, Kimberlee, Pacific Grove.
- Qualitative methods for health research, 2nd Edition by Judith M. Green and Nicki Thorogoo.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand a general definition of research design.
 - Familiarize with ethical issues in educational research, including those issues that arise in using quantitative and qualitative research.
 - Learn the primary characteristics of quantitative research and qualitative research.
 - Familiarize with how to write a good introduction to an educational research study and the components that comprise such an introduction.
 - Define the meaning of a variable and be able to identify independent, dependent and mediating variables.
 - Define theory use in quantitative research.
 - Define a central phenomenon in qualitative research.
 - Learn the steps in the process of quantitative data collection.
 - Learn the various types of quantitative sampling and find which one presents the most rigorous approach to use.
 - Learn the types of descriptive statistics typically reported in educational research studies.
 - Learn how to conduct a statistical test of a hypothesis and which criteria for selecting appropriate statistical test.
 - Learn the various types of validity strategies typically used in good qualitative research.
 - Learn the conventions with good APA style for scholarly writing.
- Cognitive skills (thinking and analysis)
 - Understand why educational research is undertaken and find the audiences that profit from research studies.
 - Identify the overall process of designing a research study from its inception to its report.
 - Identify a research problem stated in a study.
 - Learn about conducting a literature review for a scholarly educational study.
 - Distinguish a purpose statement, a research question or hypothesis and a research objective.
 - Distinguish between categorical and continuous measures.
 - Distinguish between a population and a sample.
 - Know various types of quantitative sampling and find which one presents the most rigorous approach to use.
 - Understand the link between quantitative research questions and data collection and how research questions are operationalized in educational practice.
 - Distinguish between the writing structure used for a quantitative study and for a qualitative study.
 - Learn how to evaluate a quantitative study and a qualitative study.

- Communication skills (personal and academic)
 - ▮ An individual assignment (demonstration and presentation) involves writing effective research reports using proper writing structure and making effective presentations.
- Practical and subject specific skills (transferable skills)
 - ▮ Design a good quantitative purpose statement and good quantitative research questions and hypotheses.
 - ▮ Design a good qualitative purpose statement and a good central question in qualitative research.
 - ▮ Write scholarly report using proper writing convention.

Option - III

ICT 4227: Digital Forensic

Course Objectives:	The objective of this course is to ● Emphasize the importance of digital forensics. ● Prepare students to conduct a digital investigation in an organized and systematic way. ● Provide theoretical and practical knowledge as well as current research on Digital Forensics.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To Describe digital forensics and relate it to an investigative process.	PLO1, PLO4
	CLO2- To Explain the legal issues of preparing for and performing digital forensic analysis based on the investigator's position and duty.	PLO8
	CLO3- To Demonstrate use of digital forensics tools and perform basic digital forensics.	PLO2, PLO3
	CLO4- To Guide a digital forensics exercise and recognize the state of the practice and the gaps in technology, policy, and legal issues.	PLO3, PLO5, PLO8

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Introduction to Computer Forensics: Understanding the need for computer forensics, Defining computer forensic.	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer Exercise
CLO1, CLO4	Computer Hardware: Understanding computer components, Digital Media, Hard disk basics Files and File Systems: Windows file systems, - FAT32 - NTFS - Forensic file images	Lecture Discussion Problem Based Learning (PBL)	Essay Type Answer MCQ Exercise
CLO2, CLO4	Forensic Tools: Forensic hardware, Hardware write/blockers, Hard drive acquisitions, Processing the scene. Forensic Software: Overview of different software packages, EnCase Introduction.	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2, CLO3	Bookmarking and Searching: Creating basic search queries, Hex, Decimal, and Binary - ASCII - Unicode GREP: Understanding GREP, Building Regular Expressions, Creating GREP keywords, Viewing and managing keywords and cases	Lecture Discussion Problem Based Learning (PBL) Demonstration	Essay Type Answer Exercise Presentation
CLO2, CLO3, CLO4	Forensic Reports: Creating a forensic report, Proper report writing, and Explaining forensics to the uneducated. Email Analysis: Viewing e-mail, Webmail, POP, IMAP. File Signature Analysis: File signatures, File extensions, Differences between, Identifying differences. Hash Analysis: Understanding hash algorithms, Hashing files, Hash libraries.	Lecture Group Assignment Problem Based Learning (PBL) Demonstration	Report Exercise Presentation

CLO4	Other Windows Artifacts: Common windows artifacts, recycle bin, My Documents, Recent files, Installed programs, Windows XP vs. Windows 7.	Lecture Discussion Group Assignment Problem Based Learning (PBL) Demonstration	Essay Type Answer Report Exercise Presentation
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Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1		√			√
CLO2	√				√
CLO3	√				√
CLO4	√			√	√

Suggested Texts:

- Practical Mobile Forensics, 4th Edition by Rohit Tamma, Oleg Skulkin, Heather Mahalik, Satish Bommisetty, 2020.
- Digital Forensics and Incident Response, 2nd Edition by Gerard Johansen, 2020.
- Cyber Security, 1st Edition by Jocelyn O. Padallan, 2019.
- File System Forensic Analysis, 1st Edition by Brian Carrier.
- Handbook of Digital Forensics and Investigation, 1st Edition by Eoghan Casey.

Intended Learning Outcomes:

- Knowledge and understanding
 - ▢ Digital Forensics methodology with a solid understanding of requirements for handling digital evidence.
 - ▢ Requirements and impact on maintaining evidence integrity and chain of custody.
 - ▢ Principles, procedures, and the basic concepts of forensic standards and best practices, e.g. forensic tool testing.
 - ▢ The overall process for establishment and maintenance of a digital forensic lab environment.
 - ▢ The role of expert witnesses and digital evidence in the context of legal proceedings.

- The role of policies, standards and guidelines for controls and is capable of applying his/her knowledge in case studies.
- Legal, privacy and ethical aspects of digital forensics investigations.
- Cognitive skills (personal and academic)
 - Forensic acquisition of digital evidence from computer and network media.
 - Live system forensics and evaluation of order of volatility.
 - Evidence analysis with timeline analysis and forensic reconstruction.
 - Scientific documentation of forensic acquisition and analysis.
 - Applying forensic principles on practical case-studies.
 - Performing stakeholder analysis, risk assessment and forensic triage on limited case-studies.
 - Evaluating the applicability of forensic methods and tools for various controls given a certain scope and policy for the control.
- Communication skills (personal and academic)
 - Capability of analyzing business, legal, ethical and case-specific requirements for planning and conducting a digital forensics investigation.
 - Understanding of forensic analysis and incident response processes.
 - Working independently and familiarity with digital forensics terminology.
 - Capability of discussing professional problems such as documentation, decision making processes, implementation plans, operations, reviews and corrective actions, with forensic experts, IT specialists and general managers.
 - Learning skills to continue acquiring new knowledge and skills in a largely self-directed manner.
 - Ability to contribute to innovative thinking and innovation processes.

ICT 4229: Communication Management

Course Objectives:	The objective of this course is to ● Acquire the knowledge about network management standards (OSI and TCP/IP) ● Acquire the knowledge about various network management tools and the skill to use them in monitoring a network ● Analyze the challenges faced by Network managers ● Evaluate various commercial network management systems and open network management systems. ● Analyze and interpret the data provided by a NMS and take suitable action.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping

	CLO1- To Understand comprehensive view of the network management problem, including its concepts, scope, challenges, and techniques.	PLO1, PLO2
	CLO2- To Understand the network management concepts in concrete applications and scenarios such as MPLS, VPN, VLAN, etc.	PLO5, PLO6
	CLO3- To Interpret application on emerging networking technologies such as Software Defined Networking for managing future IT infrastructures.	PLO5, PLO11, PLO12

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	Network Management: Introduction to Network Management, Network Management Architecture, Concepts, Managers and agents. TMN (Telecommunications Management Network), Network management goals and dimensions.	Lecture Assignment	Short Answer Exercise
CLO1, CLO2	Network Management System: Configuration management, Fault management Performance management, Security management, Account management. Fault Tolerance.	Lecture Assignment Demonstrate	Short Answer Identification Exercise
CLO1, CLO2	Network Management Functions Configuration: Configuration Management and Auto-discovery, Configuration Database and Reports, Abstract Syntax Notation One (ASN.1).	Lecture Assignment	Short Answer Discussion
CLO1. CLO2	Network Management Functions – Security: Security Management, Protecting Sensitive Information, Host and User Authentication, Key Management.	Lecture Assignment	Short Answer presentation
CLO1, CLO2.	Network Management Functions – Fault: Fault Management, Fault Identification and Isolation, Event Correlation Techniques.	Lecture Assignment Demonstrate	Short Answer presentation
CLO1, CLO2	Simple Network Management Protocol - SNMP v1: Structure of Management Information, Std. Management Information Base (MIBs), SNMPv1 Protocol.	Lecture Assignment	Short Answer Presentation

CLO2	Simple Network Management Protocol - SNMP v2: Version 2 Protocol Specification, Version 2 MIB Enhancements, MIB-II, Case Diagrams.	Lecture Assignment	Short Answer presentation
CLO2, CLO3	Simple Network Management Protocol - SNMP v3: Version 3 Protocol & MIB, User Based Security Model, View Based Access Model.	Lecture Assignment	Short Answer Discussion
CLO2, CLO3	Remote Network Monitoring RMON 1: Statistics Collection, Alarms and Filters, Remote Network Monitoring RMON 2, Monitoring Network Protocol Traffic, Application-Layer Visibility, Management Tools, Systems and Applications, Test and Monitoring Tools, Integrating Tools, Development Tools, Web-based Enterprise Management. Management Component: Role of SNMP, Role of SMI, Role of MIB. SMI, Name, Type, Encoding Method.	Lecture Assignment	Short Answer presentation
CLO3	Simple Network Management System Protocol (SNMP): Concept, Management Component, Structure of Management Information, Management Information Base (MIB), Lexicographic Ordering, SNMP, Message, UPD port, Security.	Lecture Assignment	Short Answer Discussion

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√			√	√
CLO3	√	√			√

Suggested Texts:

- Network Management Principles and Practice, 2nd Edition by Mani Subrahmanian, Publisher: Pearson Education, 2010.
- Network management, 1st Edition by Morris, Publisher: Pearson Education, 2008.
- Principles of Network System Administration, 1st Edition Mark Burges, Publisher: Wiley DreamTech, 2008.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand networking stack and interdependencies among different layers
 - Define network management (e.g. configuration, monitoring, problem discovery, traffic policy, etc.)
- Cognitive skills (thinking and analysis)
 - Identify and compare approaches to automatically managing networks
 - Analyze network problems to determine the root cause and relevant layer
 - Describe how network problem discovery can be automated
 - Identify and describe common network problems
 - Explore and analyze classic and recent approaches to solving common network problems
- Communication skills (personal and academic)
 - Develop research skills related to finding and reading current literature
 - Write a review of current literature on an aspect of network management
 - Select a technical paper and summarize it for the class in a brief presentation
- Practical and subject specific skills (Transferable Skills)
 - Practice managing a small-scale network, including writing scripts
 - Apply recent tools to managing a small-scale network, including some programming

ICT 4231: Microwave Engineering and Satellite Communication

Course Objectives:	The objective of this course is to ● Introduce about RF/microwave research and construction techniques. ● Discuss scattering parameters, device, and machine behavior. ● Use passive device, active device, and two-port network signal transmission in microwave subsystems. ● Design protocols as well as strategies for evaluating system results. ● Explain the link design and link carrier-to-noise ratio performance factor. ● Discuss about basic satellite terminology, provide students with a solid foundation in orbital dynamics and/satellite launches, enlighten students on various access networks and earth station developments, ● Introduce students to global positioning system and data packets.
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Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To State the basic concept of microwaves, microwave measurements, Microwave link, Microwave antenna, Radar.	PLO2
	CLO2- To Describe the construction, working principle, applications of basic microwave components and devices, and Time Domain Reflectometry (TDR) Systems.	PLO2, PLO3
	CLO3- To Explain the working principle and applications of microwave tubes amplifier and oscillators.	PLO2, PLO5
	CLO4- To Analyze and design basic microwave amplifiers, RF filters, RF oscillator, and mixer models.	PLO2, PLO3
	CLO5- To Understand satellite terminology, provide foundation in orbital dynamics and/satellite launches, enlighten students on various access networks and earth station developments.	PLO4, PLO12
	CLO6- To evaluate the link design and performance of the microwave/satellite communication systems.	PLO11

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
CLO1	An introduction to microwave engineering Transmission lines as circuit elements Smith chart analysis methods.	Lecture Assignment	Short Answer Exercise
CLO2	Impedance transforming and matching circuits Line and waveguide structures and associated components Power waves and the network scattering matrix	Lecture Assignment	Short Answer Exercise
CLO3, CLO4	Passive devices Two-port network signal transmission Active devices Communication link design	Lecture Assignment Case Study Group Discussion	Short Answer Exercise MCQ

CLO5	Communication Satellite: Orbit and Description: A Brief history of satellite Communication, Satellite Frequency Bands, Satellite Systems, Applications, Orbital Period and Velocity, effects of Orbital Inclination, Azimuth and Elevation, Coverage angle and slant Range, Eclipse, Orbital Perturbations, Placement of a Satellite in a Geo-Stationary orbit.	Lecture Assignment Problem Based Learning (PBL)	Short Answer Identification Group Exercise
CLO5, CLO6	Satellite Sub-Systems: Attitude and Orbit Control system, I I & C subsystem, Attitude Control subsystem, Power systems, Communication subsystems, Satellite Antenna Equipment. Satellite Link: Basic Transmission Theory, System Noise Temperature and G/T ratio, Basic Link Analysis, Interference Analysis, Design of satellite Links for a specified C/N, (With and without frequency Re-use), Link Budget.	Lecture Assignment Case Study Group Discussion	Short Answer Discussion Exercise MCQ
CLO5	Propagation effects: Introduction, Atmospheric Absorption, Cloud Attenuation, Tropospheric and Ionospheric Scintillation and Low angle fading, Rain induced attenuation, rain induced cross polarization interference. Multiple Access.	Lecture Assignment	Short Answer Discussion
CLO5, CLO6	Earth Station Technology: Transmitters, Receivers, Antennas, Tracking Systems, Terrestrial Interface, Power Test Methods, Lower Orbit Considerations. Satellite Navigation and GPS Systems: Radio and Satellite Navigation, GPS Position Location Principles, GPS Receivers, GPS C/A Code Accuracy, Differential GPS.	Lecture Assignment	Short Answer Assignment MCQ

Assessment Pattern:

CLO	Continuous Assessment				Term Final
	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2				√	√

CLO3	√			√	√
CLO4		√			√
CLO5	√				√
CLO6		√			√

Suggested Texts:

- Microwave Engineering, 4th Edition by David M. Pozar, Publisher: John Wiley & Sons, 2011.
- Microwave Engineering, 1st Edition by Sanjeeva Gupta.
- Satellite Communications, 2nd Edition by Timothy Pratt, Charles Bostian, Jeremy Allnutt, Publisher: John Wiley & Sons, 2013.
- Principle of Carriers Communication, 1st Edition by N. Biswas.
- Satellite Communications Engineering, 2nd Edition by Wilbur, L. Pritchard, Robert A. Nelson and Heuri G. Suyderhoud, Publisher: Pearson.
- Principles of Communication Engineering, 2006 Edition by A.K. Chhabra.
- Microwave communication system, 1st Edition by V.K. Mourya.

Intended Learning Outcomes:

- Knowledge and understanding
 - Acknowledge the principles of operation and the main features of microwave engineering & satellite communications and their applications.
 - Understand the impact of engineering solutions in a global, economic, environmental and societal context.
- Cognitive skills (Thinking and analysis)
 - Be able to identify, formulate, and solve engineering problems.
 - Be able to understand Impedance transforming and matching circuits, line and waveguide structures and its associated components, power waves and the network scattering matrix.
 - Be able understand operation of satellite subsystems.
 - Be able to compute various types of losses in microwave and satellite systems.
 - Be able to analyze Active and Passive devices Two-port network signal transmission of microwave systems and effects of propagation in satellite communication.
- Communication skills (Personal and academic)
 - An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations and giving and receiving clear instructions.
 - Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.

- Practical and subject specific skills (Transferable skills)
 - Be able to apply design microwave and satellite communication link.

ICT 4233: Multimedia Communication

Course Objectives:	The objective of this course is to ● Understanding the multimedia communications systems, application and basic principles. ● Analysis of the multimedia streaming. ● Performing and establishing multimedia communication terminals. ● Presentation of multimedia communications.	
Course Learning Outcomes (CLOs):	On completion of the course students will be able:	CLO-PLO Mapping
	CLO1- To describe technical characteristics and performance of multimedia system and terminals.	PL01, PL02
	CLO2- To design creative approach in application of multimedia devices, equipment and systems.	PL03
	CLO3- To carry out experiments and measurements on the multimedia systems in laboratory conditions on real components and equipment.	PL04, PL011
	CLO4- To interpret and analyze measurement results obtained on the multimedia system and components.	PL04, PL05
	CLO5- To describe the development process and applications of the multimedia systems.	PL09, PL011
	CLO6- To test multimedia communication systems and equipment in real conditions.	PL06, PL07

Course Contents:

CLO	Course Contents	Teaching Learning Strategy	Assessment Strategy
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CLO1	Introduction to Multimedia Communication. Information Theory: Info theoretical foundation for lossless coding: entropy, conditional entropy, relative entropy, mutual information, prefix coding, kraft-McMillan inequality. Entropy Coding: Shannon-Fano and Huffman	Lecture Assignment	Short Answer Exercise
CLO1	Variable Length Coding in JPEG and H.264 and JPEG/MPEG	Lecture Assignment	Short Answer Exercise
CLO1, CLO2	Arithmetic Coding Transforms: KL transform (de-correlation interpretation), PCA (geometry interpretation), SVD, DCT, and Graph Fourier Transform (GFT)	Lecture Assignment	Short Answer presentation
CLO3, CLO4	Scalar and Vector Quantization schemes in image coding. CABAC in HEVC.	Lecture Assignment	Short Answer presentation
CLO2, CLO3, CLO5	Video Signal Processing: YCbCr color space sampling, block based motion model, sub-pixel resolution motion estimation, fast algorithms in motion estimation; MV Prediction, Intra Prediction, Deblocking, SAO and Scalability.	Lecture Assignment	Short Answer presentation
CLO2, CLO3, CLO4	Video Coding Standard and Systems: HEVC. Rate-Distortion Optimization: Lagrangian Method; Video Coding Mode Decision and Rate Control	Lecture Assignment Problem Based Learning (PBL)	Short Answer Exercise
CLO3, CLO4, CLO6	Rate Control in HEVC; Deep Learning in Compression, Quality of Experiences (QoE), MPEG System I: ISO BMFF and DASH; Media Transport I: Congestion Control; Media Transport II: Error Control	Lecture Assignment Problem Based Learning (PBL)	Short Answer Group Discussion Exercise

Assessment Pattern:

CLO	Continuous Assessment	Term Final
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	Class Test	Assignment	Project	Presentation	
CLO1	√				√
CLO2	√				√
CLO3		√			√
CLO4	√	√			√
CLO5	√			√	√
CLO6	√			√	√

Suggested Texts:

- Introduction to Multimedia Communications: Applications, Middleware, Networking, 1st Edition by Kamisetty Rao, Zoran Bojkovic, Dragorad Milovanovic, 2006.
- Multimedia Communication Systems—Techniques, Standards And Networks, 1st Edition by Dragorad A. Milovanovic, 2012.
- Multimedia, 1st Edition by Robert Burnett, Anna Brunstrom, Publisher: Wiley.
- Wireless Multimedia Communication System, 1st Edition by K.R. Rao, Publisher: CRC Press, 2014.

Intended Learning Outcomes:

- Knowledge and understanding
 - Understand the fundamental concept of multimedia communication and entropy theory.
 - Understand the coding and performance metric of multimedia communication systems.
- Cognitive skills (thinking and analysis)
 - Be able to explain entropy coding and arithmetic coding.
 - Be able to apply entropy transformation and apply quantization schemes in image coding.
 - Be able to investigate video signal processing and video coding scheme.
 - Be able to apply deep learning in image/video compression.
 - Be able to design and develop congestion and error control
- Communication skills (personal and academic)

- ▯ An individual assignment (demonstration and presentation) involves writing effective reports and designing documentation, making effective presentations, and giving and receiving clear instructions.
- ▯ Group assignment (to be done in a group) - where students will be assessed in terms of their effectiveness as individual or leader in a team.
- Practical and subject specific skills (Transferable Skills)
 - ▯ Be able to use incorporating digital modulation and coding schemes in communication systems.

ICT 4233: Contemporary Course on Information and Communication Technology

The content and detail of this course will be designed by the course teacher.